Seven ordered points and the nonexistence of arithmetic ball quotient compactifications

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Abstract

We prove that the ordered GIT quotient of seven points on $\mathbb{P}^1$ with equal weights is not isomorphic to the Baily–Borel compactification of any arithmetic complex ball quotient. Combining this result with our companion paper and the known positive examples, we conclude that, for $n \geq 6$, the corresponding quotient of n points is isomorphic to the Baily–Borel compactification of an arithmetic complex ball quotient precisely when $n = 6, 8$. We allow any arithmetic lattice and any algebraic isomorphism, as well as torsion and divisorial ramification. We combine arithmetic reductions with identities involving Chern classes of fixed subvarieties and intersection inequalities for nef divisors on the GIT quotient. In the final comparison, we include all fixed components and need no bound on the degree of an individual fixed curve.

1 Introduction

Let

$$X_n = (\mathbb{P}^1)^n /\!/_{\mathcal{O}(1,\dots,1)} \mathrm{SL}(2,\mathbb{C})$$

be the ordered GIT quotient, in which the symmetric group is not divided out, and put $X = X_7$. In this paper, we prove the following theorem.

Theorem 1.1. There is no arithmetic lattice $\Gamma \subset \mathrm{PU}(1,4)$ for which

$$X \simeq (\mathbb{B}^4/\Gamma)^{\mathrm{BB}}$$

as algebraic varieties.

The theorem permits arbitrary torsion and divisorial ramification, and neither the branch divisor nor the action on labels is prescribed. Our companion paper [15], cited in its public version arXiv:2610.23228v1, excludes such isomorphisms for every admissible weight vector on at least nine labels. Together with the Deligne–Mostow realizations at six and eight points [2, 8, 9], this leaves seven points as the last case for equal weights. Theorem 1.1 therefore gives the following classification.

Corollary 1.2. For every $n \geq 6$, the variety X_n is the Baily–Borel compactification of an arithmetic ball quotient if and only if $n = 6, 8$.

The result concerns abstract algebraic isomorphisms, rather than a prescribed period map; it makes no assertion about nonarithmetic lattices or other compactifications.

We now outline the proof; Sections 2 and 3 fix the notation. The geometry that controls the automorphic class is supplied by the 35 coincidence planes. Each is a copy of $\mathbb{P}^2$ with normal bundle $\mathcal{O}(-1)^{\oplus 2}$, so Taylor coefficients in the normal directions bound the restriction of that class without lifting label permutations. Sections 4 and 5 exclude cocompact realizations and nonuniform realizations outside the Gaussian field. Section 6 constructs a self-dual lattice preserved by the original group and proves that $L - 3H/2$ is nef. For the group $G(4,2,2)$, this

last step uses a finite cover of a coincidence plane of degree three: its trace kernel is a twist of the normal bundle, and its cubic coefficients count the G_{31} points (Lemma 6.7).

Two results organize the final comparison. Proposition 7.1 expresses the classes of fixed strata in terms of the automorphic class and the Chern classes of the coarse space, in arbitrary dimension under the hypotheses on finite volume in its statement. Proposition 7.3 gives the same linear relations for finite quotients of $\mathbb{P}^3$ and polarized abelian threefolds; these describe interior points and cusps, respectively. We prove these identities here and use [15, Propositions 2.2–2.4 and 3.1] for the remaining inputs. In Section 7, the resulting linear equalities and nef inequalities have no common solution. An explicit rational linear combination proves this incompatibility, retaining every fixed component, full normalizer, inertia character and cusp width, without bounding the degree of an individual curve.

2 Geometry and preliminary comparisons

Put $P_7 = (\mathbb{P}^1)^7$, and let P_7^{s} be the locus where at most three labels coincide at any point. The effective diagonal action of $\mathrm{PGL}(2,\mathbb{C})$ is free on P_7^{s}, and stability equals semistability. The principal bundle $P_7^{\mathrm{s}} \to X$ defines

$$\pi_7 : \mathcal{P}_7 = (P_7^{\mathrm{s}} \times \mathbb{P}^1)/\mathrm{PGL}(2,\mathbb{C}) \longrightarrow X, \qquad [(p,z)] \longmapsto [p],$$

with sections $\sigma_i([p]) = [(p, p_i)]$. We put $H = -K_X$; then

$$H = (2,\dots,2), \qquad \mathrm{Pic}(X) = \{b \in \mathbb{Z}^7 : \textstyle\sum_i b_i \in 2\mathbb{Z}\}.$$

For each of the 35 sets I of three labels, the coincidence plane Z_I of these labels is $\mathbb{P}^2$, its normal bundle is $\mathcal{O}_{\mathbb{P}^2}(-1)^{\oplus 2}$, and $H|_{Z_I} = \mathcal{O}_{\mathbb{P}^2}(1)$. These assertions follow from the free quotient and its marked sections, as in [15, Proposition 2.5, Equation (2.2), and Proposition 2.6].

Suppose that $X \simeq (\mathbb{B}^4/\Gamma)^{\mathrm{BB}}$ for an arithmetic lattice $\Gamma \subset \mathrm{PU}(1,4)$ as in Theorem 1.1, and identify $\mathbb{B}^4/\Gamma$ with the open part of X. We realize $\mathbb{B}^4$ as the space of negative lines in a Hermitian space of signature $(1,4)$. Let $\mathcal{L}$ be the automorphic rational line bundle on $\mathbb{B}^4/\Gamma$, normalized by $K_{\mathbb{B}^4} = \mathcal{L}^5$, and put $L = c_1(\mathcal{L})$. We write $\chi_{\mathrm{orb}} = \chi(\mathbb{B}^4/\Gamma')/[\Gamma : \Gamma']$, where $\Gamma' \subset \Gamma$ is a torsion-free subgroup of finite index; proportionality gives $\chi_{\mathrm{orb}} = 5\int_X L^4$. The full normalizer of a fixed ball is its stabilizer in Γ, and its effective image is the quotient by the pointwise stabilizer [15, Lemma 2.12]. Let C be the smooth projective normalization of the Baily–Borel compactification of a fixed disk modulo the effective image of its full normalizer in Γ, and let $f : C \to X$ be the induced finite map. We set

$$a = \deg f^*H, \qquad t = \deg f^*L, \qquad x = g(C) - 1.$$

For an invariant direction of polynomial degree e and dimension one preserved by the full normalizer, E_e denotes its equivariant bundle with all finite inertia characters, extended to the coarse curve C using the local invariant coordinates given below at each upper point and cusp. Its orbifold degree is $-et$; its degree on C adds the corrections at these points computed in these coordinates. We write S for the reduced cusp divisor on C, put $s = \deg S$, and use subscripts for specified subsets of cusps. When R counts the points of C with a specified type of larger stabilizer, D_R denotes the reduced divisor of these points. We put $\sigma = \sum_z (1 - e_z^{-1})$, where z runs over the points of C with a stabilizer that is a direct product and e_z is the order of the effective inertia on C at z.

We use [15, Proposition 2.2, Proposition 2.3, Proposition 2.4] for quotients of f^*T_X. For a negative target, every homomorphism factors through differences of coincident sections. Write the pullback of $\mathcal{P}_7$ by f as $\mathbb{P}_C(V)$, the projective bundle of quotients of dimension one, with quotient maps $V \twoheadrightarrow L_i$. We write $[ij]$ for their determinant, a section of $(\det V)^{-1} \otimes L_i \otimes L_j$.

Suppose that the labels in I, with $|I| = m$, coincide and their common relative tangent line has degree $-e < 0$. For $j \notin I$, let D_j be the zero divisor of $[ij]$, where $i \in I$, and put $\delta = \sum_{j \notin I} \deg D_j$. Then

$$2 \le m \le 3, \qquad a = (7 - 2m)e + 2\delta, \qquad \delta \ge 0.$$

The remaining relative tangent lines generate the kernel, whose exterior powers are bounded by [15, Proposition 2.3]. A quotient of rank 2 is always required to be surjective at every point, not just at the generic point. The nef criterion of [15, Proposition 2.6] must be checked on every Z_I; it does not identify the fixed balls over different planes.

The original full arithmetic group and its normalized lift are those of [15, Proposition 2.11]. Full stabilizers are taken in the original group itself, not in a subgroup of finite index, and an element, reflection, or local datum is actual if it occurs in this group, not only in the classification of finite reflection groups. Let c be the order of the finite scalar kernel of the map from the normalized lift to Γ. For each fixed ball we use its full normalizer, the finite morphism of [15, Lemma 2.12], and the restriction of $\mathcal{L}$. At a cusp, P denotes the full stabilizer, N its Heisenberg lattice, and $F = P/N$ its finite horizontal group. The horizontal torus A and its positive line bundle are defined with the least positive central period of this N, with the actual action on the fiber. For a smooth cusp, the horizontal holonomy is irreducible. For its actual linearization, the invariant sections form a polynomial ring, whose projectivization is generally a weighted projective space. The least positive central period gives its faithful linearization and the grading of its invariant section ring, as in [15, Lemma 2.16]. The lift fixing the point in [15, Lemma 2.17] applies to the kernel of the actual fiber character. The degree of an invariant section in this grading is distinct from the polynomial degree e of an invariant of the normal representation. In a fixed curve coordinate the ambient central coordinate satisfies $Q = q^{\kappa}$, where κ is its actual relative width. We use [15, Proposition 3.4] and [15, Lemma 3.5] only after specifying the larger stabilizers, the full horizontal stabilizers at the ends, the pointwise fixed groups lifted from them as kernels of finite characters, and the widths given by these characters.

The dependencies on the companion paper are summarized in Table 1. All statement numbers in this table refer to arXiv:2610.23228v1; the hypotheses stated there remain in force.

Lemma 2.1. For $X = X_7$, let $\mathcal{A}$ be the adjoint bundle of the principal bundle $P_7^{\mathrm{s}} \to X$, with fiber $\mathfrak{sl}_2(\mathbb{C})$. Put $\psi_i = c_1(\sigma_i^* T_{\mathcal{P}_7/X})$ and $p = -c_2(\mathcal{A})$. Then, in the rational Chow ring of X,

$$H = \sum_i \psi_i, \qquad \psi_i^2 = p, \qquad c_2(T_X) = \frac{H^2 - 5p}{2}, \qquad c_3(T_X) = \frac{H^3 - 13Hp}{6}.$$

Proof. The section σ_1 represents the universal family as $\mathbb{P}_X(\mathcal{V})$, where $\mathcal{V} = \pi_{7*}\mathcal{O}_{\mathcal{P}_7}(\sigma_1(X))$ has rank 2. For each quotient $\mathcal{V} \twoheadrightarrow \mathcal{L}_i$, the relative tangent line has first Chern class $2c_1(\mathcal{L}_i) - c_1(\mathcal{V})$, whose square equals $c_1(\mathcal{V})^2 - 4c_2(\mathcal{V}) = -c_2(\mathrm{End}_0(\mathcal{V})) = p$. The tangent sequence is $0 \to \mathcal{A} \to \bigoplus_i \sigma_i^* T_{\mathcal{P}_7/X} \to T_X \to 0$. Since $c_1(\mathcal{A}) = c_3(\mathcal{A}) = 0$, expanding $c(T_X) = \prod_i (1 + \psi_i)/(1 - p)$ through degree 3 proves the formulas. □

3 The automorphic degree on the coincidence planes

Throughout this section, $X = X_7$, $H = -K_X$, and $\mathcal{L}$ is the automorphic line bundle. For a curve $f : C \to X$, we write $a = \deg f^* H$ and $t = \deg f^* \mathcal{L}$. The degree of $\mathcal{L}$ is understood on the orbifold until an indicated power descends to X. For a fixed curve, $\mathcal{N}$ denotes $f^* T_X / (df(T_C))^{\mathrm{sat}}$. For an invariant direction of polynomial degree e preserved by the full normalizer, E_e denotes its line bundle with the finite inertia characters included; its degree is stated in each calculation. In the tables, a string such as 1112 abbreviates the tuple $(1, 1, 1, 2)$. The rank of a finite linear

Source	Hypotheses and conclusion used	Use in this paper
Propositions 2.2–2.4	Complete curves in the stable quotient; homomorphisms to negative lines factor through coincident sections, with the stated kernel and determinant bounds.	Tangent and normal quotients.
Propositions 2.5–2.6	The marked family and the coincidence subvarieties; the nef criterion requires every relevant coincidence plane.	The classes of X_7 and the estimate for $L - 3H/2$.
Proposition 2.11	A hypothetical arithmetic realization; the integral model and scalar normalization retain the original full group.	Arithmetic reductions.
Lemma 2.12	A fixed ball with its full generic stabilizer; its full normalizer has finite volume and a finite map, with the original automorphic bundle restricted.	All fixed components; its additional product hypothesis is required for the stated ordinary normal bundle.
Proposition 2.14 and Lemma 2.15	The product hypothesis for persistence, and nonuniformity for embedding into the horizontal group of an actual cusp of the original lattice, preserving the reflection representation and orders, with the rank bounds stated there.	Reduction of the possible stabilizers.
Lemmas 2.16–2.17	A smooth cusp with its actual central period and fiber action; a finite lift is obtained for the kernel of the fiber character.	Degrees of invariant theta sections, finite characters, and the relative width.
Propositions 3.1, 3.4 and Lemma 3.5	Full normalizers, all larger stabilizers, the full horizontal stabilizers at the ends of the curve, the pointwise fixed groups lifted from them as kernels of finite characters, and these characters, whose images give the widths.	Determinant and monomial comparisons, after their hypotheses are checked.
Equation 3.13	The same restricted automorphic class on a fixed ball.	The Chern polynomial used in Proposition 7.1.

Table 1: Results of the companion paper used in this paper.

group is the codimension of its fixed subspace. We use G_j for the Shephard–Todd group numbered j, and A_j, B_j for the usual Weyl groups. We write $\mu_b = \{\zeta \in \mathbb{C} : \zeta^b = 1\}$, and $I_2(m)$ denotes the dihedral Coxeter group of order $2m$. For positive integers m, p, r with $p \mid m$, $G(m, p, r)$ consists of the monomial matrices of size r whose nonzero entries lie in μ_m and have product in $\mu_{m/p}$.

Lemma 3.1. Suppose that X is the Baily–Borel compactification of an arithmetic ball quotient, and let $Z \simeq \mathbb{P}^2$ be a coincidence plane. Write $L|_Z = \lambda c_1(\mathcal{O}_Z(1))$. If the nontrivial part of the normal representation of the full generic stabilizer has rank 0 or 1, then $\lambda \geq 1$. If it has rank 2 and its least positive invariant degree is q, then $q\lambda \geq 1$.

Proof. On a neat cover defined by a normal subgroup, resolve a component above Z, with maps $h : S \to Y$ and $\tau : S \to Z$. Let G_0 be the stabilizer of the normalized component in the deck group, and let K_0 be the kernel of its action on that component. The quotient G_0/K_0 acts faithfully and transitively on a general fiber, so the morphism from the normalized component to Z restricts over a suitable dense open subset of Z to a finite Galois cover with group G_0/K_0. Let R be the common divisor of the maximal minors of dh. After removing R, the Gauss equation gives a positive current in $c_1(K_S - R - 3h^*\mathcal{L})$, whose potential is locally bounded above. On the inverse image of degree e of a general line avoiding cusps and exceptional images,

$$(K_S - R) \cdot \tau^*\mathcal{O}_Z(1) \geq 3\lambda e. \tag{3.1}$$

For ranks $0, 1$, apply [15, Lemma 3.10] with $k = c = 2$. The normal bundle is $\mathcal{O}_Z(-1)^{\oplus 2}$, the set of cusps on Z is finite, and the full normalizer retains the restricted $\mathcal{L}$ by [15, Lemma 2.12]. Thus $\lambda \geq 1$.

For rank 2, the fixed ball maps finitely to the punctured plane. The full normalizer acts with finite image on the positive normal space U, and $\underline{U}$ denotes the associated flat bundle on the neat cover. The fixed ball has normal bundle $\mathcal{L}^{-1} \otimes \underline{U}$ [15, Lemma 2.12]. Let T be the full generic stabilizer, put $R_T = \mathrm{Sym}(U^\vee)^T$, and let $R_{T,+}$ be its ideal of positive degrees. We denote by $\underline{U}_q$ the bundle associated with the dual of $(R_{T,+}/R_{T,+}^2)_q$, retaining the finite action of the full normalizer. If q is the least positive invariant degree and $k = \operatorname{rank} \underline{U}_q$, the first normal Taylor coefficients of the plane ideal give a map $\mathcal{L}^{-q} \otimes \underline{U}_q \to \tau^*\mathcal{O}_Z(-1)^{\oplus 2}$ of generic rank k. Kill the finite monodromy on a compact inverse line; a negative coefficient degree would make this map zero. Thus $q\lambda \geq 1$. □

The following two assertions concern a nonuniform arithmetic realization over an imaginary quadratic field K. We put $w_K = |\mu(K)|$, where $\mu(K)$ is the group of roots of unity in K; thus $w_K \in \{2, 4, 6\}$.

Lemma 3.2. Let $Z \simeq \mathbb{P}^2$ be a coincidence plane defined by three labels, and put $L|_Z = \lambda c_1(\mathcal{O}_Z(1))$. If the normal representation of its full generic stabilizer has rank 0, then $\lambda \geq 1$. If it has rank 1 and its image has order b, then $(1 - b^{-1})\lambda \geq 1/3$. If it has rank 2 and the least positive invariant degree is q, then $q\lambda > 1$. In all cases $w_K L - 2H$ is nef, and hence $a \leq w_K t/2$ on every curve.

Proof. By Lemma 3.1, only the case $q\lambda = 1$ in rank 2 remains. Retain its notation; since $\mathcal{L}^c$ descends, $q \mid c$. The bundle $\mathcal{S} = \mathcal{L}^q \otimes \tau^*\mathcal{O}_Z(-1)$ is torsion, with an equivariant trivialization of its cth power. The bundle $\mathcal{F} = \mathcal{S}^{-1} \otimes \underline{U}_q$ is finite flat. The central scalar ζ acts by ζ^q on both factors before taking their quotient; thus $\mathcal{F}$ and the map $\mathcal{F} \to \mathcal{O}_M^{\oplus 2}$ are equivariant for the actual stabilizer of the component, acting projectively. Constancy on compact inverse lines after a finite cover gives rank k at every interior point; the source $\mathcal{F}$ of this injective map then has trivial point inertia. It descends to the simply connected punctured plane as a trivial finite flat bundle, and Hartogs's theorem extends its constant rank k matrix to Z. [15, Lemma 2.15] supplies an actual cusp of this fixed ball. In a unipotent cover with central coordinate Q and horizontal normal coordinates $y \in U$, the plane equations and their Taylor coefficients of degree

q are divisible by Q, since the zero section contracts to the cusp. The trivialization of $\mathcal{S}^c$ is pulled back from $(\mathcal{L}^c|_Z)^q \simeq \mathcal{O}_Z(c)$ on the whole plane, using the Baily–Borel extension of [15, Proposition 2.11]. In the frame of $\mathcal{L}$ on a Siegel domain at the cusp and a regular frame of $\mathcal{O}_Z(-1)$, this trivialization is a unit at $Q = 0$. Once the finite coefficient action on the same cover is killed, its local cth root is a unit, so the matrix of Taylor coefficients of degree q still vanishes at $Q = 0$. The constant matrix of rank $k > 0$ therefore vanishes along a vertical ray, a contradiction.

Finally $q \leq w_K$ in every possible group of rank 2. For monomial groups this follows from their degrees $m, 2m/p$ and their character fields; a dihedral group has degree 2. For primitive groups, [15, Lemma 2.15] excludes G_5, G_8, G_{12}: the primitive groups of rank 3 over quadratic fields are G_{24}, G_{25}, G_{26} and their only primitive parabolic group of rank 2 is G_4, of degrees $4, 6$ [7, Table 1], [10, Table 4]. Thus the remaining primitive case has $K = \mathbb{Q}(\omega)$ and $q = 4 \leq 6$. The integer $w_K\lambda$ is strictly greater than 1 in rank 2 and at least 2 in ranks 0 and 1. [15, Proposition 2.6] applied to the 35 planes gives the asserted nef inequality.

□

Lemma 3.3. In the setting of Lemma 3.2, if the full generic stabilizer has rank 0 or 1, then $\lambda > 1$.

Proof. Suppose $\lambda = 1$ in rank 0 or 1. The equality case of [15, Lemma 3.10], with the same $k = c = 2$, makes the regular inverse image locally a totally geodesic ball of dimension 2, inside the generic fixed ball in rank 1. Let $B = \mathbb{B}(W)$ be this ball in the universal cover. The identity theorem makes B an entire irreducible component of the closed inverse image of Z. Local finiteness of these components proves properness of the quotient M_0 of B by its stabilizer; its generically injective immersion thus normalizes the component and maps finitely to the punctured plane. Remove deep cusp neighborhoods with compact complement [6]. Their intersections with B are horoballs if the isotropic vector lies in W, and bounded geodesic balls or empty otherwise [17]. In the latter case write the isotropic vector of the cusp as $\rho = \rho_W + \rho_U \in W \oplus W^\perp$; then $h(\rho_W, \rho_W) = -h(\rho_U, \rho_U) < 0$ and the height is $-h(z, z)/|h(\rho_W, z)|^2$. Properness over a compact regular horosphere gives finitely many cuts, with compact horizontal cross sections for the horoballs. Finite volume follows from $\int_{u_0}^{\infty} u^{-3}\, du < \infty$ on the horoballs, the boundedness of the other cuts, and the compact complement.

Borel density makes the stabilizer on W irreducible [1]. The K-algebra of its actual linear lifts is semisimple, since its complexification acts faithfully and completely reducibly on $W \oplus W^\perp$. Its unique central idempotent of rank 3 projects onto W; the positive complement has dimension 2, so Galois invariance of that rank defines both summands over K. The preserved lattice makes the normal action of the full stabilizer finite, and the original full stabilizer gives a finite birational map onto the normal punctured plane, hence an isomorphism.

For rank 1, let U_0 be the normal axis fixed by the generic cyclic group; for rank 0 let it be the entire positive normal space. The first Taylor coefficients of the plane ideal give a generically injective map $\mathcal{L}^{-1} \otimes \underline{U}_0 \to \tau^*\mathcal{O}_Z(-1)^{\oplus 2}$. Apply the argument excluding equality in Lemma 3.2 with $q = 1$ and $\underline{U}_q = \underline{U}_0$. Let c be the order of the full scalar kernel and put $\mathcal{S} = \mathcal{L} \otimes \tau^*\mathcal{O}_Z(-1)$. The fixed ball is now defined over K, its full normal action is finite, and the two factors have the same central scalar character. The first derivatives also exclude an additional inverse branch by the implicit function theorem. Thus their quotient descends to a nonzero constant matrix on the punctured plane. Meyer's theorem in six variables supplies an actual cusp, and the same trivialization of $\mathcal{S}^c$ on the whole plane is a unit in the frame of $\mathcal{L}$ on a Siegel domain at this cusp. The first Taylor coefficients are divisible by the central coordinate Q, giving the same contradiction.

□

A coincidence plane with full generic stabilizer of rank 2 is the compactification of its fixed ball quotient: [15, Lemma 2.12] gives a finite birational map, and the plane is normal.

4 Seven points and cocompact lattices

Throughout this section we assume that $X = X_7$ is the quotient of $\mathbb{B}^4$ by a cocompact arithmetic lattice Γ. We retain $H = -K_X$ and $L = c_1(\mathcal{L})$ with $K_{\mathbb{B}^4} = \mathcal{L}^5$. The branch divisor, its indices, and the isomorphism with X are arbitrary. [15, Proposition 2.11] gives a normalized lift generated by reflections over a CM field K; let F be its maximal real subfield and D_K, D_F their absolute discriminants. Every stabilizer below is the full stabilizer, and every quotient of a fixed ball uses the full normalizer. On each fixed plane, d_k is the total degree of the branch components of ramification index k. Lemma 4.3 first forces a factor of rank four. The normal quotient comparisons eliminate the remaining groups except F_4; Lemmas 4.11, 4.14, and 4.17 then force an F_4 point and exclude G_{29}. Lemma 4.16 yields $L \equiv H$, contradicting the arithmetic volume calculation in Lemma 4.13.

For a fixed curve $f : C \to X$ we put $x = g(C) - 1$, $a = \deg f^*H$, and $t = \deg f^*L$. When df vanishes on a divisor D, we use $N = f^*T_X/T_C(D)$.

4.1 Finite stabilizers

On a fixed curve or plane with full generic stabilizer T, an upper point is a point whose full stabilizer, an upper group, properly contains T; these points form the upper locus. The parabolic classifications [7, Table 1], [10, Theorem 3.6 and Tables 5–9] give the irreducible upper groups in Table 2; products $T \times \mu_b$ retain every positive integer b before reflection orders are restricted. The full list of groups of rank 4 is A_4, $G(m,p,4)$, $F_4, H_4, G_{29}, G_{31}, G_{32}$.

T	irreducible upper groups	effective inertia
$G(m,p,3),\ m \geq 6$	$G(m,p,4)$	m
$G(m,1,3),\ 3 \leq m \leq 5$	$G(m,1,4)$	m
$G(5,5,3)$	$G(5,5,4)$	5
$G(3,3,3)$	$G(3,3,4), G(3,1,4)$	$3,3$
B_3	B_4, F_4, G_{29}	$2,2,4$
$G(4,2,3), G(4,4,3)$	$G(4,2,4), G_{31};\ G(4,4,4), G_{29}$	4
G_{25}, H_3	$G_{32};\ H_4$	$6;\ 2$
A_3	$A_4, D_4, F_4, H_4, G_{29}, G_{31}$	

Table 2: Irreducible upper groups and orders of effective inertia.

After Lemma 4.3, products of rank 3 are $T \times \mu_2$, and also $G_{25} \times \mu_3$; the rank 2 factors are products, $A_2, B_2, I_2(5), G(4,2,2), G_4$. A factor with reflection orders $2,3$ cannot enter rank 4; on a fixed plane it is $G(6,2,2)$. For $G(m,p,2)$, the invariants $S = x^m + y^m$, $P = (xy)^{m/p}$ give the branch equation $S^2 - 4P^p = 0$ and, for $p < m$, $P = 0$ of index m/p. For the full effective local group J on the plane at a point, not in the ambient quotient, a mirror with J-orbit of size ρ and pointwise stabilizer of order b has effective normalized order $|J|/(b\rho)$, separately on every branch of a self-intersection [29, Table 1]. The effective normalized orders for $A_2, I_2(5), B_2, G(4,2,2), G(6,2,2), G_4$ are $1,1,2,4,6,2$. The groups $A_2, I_2(5), G_4$ give cusps $(2,3), (2,5), (2,3)$; B_2 gives two branches of contact 2, and $G(4,2,2), G(6,2,2)$ give ordinary triple points. A product $\mu_b \times \mu_c$ gives a node of effective normalized orders c, b. For planes of degree below $1/2$, the only pairs (T,G) with irreducible upper group G are $(G(4,2,2), G(4,2,3))$, (G_4, G_{25}), (μ_3^2, G_{25}). Their degree corrections and normal determinant orders are $(1/2, 2)$, a nonimmersed axis (v^2, v^3), and $(1/2, 3)$, respectively. The further product $G_4 \times \mu_3$ for μ_3^2 has the same nonimmersed cusp. At G_{32} the effective local group $G(6,1,2)$ on the plane has degrees

6, 12; ambient degrees 12, 18, 24, 30 omit its degree 6 cotangent direction. Thus this point is excluded from a smooth fixed plane, and a cyclic factor contained in a larger irreducible factor meets a point with an excluded upper group. This proves the assertions on products of groups used in Lemma 4.4.

4.1.1 Monomial groups

Put $k = mt$, $D = 3m + 3m/p$ on a $G(m, p, 3)$ axis. The degree m direction is unique except for $p = 3$. For $m \geq 6, p = 3$, coordinate mirrors have squared normalized inner products in $\{0, 1/2\}$, but diagonal mirrors have a pair with value $\cos^2(\pi/m) \notin \{0, 1/2\}$. Thus the full normalizer preserves the coordinate mirrors and their elementary symmetric invariant of degree m. The character at every larger monomial group and product is trivial, so $k \in \mathbb{Z}_{>0}$. The equations of [15, Proposition 3.4] are $2x + (1 - m^{-1})R + \sigma = 2t$ and $a = -2x - Dt + (2 + p^{-1})R$, where $\sigma = 0$ or $\sigma \geq 1/2$. For $m \geq 6$, and for $3 \leq m \leq 5, p = 1$, the inequality $D > 2(2 + p^{-1})/(1 - m^{-1})$ gives $g = 0$. The finite test uses Table 3; h is the target associated with the invariant of greatest degree. In the first row $(k, R, a, h) = (1, 2, 2, -1)$ contradicts $a \geq 3$. In the second row $k \leq 2$,

Group	Larger group	h	Finite inequalities or surviving row
$G(m, 1, 3)$, $m \geq 6$	$G(m, 1, 4)$	$R - 3k$	$2k \leq R \leq 2(k + m)/(m - 1)$; $k(m - 2) \leq m$.
$G(m, p, 3)$, $m \geq 6, p > 1$	$G(m, p, 4)$		$R > (3k - 2)/2$; $k < (6m - 2)/(3m - 7)$.
$G(5, 5, 3)$	$G(5, 5, 4)$	$-3t + R/5$	$(k, R, a, t) = (1, 3, 5, 1/5)$.
$G(m, 1, 3)$, $3 \leq m \leq 5$	$G(m, 1, 4)$	$R - 3k$	$1 \leq k \leq \lfloor m/(m - 2) \rfloor$; $2k \leq R \leq \lfloor 2(k+m)/(m-1) \rfloor$; $a = 2 - 6k + 3R$.

Table 3: Finite tests for the imprimitive groups of rank 3.

except $m = 6, k = 3$ where $a \leq 0$; substituting $k = 1, 2$ leaves no positive integral a with $\sigma = 0$ or $\sigma \geq 1/2$. For the other rows, [15, Lemma 3.5] applied to coefficient degrees $R - 2k, R - k, R$ and the zeros at all R upper points gives Hom dimension 1, attained by the projection of the inverse differential to the coordinate of greatest invariant degree; at $G(5, 5, 4)$ the other degrees are $1, 2 < R$. The tangent sequence in the proof of Lemma 2.1 excludes degree 0 for odd a and degree 1 for even a: negative relative tangent degrees contribute at least $2, 4$, respectively, and otherwise, for even a, evaluation along the constant sections is a constant matrix, so the Hom space is its kernel tensored with $H^0(\mathcal{O}_{\mathbb{P}^1}(1))$, of even dimension. Together with $a \geq -3h$ for $h < 0$, this excludes $G(5, 5, 4)$ and leaves only $(m, k, R, a, t, \sigma) = (3, 2, 5, 5, 2/3, 0)$, of relative tangent degrees $(-1, -1, 3, 1, 1, 1, 1)$.

Lemma 4.1. For an ample GIT class D of multidegree (b_i) and half sum q, one has

$$D^4 = 3q^4 - 2\sum_i (q - b_i)^4 + \sum_{i<j} (q - b_i - b_j)^4.$$

If its degree on a curve of relative tangent degrees $(-1, -1, 3, 1, 1, 1, 1)$ is $2/3$, then $D^4 \leq 28/81$.

Proof. Extract the coefficient of k^4 in $\sum_{|I| \leq 3} (-1)^{|I|} \binom{k(q - \sum_{i \in I} b_i) + 5 - |I|}{5}$. Concavity of $(D^4)^{1/4}$ permits averaging to (x, x, z, y, y, y, y), where $r_0 = x - y$, $y = 2/3 + r_0 - 3z/2$, $x = 2/3 + 2r_0 - 3z/2$, $0 < z < 1/3$, $|r_0| < z/2$. For $u = 3z$, $v = r_0/z + 1/2$, the coefficients of $648D^4$ in $B_i(u)B_j(v)$, $B_j(T) = \binom{4}{j} T^j (1 - T)^{4-j}$, have maximum 224 at $(i, j) = (2, 4)$. This nonnegative basis sums to 1, giving $D^4 \leq 28/81$. □

Lemma 4.2. For any cocompact arithmetic lattice defined over K/F, put $h_{K,5} = |\operatorname{Cl}(K)[5]|$.

Then

$$\chi_{\mathrm{orb}} \geq \frac{5}{h_{K,5}} D_F^{12} (D_K/D_F^2)^7 \left(\frac{288}{(2\pi)^{14}} \right)^{[F:\mathbb{Q}]}.$$

Proof. For split nonhyperspecial places S_{split}, the maximal normalizer index is at most $5^{1+|S_{\mathrm{split}}|} h_{K,5}$ and each local factor exceeds 5. Cancel these factors in Prasad's formula, retaining factors 5 from the compact dual and central covering [21, Sections 2.3 and 2.5–2.9, equations (3), (5), (6), (8)]. □

The integral ample class cL, for scalar order c in [15, Proposition 2.11], has positive integral degree on every coincidence line. Complementary triples therefore make the multidegrees of $cL - H$ and its degrees on all Z_I nonnegative; [15, Proposition 2.6] yields $a \leq ct$. For the last $G(3,1,4)$ row the fixed signature is $(0; 3, 3, 3, 3, 3)$, of area $4/3$. Odd generators lie in an Eichler norm one group of genus zero because the normalizer quotient is an elementary abelian 2-group [11, Theorem 2.1 and Section 3]. Voight's exhaustive list of 1209 groups of genus at most 2 gives $[F : \mathbb{Q}] \leq 7$ [31, Theorem in the Introduction and Section 4]. Here $c \geq 8$, $\zeta_3, \zeta_c \in K$, and $m = \mathrm{lcm}(3, c)$ lies in $\{3, 6, 9, 12, 15, 18, 21, 24, 30, 36, 42\}$. For each row corresponding to a compact orbifold curve of genus zero and area μ, require integral $n = (4/3)/\mu$, an embedding $\mathbb{Q}(\zeta_m + \zeta_m^{-1}) \hookrightarrow F$, and cycles $n = ex_e + (e/3)y_e$ at each cone point, with $x_e, y_e \geq 0$, $y_e = 0$ if $3 \nmid e$, and $\sum y_e = 5$. Testing embeddings by defining polynomials gives the following rows. The rows $(D_F, \text{signature}, n)$ are $(12, (0; 3, 3, 6), 8)$, $(81, (0; 2, 3, 9), 24)$, $(81, (0; 3, 3, 9), 6)$ and $(1125, (0; 3, 3, 15), 5)$. The defining polynomials are $T^2 - 3$, $T^3 - 3T + 1$, $T^4 - T^3 - 4T^2 + 4T + 1$, giving $K = \mathbb{Q}(\zeta_{12}), \mathbb{Q}(\zeta_9), \mathbb{Q}(\zeta_{15})$. In the first field $c = 12$; traces contain every reflection root of unity, but an order 4 or 12 reflection would remain in a factor of rank 2 with equal degrees $q \mid 12$, where no such reflection exists. All orders therefore divide 6, and the generated determinant lies in μ_6, giving $c \mid 30$, a contradiction. The other two fields have class number 1; Lemma 4.2 gives lower bounds greater than 6 and 270, contrary to $\chi_{\mathrm{orb}} = 5L^4 \leq 140/81$.

4.1.2 The group $G(3,3,4)$

Once $G(3,1,4)$ is excluded, the $G(3,3,3)$ axis has only $G(3,3,4)$ and product upper groups. The normalizer preserves the direction of degree 6 but may mix the directions of degree 3. [15, Proposition 3.4] gives $k = 6t \in \mathbb{Z}_{>0}$, $g = 0$, $k \leq 10$, $a = 2 - 2k + 7R/3$, $\sigma = 2 + k/3 - 2R/3$, leaving exactly

$$(k, R, a, \sigma, h) = (2, 3, 5, 2/3, 1), (3, 3, 3, 1, 0), (4, 3, 1, 4/3, -1), (6, 6, 4, 0, 0), \quad h = R - k.$$

The projection of the inverse differential to the coordinate of greatest invariant degree is everywhere surjective onto $\mathcal{O}(h)$. The invariants $v + S_1, vS_1 + S_2, vS_2 + P^3, uP$, with $v = u^3$, $S_1 = \sum y_j^3$, $S_2 = \sum_{i<j} y_i^3 y_j^3$, $P = y_1 y_2 y_3$, give $dS_1 = dg_2/v$, $dP = dg_4/u$ on its kernel. Hence, once finite monodromy is killed, every line subbundle of the kernel has degree at most $R - k/2$; no splitting on C is assumed. For $k = 3$, odd relative tangent degrees of sum 3 force negative labels: three give degree $7 > 3/2$ in the kernel, two prevent surjection to $\mathcal{O}$, and one requires $a \geq 5e$.

Apply the preceding conditions on n, x_e, y_e to the compact genus zero rows in Voight's table, with area $2/3$ and four order 3 points, or area 2 and six order 3 points; the area of an orbifold curve of genus g with cone orders e_i is $-\chi_{\mathrm{orb}} = 2g - 2 + \sum_i (1 - 1/e_i)$. The second test gives the fields below; each pair $([F : \mathbb{Q}], D_F)$ identifies a unique defining polynomial in that

table.

$[F:\mathbb{Q}]$	D_F
1	1
2	5, 8, 12, 13, 17, 24, 28, 29, 33, 60
3	49, 81, 148, 169, 229, 257, 316, 321, 361, 469
4	725, 1957, 2000, 2304, 2777, 3981, 4205, 4352
5	24217, 36497, 38569

.

The first test deletes $24, 29, 33, 60, 321, 361, 469, 2304, 3981, 4205$, leaving 22 fields. For either test, N is the required area divided by the area of the tabulated row, and each cone order e permits precisely the integers $0 \le y_e \le 4$ or $0 \le y_e \le 6$ satisfying $n-(e/3)y_e \in e\mathbb{Z}_{\ge 0}$ and $y_e = 0$ if $3 \nmid e$; the sum of the y_e is 4 or 6. These bounded conditions specify every row used in the test. For $k = 2$, use the first list. The conditions $c \ge 15$, $K = F(\zeta_3)$, $F \ne \mathbb{Q}$, $|\mu(K)| \ge 15$ leave only $K = \mathbb{Q}(\zeta_9)$, $D_F = 81$. Three negative labels violate the kernel bound $13 > 2$; otherwise Lemma 4.1 for $2L$, or averaging with $y = 1/9 + z/6$, $0 < z < 1/3$, gives $L^4 \le 7/324$ or $31/1944$. Every ideal class of $\mathbb{Q}(\zeta_9)$ has a representative of norm at most 5 by Minkowski's theorem. The primes $2, 5$ have residue degree 6, and the unique prime above 3 is generated by $1 - \zeta_9$; hence the class number is 1. Both contradict the arithmetic bound greater than 6. For $k = 6$ the kernel bound excludes negative degrees by $10 > 3$; immersion then gives $N = \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(2)$, $t = 1$. Average to $(z, z, z, z, z, 1/2, 1/2)$, $1/3 < z < 1$, giving $L^4 \le 27/8$. The same conditions with area 2 and six order 3 points give 32 polynomials. For $K = F(\zeta_3)$ put $d = [F : \mathbb{Q}]$, $\Delta = D_K/D_F^2$, $\varpi = 288/(44/7)^{14}$, and

$$M_K = \left\lfloor \left(\frac{4}{3}\right)^d \frac{(2d)!}{(2d)^{2d}} (\lfloor \sqrt{D_K} \rfloor + 1) \right\rfloor, \qquad U_K = \sum_{1 \le m \le M_K} \prod_{p^a \| m} \binom{2d+a-1}{a}.$$

Minkowski's theorem and the coefficients of ζ_K give $h_{K,5} \le h_K \le U_K$. Every ideal class has a representative of norm at most M_K, and the number of ideals of norm p^a is at most $\binom{2d+a-1}{a}$. For the listed fields, $\Delta = 3^d$ except that $\Delta = 1$ at $D_F = 12, 24, 33, 60, 2304$, $\Delta = 3$ at $81, 321$, and $\Delta = 9$ at 3981. The lower bound $5D_F^{12}\Delta^7\varpi^d/U_K$ exceeds $135/8$ in the following cases; the last column bounds its integer part from below for every indicated field.

d	D_F	lower bound
2	13, 17, 28, 29, 60	52
3	49, 148, 169, 229, 257, 316, 321, 361, 469	443
4	725, 1957, 2000, 2304, 2777, 3981, 4205, 4352	1343
5	24217, 36497, 38569	10^{19}

.

For $D_F = 33$, one has $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ and $M_K = 5$. The prime 2 has two primes of residue degree 2, the prime 3 has two primes of ramification index 2 and residue degree 1, and the prime 5 has residue degree 2. Thus the prime ideals of norm at most 5 have norm 3 or 4. Each prime of norm 4 is principal, generated by a conjugate of $(\sqrt{-3} + \sqrt{-11})/2$, whose norm is 4. The square of each prime of norm 3 is principal, generated by a conjugate of $(1 + \sqrt{-11})/2$. Thus $h_{K,5} = 1$, and the volume lower bound exceeds $30 > 135/8$. Consequently $F \ne \mathbb{Q}$, $c \ge 4$ and the volume bound leave the following fields, where $\rho_{K/F}$ is the rational number of Lemma 4.12:

D_F	D_K	$\lvert\mu(K)\rvert$	$\rho_{K/F}$
5	225	6	31/6075
8	576	6	201641/124416
12	144	12	23/497664
24	576	6	551/2560
81	19683	18	18628987/12597120

.

The numerator divides $(cL)^4$ and is coprime to $5c$ because the maximal normalizer index is a power of 5. For $D_F = 81$ it exceeds $18^4(27/8) = 354294$. Otherwise $c = 6$, also using $c \mid 30$ for $D_F = 12$; positive degrees of cL on the planes Z_I and moving weight sum 6 leave $(2,3,3,3,3;3,3)$, $(3,3,3,3,4;3,3)$, $(4,4,4,4,4;2,4)$, $(4,4,4,4,4;3,3)$, of volumes 586, 880, 1450, and 1684. None has the required divisibility, with special values from Section 4.7.1.

For $k = 4$ the negative target gives a line $C \subset Z$, $N = \mathcal{O}(1) \oplus \mathcal{O}(-1)^{\oplus 2}$, and kernel of the projection to the coordinate of greatest invariant degree $\mathcal{O}(1) \oplus \mathcal{O}(-1)$. The orbifold bundle in polynomial degree 3 twisted by $\mathcal{O}_C(2)$ is unitary flat with coarse image $\mathcal{O} \oplus \mathcal{O}(-2)$; Bochner makes the trivial summand parallel. The characters $(1, \zeta_3^2)$, the trivial product action, and the parallel normalizer-invariant lines $P(S_1^3 - 27P^3) = 0$ select a plane fixed by S_3. Its compact quotient Y has maps $(U, V) \mapsto (U, V, 0, 0)$, $U = u^3 + z^3$, $V = u^3 z^3$, at upper points, and $(u, z) \mapsto (u^b, z^3)$ at stabilizers $T \times \mu_b$. Hence $N_{C/Y} = \mathcal{O}(-2)(D_R) = \mathcal{O}(1)$; the vanishing maps $\mathcal{O}_C(1)^{\oplus 2} \to \mathcal{O}_C(-j)$ and Krull intersection give $Y = Z$, $\lambda_Z = 2/3$. The product contribution $4/3$ and three index 2 contacts with C leave $\Delta_Z = 2C/3 + D/2 + 2E/3$, $(\deg D, \deg E) = (6, 2)$, or $\Delta_Z = 2C/3 + D/2 + 5A/6$, $(\deg D, \deg A) = (7, 1)$. The general groups are $G(3,3,3)$ on C and $S_3 \times \mu_b$ elsewhere. [10, Theorem 3.12 and Tables 5–9], applied to the full normalizers, excludes A_4, B_4 by effective degrees $(1,2), (2,2)$ and $D_4, G(4,2,4), G(4,4,4)$ by normalizer/reflection orders $(2,1), (8,4), (4,1)$; the same defect excludes G_{29}, G_{31}. For H_4, degree 6 of $I_2(6)$ is absent from $2, 12, 20, 30$; G_{32} lacks order 2 reflections. The S_3 types in $G(3,3,4)$ have degrees $(1,3)$, excluded, and $(3,6)$, giving the three points on C; in F_4, containment in B_3 contradicts general group $S_3 \times \mu_2$. Any rank 3 factor absorbing S_3 must act on its fixed line, since otherwise immersion requires degree 1; it therefore occurs only on C as $G(3,3,3)$. Off C the irreducible factors of $S_3 \times J$ with reflections of two orders are $G(6,2,2), G_7$ in the first divisor and $G(12,2,2)$ in the second, of effective normalized orders $6, 12, 12$. For A, Bézout and proportionality give $m + 2k = 7$, $-2 + 2/3 + m/2 + 11k/12 = 4/3$, hence $(k, m) = (10, -13)$. For irreducible E they give $m + 2k = 12$, $-2 + 4/3 + m/2 + 5k/6 = 8/3$, hence $(k, m) = (16, -20)$. If E splits, its crossing is off C; a line gives $m + 2k = 6$, $-2 + 4/3 + m/2 + 5k/6 = 4/3$, hence $m = -6$, off D. On D, $J = G_7$ gives $m + 2k = 5$, $-2 + 2/3 + 11/12 + m/2 + 5k/6 = 4/3$, hence $m = -4$.

Lemma 4.3. An irreducible factor of rank 4 occurs in some point stabilizer. The possible groups of rank 4 are $A_4, B_4, D_4, G(4,2,4), G(4,4,4), F_4, H_4, G_{29}, G_{31}, G_{32}$. Every smaller irreducible factor enters one of these groups as a parabolic subgroup, so every reflection has order 2 or 3.

Proof. Suppose first that no point stabilizer has an irreducible factor of rank 4. Follow an irreducible factor through successive absorptions, that is, points at which it becomes a parabolic subgroup of a larger irreducible factor; enlargements of the same rank already lie in its full stabilizer. If the rank stops below 4, [15, Lemma 2.12, Proposition 2.14] exclude ranks $3, 1$ and give $Y = \mathbb{P}^2$, $N_{Y/X} = \mathcal{O}(-1)^{\oplus 2}$ in rank 2. Equal degrees q occur only for $G(m,2,2)$, even $m \geq 4$, and G_7, G_{11}, G_{19}, with $q = m, 12, 24, 60$. At a point x of Y, this factor is a direct factor of the full stabilizer of x, since an absorption would give an irreducible factor of rank at least 3. The effective local groups on the plane are therefore products of cyclic groups or the irreducible groups of rank 2 with equal basic degrees listed above. Their branch germs are nodes or ordinary triple points of types $[2,2,m/2]$, $[2,3,3]$, $[2,3,4]$, and $[2,3,5]$, by [29, Table 1]. Several local branches may belong to the same irreducible curve. Write $\Delta_Y = \sum_i \beta_i D_i$, where $\beta_i = 1 - b_i^{-1}$, and let d_i, g_i, and δ_i be the degree, the genus of the normalization, and the sum of the delta invariants of D_i. In [29, equations (5), (7), and (8)], identifying r local branches of index b_i changes θ_x by $-3\beta_i\binom{r}{2}$. Here Γ_x is the effective local group on the plane, and ρ_1, ρ_2 count the mirrors above two of these branches. Their intersection multiplicity is one, so pulling back gives $b_i^2 \rho_1 \rho_2 = |\Gamma_x|$. Use $\mathcal{P}(Y, \Delta_Y) = \mathrm{Prop}(Y, \Delta_Y) - \frac{1}{2}\sum_i \beta_i \, \mathrm{prop}_{(Y,\Delta_Y)}(D_i)$ with the surface and curve invariants of [29, Definition 3.8 and Remark 3.10]; both vanish for a compact

ball quotient orbifold. Thus [29, Propositions 3.7 and 3.9] gives

$$0 = \mathcal{P}(Y, \Delta_Y) = \frac{3}{2} \sum_i \beta_i \big(d_i(d_i - 1) - 2\delta_i \big) = 3 \sum_i \beta_i (d_i - 1 + g_i).$$

Every component is consequently a line, so the hypotheses of [29, Theorem 1.1] hold. Its two arrangements have automorphic degrees $1/6$ and $1/2$. Since $q\lambda = 1$ and this factor is irreducible, $q = 6$ and the arrangement is the complete quadrilateral. A point of type $[2, 3, 3]$ supplies a factor G_7 that is never absorbed, whose fixed plane has degree $1/12$, contradicting the same conclusion. This proves the first assertion. Sections 4.1.1–4.1.2 exclude every other rank 4 group. For a smaller factor never entering rank 4, the maximal 2-adic reflection order and Section 4.1 exclude G_7, G_{11}, G_{19} and leave $q = 2s$ with odd $s \geq 3$. An order divisible by 4 gives $3/q = \deg(K_Y + \Delta_Y)$ a denominator of larger 2-adic valuation than its branch coefficients have. For index $b \geq 5$, branch degree d and ν nodes, proportionality and Bézout give

$$6d - 4 + 2b\lambda d \leq 2d^2 + 2b\lambda d - 4\nu \leq d(3 + 3\lambda - (1 - b^{-1})d), \quad \lambda = 1/(2s),$$

forcing $d = 1$, $b/s \in \mathbb{Z}_{>0}$, $b(2b - 3) \leq 2s$, a contradiction. Thus $3d_2 + 4d_3 = 18 + 9/s$, $s = 3, 9$. Since the degree of each component of ramification index 3 is divisible by 3, we exclude $q = 18$ and $(q, d_2, d_3) = (6, 7, 0)$. For $(d_2, d_3) = (3, 3)$, a component of degree $1 \leq d \leq 3$, with c cusps, satisfies $2d^2 + d - 9p_a(d) \leq 2d^2 + d - 9c - 4\nu \leq 3d/2$, where $p_a(d) = (d-1)(d-2)/2$; the lower bounds $3, 10, 12$ exceed $3/2, 3, 9/2$. □

Lemma 4.4. For every coincidence plane $Z_I \simeq \mathbb{P}^2$, $|I| = 3$, write $L|_{Z_I} = \lambda_I c_1(\mathcal{O}_{Z_I}(1))$. Then $\lambda_I \geq 1/2$ and $2L - H$ is semiample. In particular, $a \leq 2t$ for every complete curve.

Proof. Lemma 3.1 gives $\lambda \geq 1$ in ranks $0, 1$ and $q\lambda \geq 1$ for the least positive invariant degree q in rank 2, with regular maps also at larger stabilizers. For $\lambda < 1/2$, Section 4.1 leaves the following possibilities, with d the total degree of the upper locus:

generic stabilizer	upper group	degree equation
$G(4,2,2)$	$G(4,2,3)$	$8\lambda = 2 + 3d/2, \quad d = 0, 1$
G_4	G_{25}	$10\lambda = 2$ if no upper locus
μ_3^2	G_{25}	$6\lambda = 2 + 5d/2, \quad d = 0$

In the first row the upper locus has index 4, and with the degrees d_2', d_3' of the other branch components of indices $2, 3$, $3\lambda + 3 = d_2'/2 + 2d_3'/3 + 3d/4$; the invariants $v + S, vS + P^2, u^2P$, $v = u^4$, $S = y_1^4 + y_2^4$, $P = (y_1y_2)^2$, give $d_2'/2 + 2d_3'/3 = (60 - 3d)/16$, impossible after multiplication by 6. For G_4, the axis $(v^2, v^3, 0)$ at an upper point is nonimmersed; an empty upper locus contradicts $4\lambda \geq 1$. For μ_3^2, the normal differential $\left(\begin{smallmatrix} 0 & -v \\ 24v^2 & 0 \end{smallmatrix}\right)$ at an upper point gives $\lambda = 1/3$ and $(d_2, d_3) = (8, 0), (4, 3), (0, 6)$. An index 3 component of degree $d \leq 6$ requires $9c + 4\nu = 2d(d + 1)$ and $c + \nu \leq (d - 1)(d - 2)/2$. For $d \leq 5$ these are incompatible; for $d = 6$, the only pairs $(c, \nu) = (0, 21), (4, 12), (8, 3)$ still violate the bound. Hence $d_3 = 0$, but the effective normalized orders $1, 2, 4$ make each index 2 degree divisible by 3, contrary to $d_2 = 8$. For multidegrees (w_i) of $2L - H$, the degrees $d_I = 2\lambda_I - 1 \geq 0$ on the planes Z_I give $w_i = d_I + d_J \geq 0$ by complementary triples; [15, Proposition 2.6] proves semiampleness. □

4.2 Normal differentials at larger stabilizers

The following finite recipe uses invariant coordinates on the quotient germs. Expand $F_j(uv_0 + y) = F_j(uv_0) + \sum_i c_{ji}(u) g_i(y)$ modulo $(g_1, \ldots, g_s)^2$, for ambient and normal basic invariants F_j, g_i. Adjoin the tangent column, remove its common factor, and eliminate the saturated tangent direction; the resulting matrix $M(v)$ is the normal differential, with columns indexed by normal directions adapted to the finite characters and rows by the normal components of the

targets from the F_j. Its entries are taken in the coarse frames specified below, local generators of the invariant sections: if $u \mapsto \zeta_e u$ and $e_i \mapsto \zeta_e^{a_i} e_i$ with $0 \le a_i < e$, the frame over $v = u^e$ is $u^{a_i} e_i$, with degree correction $-a_i/e$ [15, Lemma 3.3]. Cofactors then give the rows of $M(v)^{-1}$ and their restrictions to successive kernels; each row is a rational homomorphism extracting one normal component, regular after twisting the target by its poles, and its surjectivity is checked at every point. Every remaining operation is holomorphic and invertible, so the conclusions hold under arbitrary analytic isomorphisms of quotient germs. Table 4 gives all inputs for the larger groups; its last column gives the determinant order, the pole order of the inverse row associated with the invariant of greatest degree, and the pole order of the preceding inverse row on the kernel of that row. For $G_{25} \subset G_{32}$ use $r_j z = z - (Cz)_j \varepsilon_j$, with

$T \subset G$	degrees of G	degrees of T	orders
$G_{25} \subset G_{32}$	$12, 18, 24, 30$	$6, 9, 12$	$(8, 3, —)$
$A_3 \subset A_4$	$2, 3, 4, 5$	$2, 3, 4$	$(3, 1, 1)$
$A_3 \subset D_4$	$2, 4, 4, 6$	$2, 3, 4$	$(3, 1, 1)$
$B_3 \subset B_4$	$2, 4, 6, 8$	$2, 4, 6$	$(3, 1, 1)$
$B_3 \subset F_4$	$2, 6, 8, 12$	$2, 4, 6$	$(7, 3, 2)$
$H_3 \subset H_4$	$2, 12, 20, 30$	$2, 6, 10$	$(22, 10, 7)$
$G(4,4,3) \subset G_{29}$	$4, 8, 12, 20$	$4, 3, 8$	$(7, 3, 3)$

Table 4: Degrees and orders for the pairs $T \subset G$.

$$C = \begin{pmatrix} 1-\omega & \omega^2 & 0 & 0 \\ -\omega^2 & 1-\omega & -\omega^2 & 0 \\ 0 & \omega^2 & 1-\omega & \omega^2 \\ 0 & 0 & -\omega^2 & 1-\omega \end{pmatrix}, \qquad \omega^3 = 1, \quad \omega \neq 1.$$

The first three reflections fix $v_0 = ((2\omega+1)/3, 1, (4\omega+2)/3, 1)$ and act normally on $z_4 = 0$. Use the 240 forms in $e_4^* G_{32}$ and $(1,2,3)G_{25}$ for the respective power sums; separate the square of degree 6 in degree 12. Normalizing the first coefficient row to $(1,1,1,1)$ gives determinant $-19489164500000/1162261467$. On the axis, the first invariants are $(56/9)v^2, (1456/243)v^3$, $v = u^6$, and the frame in degree 9 is $u^3 \partial_{f_9}$. Thus $M(v) = \operatorname{diag}(v^2, v^3, v^4) A \operatorname{diag}(1, 1, v^{-1})$, $\det A \neq 0$, with a simple tangent zero.

For $A_3 \subset A_4$ use $(u, u, u, u, -4u)$ and $\sum y_i = 0$; the power sums have values $(20u^2, -60u^3, 260u^4, -1020u^5)$. For $A_3 \subset D_4$ use (u, u, u, u), $v = u^2$, and character frame $u\partial_{q_3}$. The matrices are

$$M_A = \begin{pmatrix} 15u/2 & 1 & 0 \\ -20u^2 & 4u & 1 \\ 275u^3/2 & 10u^2 & 5u \end{pmatrix}, \quad M_D = \begin{pmatrix} 4v & 4v & 1 \\ 0 & 4v/3 & 0 \\ 0 & 8v^2 & 12v \end{pmatrix}.$$

The determinants are $\det M_A = 625u^3/2$ and $\det M_D = 64v^3$. The row associated with the invariant of greatest degree and the next row restricted to its kernel remain nonzero after pole correction at every stalk; the last zero has order 1, and the A_4 tangent zero is simple. All three D_4 types are conjugate by the root preserving transformations $\frac{1}{2}\left(\begin{smallmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{smallmatrix}\right)$ and $\operatorname{diag}(-1, 1, 1, 1)$.

For $B_3 \subset F_4$ take the short roots $\{\pm\varepsilon_j\} \cup \{\frac{1}{2}\sum \epsilon_j \varepsilon_j : \epsilon_j = \pm 1\}$ and their power sums. On $u\varepsilon_1$, $v = u^2$, elimination gives rows $(-3v^2, 15v, 18)$, $(0, -15v^2, 45v)$, $(0, 0, -405v^3/8)$. For $u(\varepsilon_1 + \varepsilon_2)$ multiply the rows by $-8, 16, -64$; the values $6v, 12v$ of invariants F_j on the axis prove immersion, and the last zero order is 2. For $B_3 \subset B_4$ use $\left(\begin{smallmatrix} v & 1 & 0 \\ 0 & v & 1 \\ 0 & 0 & v \end{smallmatrix}\right)$, of last zero order 1. For $H_3 \subset H_4$ use the Gram matrix of diagonal 2 and successive off-diagonal entries $-\phi, -1, -1$,

$\phi = (1+\sqrt{5})/2$, and axis $(1+\sqrt{5}/2, (3+\sqrt{5})/2, (5+\sqrt{5})/4, 1)$. The 120 ambient and 30 normal root sums give coefficients of exponent $(d-e)/2$ in $v = u^2$. The nonzero pivots are $-(21\sqrt{5}+63)v^5/512$, $(3740\sqrt{5}+8415)v^7/1024$, $(22297275\sqrt{5}+49864815)v^{10}/1048576$.

For $G(4,4,3) \subset G_{29}$ use $r_j z = z - (Bz)_j \varepsilon_j$, with

$$B = \begin{pmatrix} 2 & -1 & -i & 0 \\ -1 & 2 & i-1 & 0 \\ i & -i-1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

The first three reflections fix $((1+i)/2, 1-i/2, 3/2, 1)$; the 160 forms in $z_1 G_{29}$ give tangent and normal determinant $128957475465i/524288$. For $v = u^4$, elimination gives rows $(-81v/8, -9iv^2/4, -81/4)$, $(0, -1183iv^3/384, -3549v/16)$, $(0, 0, -4037365v^3/4096)$. The character in degree 3 is $-i$, with frame $u^3\partial_P$, and the last zero has order 1. At $G(4,4,4)$ use $\left(\begin{smallmatrix} v & 0 & 1 \\ 0 & v & 0 \\ 0 & 0 & v \end{smallmatrix}\right)$. The unique parabolic class and nonzero value, proportional to v, of the first invariant on the axis give a single smooth branch. Distinct basic degrees make the full normalizer preserve the flag.

4.3 The groups B_4 and G_{32}

Lemma 4.5. A B_4 axis contains an F_4 or G_{29} point. A $G(4,2,4)$ axis contains a G_{31} point, and a $G(4,4,4)$ axis contains a G_{29} point.

Proof. If a B_4 axis contains no F_4 or G_{29} point, its only upper groups are B_4 and products. [15, Proposition 3.4] gives $2x + R/2 + \sigma = 2t$ and $a = -2x - 12t + 3R$. These equations imply $a \le -14x$, so $C = \mathbb{P}^1$. Put $k = 2t \in \mathbb{Z}_{>0}$ and $j = R - 2k$. Then $a = 2 + 3j$, $\sigma = 2 - j/2$, and the quotient associated with the invariant of greatest degree has degree $h = j - k$. Lemma 4.4 gives $k \ge 2 + 3j$, hence $j \ge 0$ and $h < 0$. The monomial calculation in Section 4.1.1 gives Hom dimension 1, attained by the surjective projection to the coordinate of greatest invariant degree. [15, Proposition 2.2] therefore gives two coincident labels, a common bundle of degree $-(k-j)$, and $a \ge 3(k-j)$. This contradicts $2 + 3j \le k \le 2j + 2/3$.

For $G(4,p,4)$, $p = 2, 4$, assume the corresponding exceptional group is absent. The upper groups are then $G(4,p,4)$ and products, and $k = 4t$ is a positive integer. [15, Proposition 3.4] gives $g = 0$, $k \le 3$, and the three rows

$$(p, k, R, a, \sigma) = (2, 1, 3, 5, 1/4), (2, 2, 4, 3, 0), (4, 1, 3, 5, 1/4).$$

The first and third have $0 < \sigma < 1/2$; the second contradicts $a \le 2t$. □

Lemma 4.6. The group G_{32} does not occur, and every reflection has order 2.

Proof. Take the axis of G_{25} at a G_{32} point. Let R count G_{32} points and s_b count points with stabilizer $G_{25} \times \mu_b$, for $b = 2, 3$. The calculation in Section 4.2 gives effective inertia of order 6, characters $(1, -1, 1)$ on degrees $6, 9, 12$, a simple tangent zero, normal determinant order 8, and pole order 3 for the inverse row associated with the invariant of greatest degree. Consequently, with $k = 6t \in \mathbb{Z}_{>0}$,

$$2t = 2x + 5R/6 + s_2/2 + 2s_3/3, \qquad a = -2x - 27t + 17R/2, \qquad h = -12t + 3R.$$

The first equation gives $t = x + 5R/12 + s_2/4 + s_3/3$. Substitution in the equation for a gives $a = -29x - 11R/4 - 27s_2/4 - 9s_3$, hence $g = 0$. The conditions $a \in \mathbb{Z}_{>0}$ and $a \le k/3$ leave exactly

$$(R, s_2, s_3, t, a, h) = (2, 2, 1, 2/3, 1, -2), (7, 1, 0, 13/6, 3, -5).$$

Both nonzero inverse maps have target degree less than $-a$, contrary to [15, Proposition 2.2]. Lemma 4.3 now excludes every reflection of order 3. □

4.4 The plane of degree $1/2$

A compact plane quotient of degree $1/2$ has branch indices 2 and a reduced branch curve of degree 9. For nodes, cusps $(2,3)$, contacts 2, cusps $(2,5)$, and triple points, let the counts be ν, c, t_0, q_0, r. Their local groups $\mu_2^2, A_2, B_2, I_2(5), G(4,2,2)$ have orders $4,6,8,10,16$, δ invariants $1,1,2,2,3$, Milnor numbers $1,2,3,4,4$, and effective normalized orders $(2,2),(1),(2,2),(1),(4,4,4)$. For k components, $\sum g_i = 27 + k - \delta_B$, $\delta_B = \nu + c + 2t_0 + 2q_0 + 3r \le 36$. Proportionality and $\chi_{\mathrm{orb}} = 3/4$ give

$$4\nu + 8c + 12t_0 + 16q_0 + 15r = 180, \qquad 180\nu + 320c + 450t_0 + 576q_0 + 585r = 7020.$$

Thus $c = 9c_1$, $q_0 = 5q_1$, $r = 4r_1$, and

$$t_0 = 12 - 4c_1 - 8q_1 - 4r_1, \quad \nu = 9 - 6c_1 + 4q_1 - 3r_1, \quad \delta_B = 33 - 5c_1 - 2q_1 + r_1.$$

The conditions $c_1 + 2q_1 + r_1 \le 3$, $\nu \ge 0$, $\delta_B \le 36$ leave exactly the nine rows in Table 5. A line or smooth conic of degree d satisfies $n_i + 2t_i + 2r_i = d(9-d)$, $n_i/2 + t_i/2 + 3r_i/4 = d + 2$. Consequently lines have $(n_i, t_i, r_i) = (0,0,4), (2,1,2), (4,2,0)$ and conics need $t_i \ge 5$. For the union B^* of duals of nonlinear components and l lines, $D = \deg B^* = 72 - 2\nu - 3c - 4t_0 - 5q_0 - 6r$ and $\delta(B^*) = (D-1)(D-2)/2 + \delta_B - 28 - l$. A contact not involving a line, or a cusp $(2,5)$, contributes at least 2 to $\delta(B^*)$, additively even at a common dual point. For a cusp write $x = u^2$, $y = a_4u^4 + a_5u^5 + \cdots$, $a_5 \neq 0$; its dual has type $(2,5)$ after subtracting the quadratic term if $a_4 \neq 0$, and $(3,5)$ otherwise. For row 5, the conics need two mutual contacts, six line

row	(c_1, q_1, r_1)	exclusion or equality
1	$(0,0,3)$	$\delta_B = 36$: nine lines and twelve ordinary triple points.
2	$(1,0,1)$	No nodes, hence no line contact; dual upper/lower bounds $7, 8$.
3	$(0,0,2)$	$k \ge 8$: one conic and seven lines; five required contacts exceed four.
4	$(1,0,0)$	Too few nodes for a line; dual bounds $15, 16$.
5	$(0,0,1)$	$k = 7$: a cubic supplies fewer than eight contacts; the partition into two conics has impossible tangent distributions.
6	$(1,1,0)$	No contacts; dual bounds $8, 10$.
7	$(0,0,0)$	$k = 7$: at most six contacts; $k = 6$, partitions $(4,1^5), (3,2,1^4)$: at most eleven.
8	$(0,1,1)$	No contacts; dual bounds $5, 10$.
9	$(0,1,0)$	$k \ge 4$: five cusps $(2,5)$ force $(6,1,1,1)$; the three lines need six contacts rather than four.

Table 5: The nine rows for a plane of degree $1/2$ in the compact case.

contacts, and $(t_i, n_i, r_i) = (5,0,2)$. Their four triple points are distinct; distributions of lines touching two, one, or no conic are $(1,4,0), (2,2,1), (3,0,2)$. A line touching one conic joins the other's two triple points, so there are at most two; a line touching neither joins both pairs, excluding the last two distributions as well. The remaining partition $(2,2,2,1,1,1)$ in row 7 requires two contacts per conic pair. Their dual intersections are exhausted by the common tangents; a line touching two conics would give three tangent branches, an excluded local type. Only row 1 remains.

Lemma 4.7. If $\lambda_I = 1/2$, the full generic stabilizer of Z_I is μ_2^2, and every stabilizer on the fixed plane is $\mu_2^2 \times J$. The induced branch divisor consists of nine lines with twelve ordinary triple points and no other singularities. Two disjoint triples cannot both have degree $1/2$. The integral semiample bundle $\mathcal{A} = 5(2L - H)$ has strictly positive multidegrees v_i. A curve fixed by an irreducible group of rank 3 avoids every such plane.

Proof. Lemma 3.1 excludes ranks $0, 1$, and $G(4,2,2)$ would require $8\lambda = 2+3d/2$, $d = 4/3$. For the remaining groups the quadratic Taylor map to $\mathcal{O}(-1)^{\oplus 2}$ has numerically trivial coefficients, so its nonzero coefficient or determinant never vanishes. Absorption would lower its rank: nonreal factors have no quadratic invariant, and a real irreducible quadratic form cannot lie in the ideal of the plane. Thus the factors remain direct factors, and Lemma 4.3 leaves only μ_2^2. Section 4.4 gives the branch divisor. The planes of two disjoint triples meet transversely at one point and share no reflection, since otherwise they would lie in a smooth threefold. Their common point would have full stabilizer μ_2^4, inducing an excluded node. With $q = \sum v_i/2$ and $d_I = q - \sum_{i\in I} v_i$, a zero v_i gives complementary zero triples and this excluded pair; hence $v_i \geq 1$. Every factor of $\mu_2^2 \times J$ has rank at most 2, proving the final assertion. □

4.5 The group G_{31}

Lemma 4.8. Let $T = G(4,2,3)$ be the full stabilizer of a generic point of a fixed curve in a smooth finite reflection quotient of dimension 4. Assume that every upper group of the curve is $G(4,2,4)$, G_{31}, or $T \times A_1$. The coarse frames corresponding to the basic polynomial invariants have degrees $4, 6, 8$. At the first two types their inertia characters are $(1,-1,1)$, and the degree correction of the coarse frame in polynomial degree six is $-1/2$. The following orders hold, independently of arithmeticity or compactness. In the second table, the poles belong to the inverse rows in degrees eight and six, with the latter restricted to the kernel of the former. The last column records the order of vanishing of the column of the normal differential in degree four:

	ord(df)	ord(det M)
$G(4,2,4)$	0	3
G_{31}	1	10
$T \times A_1$	0	0

	pole in degree eight	pole in degree six on the kernel	zero in degree four
$G(4,2,4)$	1	1	1
G_{31}	4	4	2
$T \times A_1$	0	0	0

At G_{31}, the inverse differential has a component of pole order 5 in polynomial degree six before restriction to the kernel specified in the table.

Proof. Take $f_4 = \sum y_j^4$, $f_6 = (y_1y_2y_3)^2$, $f_8 = \sum_{j<k} y_j^4 y_k^4$. For $G(4,2,4)$, use invariants $(v + f_4, vf_4 + f_8, u^2 f_6, vf_8 + f_6^2)$, $v = u^4$, and frame $(\partial_{f_4}, u^2\partial_{f_6}, \partial_{f_8})$; removing the tangent gives M_U below. For G_{31}, the full normalizer is $T \times \mu_4$ [10, Table 8 and proof of Lemma 5.13]. From the roots and coroots in [10, Table 2], generate $I - {}^t\mathsf{A}_j\mathsf{B}_j$, $1 \leq j \leq 5$, and take power sums $8, 12, 20, 24$ of the 240 dual root forms and $4, 6, 8$ of the 192 forms in $(1,2,3)T$, subtracting a multiple of f_4^2 in degree 8. The basis, coefficient matrix, and normal matrices are

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1-i & (1+i)/2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 21/2 & -33/2 & 33 & 231 \\ 10795/2 & -41021/14 & 9690/7 & 226746 \\ 105165 & 129789/5 & 4807 & -19530841/5 \end{pmatrix}, \tag{4.1}$$

$$M_U = \begin{pmatrix} v & 0 & 1 \\ 0 & v & 0 \\ 0 & 0 & v \end{pmatrix}, \quad M_R = \operatorname{diag}(v^2, v^4, v^5) A \operatorname{diag}(1, 1, v^{-1}).$$

Here A is the lower Schur complement of C, with $A_{11} = -27$, leading minor $2070090/7$, determinant -2066330716560, and last cofactor column ${}^t(3625040628/7, -4491828, 2070090/7)$. With the automorphy factor, $u \mapsto iu$ gives characters $(1,-1,1)$; the G_{31} axis is (v^2, v^3, v^5, v^6)

up to nonzero rescaling. For $B = A^{-1}$, the kernel of the projection to the coordinate of greatest invariant degree is $c_3 = -(B_{31}v^3c_1 + B_{32}vc_2)/B_{33}$, where the middle row becomes $v^{-2}(B_{21} - B_{23}B_{31}/B_{33})c_1 + v^{-4}(B_{22} - B_{23}B_{32}/B_{33})c_2$. Its last coefficient is nonzero by $A_{11} \neq 0$, giving poles $4, 4$, rather than pole order 5 before restriction to this kernel; the saturated first column and the derivative of the axis have orders $2, 1$, giving $T_C(D_R)$. Products give the identity matrix; clearing the poles in polynomial degrees eight and six computed from (4.1), with the row in degree six restricted to the kernel of the row in degree eight, leaves primitive rows on every stalk, proving the table. □

Lemma 4.9. The group G_{31} does not occur.

Proof. On the $G(4, 2, 3)$ axis, let U, R, s count $G(4, 2, 4), G_{31}, T \times A_1$, with $R \geq 1$. Lemma 4.8 gives saturated tangent $T_C(D_R)$, $\operatorname{ord}(\det M) = 10$, and pole order 4 for the inverse row in degree eight. The inverse row in degree six has pole order 4 on the kernel of the row in degree eight, but 5 on the whole bundle. The everywhere surjective inverse rows give line quotients of degrees $h_8 = -8t+U+4R$, $h_6 = -6t+U/2+7R/2$, $h_4 = -4t+U+2R$, with $2t = 2x+3(U+R)/4+s/2$ and $a+2x-R = -18t+5U/2+19R/2$. For $V = 2t-a \geq 0$, $11h_8 = -4V-2U-R-2s < 0$; put $e = -h_8$. Three coincident labels determine J by quotienting the pulled back tangent bundle of their coincidence plane by the saturated tangent line. This quotient has degree $3a+2x-R \geq a$ by $R \leq 2a + 2x$. Both successive maps kill J, since $\deg J - h_6 = U/2 - 3e/2 + 2a > 0$, and $e \leq a$. With $b = a - e$, $\delta = e - R$, the degrees give $11\delta \geq 20b$, $x = -e/2 - \delta + b/2 < 0$, $b = e + 2\delta - 2$, $20e + 29\delta \leq 40$. The sole row $(a, e, U, R, s) = (2, 2, 6, 2, 0)$ gives a smooth conic with two tangent zeros, impossible. Thus two labels coincide, the quotient associated with the invariant of greatest degree is M, and [15, Proposition 2.3] gives $h_6 \geq e$, equivalently

$$\begin{aligned} &t = (e + U + 4R)/8, && a = (6R + U - 9e)/4 - 2x, \\ &s = (e - 2U + R)/2 - 4x, && 2R \geq U + 7e. \end{aligned} \tag{4.2}$$

Let D be the divisor where labels $1, 2$ coincide, the blow-up of $\mathbb{P}^3$ at five points, and let $\pi : D \to \mathbb{P}^3$ be its blow-down. For $j = 3, \ldots, 7$, let β_j be the degree of the zero divisor of the bracket between the common section of labels $1, 2$ and section j. For the factorization $f_D : C \to D$ of f, put $\gamma = \pi \circ f_D$, $d = \deg \gamma^* \mathcal{O}(1)$, $B_0 = \sum_{j=3}^{7} \beta_j$ for the five brackets; then $e = d - B_0$, $a = 2d + e$. The image is not a line, by $R \geq 1$ tangent zeros, so a general plane projection has positive dual degree at most $m = 2d + 2x - R$. Since $4m = 6x - 5V - s$, $m, x > 0$. For $\mathcal{A}$ of Lemma 4.7, put $q = \sum v_i/2$, $\rho = q - v_1 - v_2$; then $1 \leq v_j \leq \rho$ for $j \geq 3$, $5V = e\rho + \sum_{j=3}^{7} v_j\beta_j$, and $v_j = \rho$ implies $\beta_j = 0$. Every triple in $\{3, \ldots, 7\}$ of weight sum q gives a cross-ratio $[1j][kl] : [1k][jl]$ with no base point on C, by avoidance of the planes of degree $1/2$. The two products are sections of the same line bundle, so their ratio is independent of local frames. Its degree is $e + \beta_j + \beta_k + \beta_l > 0$, and all G_{31} points ramify, giving $R \leq 2(e + \beta_j + \beta_k + \beta_l) + 2x$. Four vanishing β_j give $R \leq 3e + 3x$ by three cross-ratios with disjoint base loci; fewer than three nonconstant ones force rational normalization. Two identical sections $j, k \geq 3$ give $R \leq 2e + 2x$, and two identical pairs would make a cross-ratio of positive degree constant. The exclusions before the finite test are

condition	inequalities	contradiction
$R \leq 3e + 3x$	$16m + 20e \leq 14x, \quad 4m + 15e \geq 16x$	$32m \leq -22x$
$\rho \geq 5$	$53m + 65e \leq 62x$	$107m \leq -22x$
$\rho \geq 3,\ v_j \geq 2$	$33m + 30e \leq 32x$	$25m \leq 0$

The first row excludes $a = 3e$ and $\rho = 1$. The eleven remaining multidegrees are parametrized below; exponents mean repetition and the first pair is ordered.

ρ	j	$(v_3, \ldots, v_7)$	(v_1, v_2)
2	$1, 3, 4$	$(1^{5-j}, 2^j)$	$(\lfloor (1+j)/2 \rfloor, \lceil (1+j)/2 \rceil)$
$3, 4$	$0, 1, 3, 4$	$(1, (\rho-1)^{4-j}, \rho^j)$	$(\lfloor (2\rho-3+j)/2 \rfloor, \lceil (2\rho-3+j)/2 \rceil)$

This follows from $v_1+v_2 = \sum_{j=3}^{7} v_j - 2\rho$ and the inequalities $\sum_{i\in I} v_i \le q$; the two rows at $\rho = 2$ with all $v_j = 2$, $j \ge 3$, have $B_0 = 0$ and are already excluded. For $\rho = 2, j = 1$, six cross-ratios give $R \le 2e+B_0+2x$. Set $S = 12s$, $Q = 6(2e+B_0+2x-R)$, and $E = 21e+5U+80x-52B_0 = 0$. Substitution of (4.2) and $a = 3e + 2B_0$ gives $828e + 7S + 106Q - 3E = -96x$. For $j = 3, 4$ use $R \le 2e + 2x$, $R \le 3e + 3x$. For $\rho = 3, 4$, $j = 0, 1, 3$, let l be the bracket degree for weight 1 and H_0 their sum for weight $\rho - 1$. Averaging gives $R \le 2e + 3H_0/2 + 2x$. Put $Q = 6(2e + 3H_0/2 + 2x - R)$ and $E_\rho = (45 - 12\rho)e + 5U + 80x - 52l - (28 + 12\rho)H_0 = 0$. Substitution of (4.2) and $a = 3e + 2(l + H_0)$ gives

$$(180\rho - 150)U + 16\rho S + (8\rho + 115)6m + (20\rho - 40)Q + 15E_\rho = -(528\rho - 1500)x.$$

The left side is nonnegative and the right side negative; $j = 4$ has four vanishing β_j and uses the first exclusion. □

4.5.1 Determinant valuations along an A_3 curve

Suppose that on an A_3 curve the kernel corresponding to the invariant of degree 2 is the quotient of the pulled back tangent bundle of a coincidence plane Z by the saturated tangent line. Let $\eta : S \to Z$ resolve an inverse image component of degree e, with $L|_Z = \lambda\mathcal{O}_Z(1)$. For S_4 invariants on $\sum y_i = 0$, the plane equations start as $F_3 = q_3+O(q^2)$, $F_4 = q_4+c(u)q_2^2+O_{>4}(y)$. The form $dF_3 \wedge dF_4$ has degree at least 5 for general rank 0. For rank 1 write $y = (v+r, v-r, z-v, -z-v)$; then $q_2 = 4v^2+2r^2+2z^2$, $q_3 = 6v(r^2-z^2)$, $q_4 = 4v^4+12v^2(r^2+z^2)+2r^4+2z^4$. The r^2 coefficients in F_3, F_4 have degrees at least $1, 2$, so their determinant with the tangential differential has degree at least 4. At a divisor with minimum normal valuation k, the leading vector satisfies $q_3(y^0) = q_4(y^0) + c(u)q_2(y^0)^2 = 0$ and $q_2(y^0) \ne 0$, since otherwise all invariants vanish. Thus η^*C has valuation $2k$, and the two determinants have valuations at least $5k, 4k$. Their saturated kernels are direct summands over the discrete valuation ring, including a critical divisor after removal of the differential's common factor. Exceptional curves have zero intersection with a general pulled back line; hence the determinant divisors contain $5\eta^*C/2$, $2\eta^*C$. Inequality (3.1) bounds their degrees by $(2\lambda - 2)e$, $(3\lambda - 2)e$, as used in Lemma 4.11.

4.6 Semiampleness of $L - H$ and the groups A_4, D_4

Lemma 4.10. Every $\lambda_I \ge 1$, and $L - H$ is semiample.

Proof. Lemmas 4.5 and 4.9 exclude $G(4, 2, 4)$. The remaining seven groups lack the $G(4, 2, 2)$ factor required at the twelve triple points of a plane of degree $1/2$ by Lemma 4.7. All reflection orders are 2, so the reduced reflection divisor $\mathscr{B}$, the sum of the distinct images of the mirrors, satisfies $5L = K_X + \mathscr{B}/2$, and $10L = \mathscr{B} - 2H$ is integral. In general rank 2, the induced indices $2, 4$ give $3\lambda = -3 + d_2/2 + 3d_4/4$, hence $12\lambda, 10\lambda \in \mathbb{Z}$ and $2\lambda \in \mathbb{Z}$. The exclusion of $\lambda = 1/2$ and Lemma 3.1 for ranks $0, 1$ yield $\lambda \ge 1$. For multidegrees v_i of $5(2L - H)$, $d_I = 10\lambda_I - 5 \ge 5$ and $v_i = d_I + d_J \ge 10$; the multidegrees $v_i - 10$ and the degrees $d_I - 5$ on the planes Z_I prove semiampleness of $10(L - H)$ by [15, Proposition 2.6]. □

Lemma 4.11. At least one point has stabilizer F_4, H_4, or G_{29}.

Proof. Otherwise Lemma 4.5 leaves A_4, D_4. On an A_3 curve count $A_4, D_4, A_3\times\mu_2$ by R_A, R_D, s. The matrices of Section 4.2 give tangent $T_C(D_A)$ and a surjective quotient Q of rank 2, with $h_3 = -3t+R_A+R_D/2$, $h_4 = -4t+R_A+R_D$. The degree calculation gives $2t = 2x+(R_D+s)/2$, $a = -2x - 9t + 4R_A + 5R_D/2$, $h_3 + h_4 = a/2 - 3t/2 - s/4 < 0$. By [15, Proposition 2.4], $Q \simeq M^{\oplus 2}$, $\deg M = -a$, with kernel given by the quotient of the pulled back tangent bundle of the coincidence plane by the saturated tangent line; hence $10a = 6t + s$, $a + 6R_A = 18x + 4s$, positive even a and $t/a \le 5/3$. If $R_A \ge 1$, generic rank 2 is excluded by the degrees $1, 2$ of the

effective images for $A_2, 2A_1 \subset A_4$. For ranks $0, 1$, Section 4.5.1 and (3.1) give $5a/2 \le 2\lambda - 2$ or $2a \le 3\lambda - 2$, contrary to $a \ge 2$, $\lambda = t/a \le 5/3$. If $R_A = 0$, the surjective map whose target has degree h_3 forces $h_3 \ge -a$, giving $12x + 5s/2 \ge 0$, $14x + 3s \le 0$. Thus $x = s = 0$, $a = t = R_D/4$, but $\deg\det Q = -a$, contrary to $Q \simeq M^{\oplus 2}$. □

4.7 The arithmetic volume

Lemma 4.12. For a cocompact arithmetic lattice generated by reflections of order 2, put $d_F = [F : \mathbb{Q}]$, $\Delta = D_K/D_F^2$, $w = |\mu_5(K)|$, and $\varpi = 288/(44/7)^{14}$. Then

$$\frac{\chi_{\mathrm{orb}}}{5w\rho_{K/F}} \in \mathbb{Z}_{>0}, \qquad \rho_{K/F} = 2^{-4d_F}\zeta_F(-1)L_{K/F}(-2)\zeta_F(-3)L_{K/F}(-4),$$

and $\chi_{\mathrm{orb}} > 5wD_F^{12}\Delta^7\varpi^{d_F}$.

Proof. For $p : \mathrm{SU}(1,4) \to \mathrm{PU}(1,4)$, choose a maximal arithmetic group above $p^{-1}\Gamma$, with principal subgroup Λ. Its quotient is a 5-group [21, Sections 2.4 and 2.9]; every normalized order 2 reflection has $-r \in \Lambda$, so $\Gamma \subset p(\Lambda)$. Prasad's formula gives $\chi_{\mathrm{orb}} = 5wj\rho_{K/F}\prod_{v<\infty} e'(P_v)$, with $j = [p(\Lambda) : \Gamma]$ and $e'(P_v) \in \mathbb{Z}_{>0}$ [21, Sections 2.3 and 2.6–2.9, equations (7)–(8)]. The functional equations give $\rho_{K/F} = D_F^{12}\Delta^7(288/(2\pi)^{14})^{d_F}$ times an Euler product greater than 1: at inert places the denominators $(1-q^{-2})(1+q^{-3})$, $(1-q^{-4})(1+q^{-5})$ are less than 1, as are those at split and ramified places. Use $2\pi < 44/7$. □

Lemma 4.13. Under the hypotheses of Lemma 4.12, $\chi_{\mathrm{orb}} \ne 770$.

Proof. Section 4.7.1 proves this by enumerating the fields and showing that $770/(5w\rho_{K/F})$ is never an integer. □

4.7.1 The value 770

Suppose $\chi_{\mathrm{orb}} = 770$ in Lemma 4.12. For $F = \mathbb{Q}$, Meyer's theorem makes the indefinite rational form in ten variables isotropic, contradicting compactness [23, Chapter IV, Section 3.2, Corollary 2]. For $d_F \ge 5$, use $D_F^{1/d_F} > 34/5$ [21, Proposition 2(d)], $(34/5)^{12}\varpi > 1$, and $5((34/5)^{12}\varpi)^5 > 770$. For $d_F = 2, 3, 4$, the discriminant bounds are $44, 230, 1226$. The following filters use the proved exhaustive small discriminant tables [21, Section 3.4, proof of Theorem 1(iv)(1)–(3)]. For $d_F = 4$, $D_F = 725, 1125$ and $\Delta \le 2, \Delta = 1$ leave only $\mathbb{Q}(\zeta_{15})$, excluded by $25\rho_{K/F} = 45814652/30375 > 770$. For $d_F = 3, 2$, substituting $\Delta_{\max} + 1$ into the lower bound of Lemma 4.12 gives

D_F	49	81	148	169	229
$\Delta_{\max}$	14	5	2	1	1

,

D_F	5	8	12	13	17	21	24	28	29	33	37	40	41
$\Delta_{\max}$	40	17	8	7	4	3	2	2	1	1	1	1	1

.

The cubic candidates are $\mathbb{Q}(\zeta_7), \mathbb{Q}(\zeta_9)$. In degree 2, the tables exclude $13, 17, 29, 37, 41$ and leave, at $21, 24, 33$, $\mathbb{Q}(\sqrt{-3}, \sqrt{-7})$, $\mathbb{Q}(\sqrt{6}, \sqrt{-3})$, $\mathbb{Q}(\sqrt{-3}, \sqrt{-11})$. At 40, the only totally complex quartic of absolute discriminant 1600 is $\mathbb{Q}(\sqrt{5}, \sqrt{-2})$, with real discriminant 5.

For $D_F = 5, 8, 12, 28$, Minkowski gives real class number 1; at 28, the only ideal of norm 2 is generated by $3 + \sqrt{7}$. Write $K = F(\sqrt{\alpha})$ with totally negative α, whose square class is a product of units and principal prime generators. An odd valuation away from 2 contributes its prime norm to Δ, and above 2 contributes exponent $2v_{\mathfrak{p}}(2) + 1$. The permitted odd prime norms are $9, 5, 11, 19, 29, 31$ at $D_F = 5$ and $9, 7, 17$ at 8, so at most one occurs. At 12 the prime above 3 has mixed signs; at 28 no odd prime is allowed, and odd valuations above 2 exceed

each bound. Signs at both embeddings leave the unique totally negative unit multiple up to squares and conjugation in each class:

F	α	Δ
$\mathbb{Q}(\sqrt{5})$	$-1, -3, -2-\phi$	$16, 9, 5$
$\mathbb{Q}(\sqrt{2})$	$-1, -3, -5-2\sqrt{2}$	$4, 9, 17$.
$\mathbb{Q}(\sqrt{3})$	-1	1
$\mathbb{Q}(\sqrt{7})$	-1	1

(4.3)

Here $\phi = (1+\sqrt{5})/2$; the square residues modulo 4 in $\mathbb{Z}[\phi]$ have pairs $(1,0), (1,1), (2,3)$, giving the relative discriminants in (4.3). For $\alpha = -5-2\sqrt{2}$, $\beta = 1+3\sqrt{2}$ gives $\beta^2 - \alpha = 8(3+\sqrt{2})$, of valuation 6 above 2, versus 2 for 2β. Hensel makes α locally square; the split Euler factor gives $\chi_{\mathrm{orb}} > 5 \cdot 8^{12} \cdot 17^7 \varpi^2 (16384/9765) > 868$. The complete twelve remaining fields are the nine biquadratic fields for

$$(u,v) = (5,-3), (5,-1), (2,-1), (2,-3), (3,-1), (-3,-7), (6,-3), (7,-1), (-3,-11),$$

and $\mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_7), \mathbb{Q}(\zeta_9)$. Only $\mathbb{Q}(\zeta_5)$ has $w = 5$, and every other field has $w = 1$. Use the exact Bernoulli sum $L(1-j,\psi) = -(f_\psi^{j-1}/j) \sum_{u=1}^{f_\psi} \psi(u) B_j(u/f_\psi)$ for primitive characters, taking the three quadratic characters or the primitive even and odd cyclotomic characters. In the listed order the reduced denominators of $770/(5w\rho_{K/F})$ are $31, 587, 19, 18331, 23, 2, 551, 113, 3995, 293, 130429, 18628987$. None is 1, contrary to Lemma 4.12.

4.8 The last three groups

Lemma 4.14. The group H_4 does not occur.

Proof. Count $H_4, H_3 \times \mu_2$ on an H_3 axis by R, s. Section 4.2 gives $2t = 2x + (R+s)/2$, $a = -2x - 18t + 22R$, and $h_6 = -6t + 7R$, $h_{10} = -10t + 10R < 0$ by $44t \geq 45R + s$. For three labels, $3(\deg J - h_6) = 8a + 4x + R > 0$ makes both quotient maps vanish on J. Their determinant and map associated with the invariant of greatest degree give $26a = 23R - 16x$, $2x - 11R \geq 0$, a contradiction. For two labels, [15, Proposition 2.3] gives $h_6 \geq e = -h_{10}$; integrality gives $e = 5b$, $b \in \mathbb{Z}_{>0}$. The complete degree conditions are $t = R + b/2$, $a = 4R - 2x - 9b$, $s = 3R - 4x + 2b$, $R \geq 8b$, but $a \leq t$ gives $R \leq 7b$. □

Lemma 4.15. Let $f : C \to X$ be an immersion from a curve of positive genus. Suppose that a semiample GIT class B has nonnegative multidegrees b_i with $\sum_{i \in I} b_i \leq \sum_i b_i/2$ for every triple I, and that f^*B has degree zero. If the normal bundle has no nonzero map to a line bundle of negative degree, then $B \equiv 0$.

Proof. For nonzero (b_i), these inequalities give $b_i < q = \sum b_i/2$ and at least six positive weights; if one weight vanishes the others are equal. Two or three negative coincident labels give a line quotient of negative degree, and one gives $\deg f^*B = e(q - b_i) + \sum_{j \neq i} b_j \beta_j > 0$. Hence all relative tangent degrees are nonnegative and at least six vanish. Three sections are distinct, since a triple coincidence and nonvanishing remaining brackets imply $a = 0$. These sections trivialize the projective line bundle, and a nonconstant label ramifies in positive genus, contrary to immersion. □

Lemma 4.16. If an F_4 point exists and no G_{29} point exists, then $L \equiv H$.

Proof. Count $B_4, F_4, B_3 \times \mu_2$ on a B_3 axis by R_B, R_F, s, and put $q_0 = R_F - s$. Section 4.2 and $a \leq t$ give $a = -14x + 4q_0 + s$, $h_4 = q_0 - 4x$, $2h_6 = 3q_0 - R_B - 12x$, $R_B \geq 15q_0 + 2s - 60x$. For $h_6 = -e < 0$ and two labels, [15, Proposition 2.3] gives $h_4 \geq e$, hence $5e + s \leq 0$. Three labels

have $\deg J - h_4 = (8a+4x+R_F)/3 > 0$; the quotient of rank 2 then gives $3R_B+5R_F = 17a+10x$, $R_F \geq 4a+2x$, hence $-3R_B \geq 3a$. For $h_6 \geq 0$ the only solution is $q_0 = R_F = 4x$, $s = R_B = 0$, $a = t = 2x$, $(h_2, h_4, h_6) = (4x, 0, 0)$. Now $g \geq 2$ and the flag has no line quotient of negative degree; apply Lemma 4.15 to $L - H$. □

Lemma 4.17. The group G_{29} does not occur.

Proof. On the $G(4,4,3)$ axis let U, R, s count $G(4,4,4), G_{29}, G(4,4,3) \times \mu_2$, with $R \geq 1$. Section 4.2 gives a single immersed branch and successive surjections, with $h_3 = -3t+U/4+9R/4$. The degree correction $-3(U+R)/4$ in degree 3 is included; the degree equations are $t = x+3(U+R)/8+s/4$, $a = -17x-27U/8+5R/8-15s/4$, $h_8 = -8x-2U-2s$. The inequality $a \leq t$ forces $h_8 < 0$: the contrary for $x = 0$ gives $U = s = 0$, $a > t$; for $x = -1$, it gives $U + s \leq 4$, $a - t \geq 3 - s/4 + R/4 > 0$. Put $e = -h_8 = 8x + 2U + 2s \in 2\mathbb{Z}_{>0}$. For three labels the image is a smooth plane curve, $x = (a-1)(a-2)/2 - 1$, and $e \leq a$. The genus bound $8x - a = a(4a-13) > 0$ excludes $a \geq 4$; integrality leaves only $a = 2$ among $a \leq 3$. For $a = 2$, the sole rows $(U, R, s, t) = (0, 6, 5, 5/2), (5, 3, 0, 2)$ contradict respectively $10(t/a) \in \mathbb{Z}$ and the next surjection, since $\deg J = 4 > h_3 = 2$ makes that surjection factor through degree -2. Thus exactly two labels coincide and $a \geq 3e$. For the class $10(L-H)$, nonzero by Lemma 4.13, let $q = \sum w_i/2$, $\rho = q - w_1 - w_2$, $\delta = t - a$. The inequalities $\sum_{i \in I} w_i \leq q$ give $\rho \geq 1$, $0 \leq w_j \leq \rho$ for $j \geq 3$, and

$$10\delta = e\rho + \sum_{j=3}^{7} w_j \beta_j, \qquad \delta \leq 8x/5 + s/10, \qquad \delta/e \leq 1/5.$$

Hence $1 \leq \rho \leq 2$, $w_1 + w_2 \leq 6$, $q \leq 8$, $w_i \leq 6$, $d_I \leq 8$. The multidegrees of $10(2H - L)$ and its degrees on the planes Z_I are at least $14, 2$, so $2H - L$ is nef and $\chi_{\mathrm{orb}} = 5L^4 \leq 5(2H)^4 = 12320$. The scalar order $c \mid 10$ cannot be $1, 2$, since integrality of cL would force $5 \mid \rho$. Thus $c = 5, 10$, $\zeta_5 \in K$; the normalized central G_{29} element has trace $1 + 4i$, giving $\mathbb{Q}(\zeta_{20}) \subset K$ and $D_F \geq 2000^{[F:\mathbb{Q}]/4}$. Lemma 4.12 gives $\chi_{\mathrm{orb}} > 25(2000^3\varpi)^{[F:\mathbb{Q}]} \geq 25(2000^3\varpi)^4 > 12320$, since $2000^3\varpi = 1192188995242382812 5/777727659178477568 > 1$. □

Theorem 4.18. No cocompact arithmetic lattice $\Gamma \subset \mathrm{PU}(1,4)$ satisfies $X_7 \simeq \mathbb{B}^4/\Gamma$.

Proof. Lemmas 4.11, 4.14, and 4.17 force an F_4 point and exclude G_{29}. Lemma 4.16 gives $L \equiv H$. Since $H^4 = 154$, proportionality gives $\chi_{\mathrm{orb}} = 5H^4 = 770$, contrary to Lemma 4.13. □

5 Nonuniform realizations outside the Gaussian field

We retain the notation of Section 3.

Proposition 5.1. If X is the Baily–Borel compactification of a nonuniform arithmetic ball quotient, then the imaginary quadratic field K defining the lattice is $\mathbb{Q}(i)$.

We apply Lemmas 3.2 and 3.3 to the 35 coincidence planes. Lemma 5.2 excludes the Coxeter possibilities once the other horizontal groups have been removed. For fields with two units the remaining primitive group is G_{24}; over the Eisenstein field the following subsections treat the primitive groups, the groups with sixth roots of unity, and finally the groups $G(3,1,3), G(3,3,3), A_3, B_3$. Thus the degree bound used at each stage is established before it is applied.

5.1 The Coxeter groups

Lemma 5.2. Suppose that the full horizontal groups are A_3, B_3, the irreducible interior groups are among A_4, B_4, D_4, F_4, and the groups fixing a ball of dimension 2 are A_1^2, A_2, B_2.

Suppose also that $\mathcal{L}^{\otimes 2}$ descends to X, $L-H$ is nef, and the finite characters on the directions of indecomposable invariants have trivial inertia except for the usual real reflection characters. Then X has no cusps.

Proof. For a coincidence plane Z of generic rank 2, put $L|_Z = \lambda c_1(\mathcal{O}_Z(1))$, $\lambda \geq 1$, $2\lambda \in \mathbb{Z}$. All inclusions use actual full parabolic stabilizers. The local determinants below give $10\lambda = 4 + 5\deg D_B$ for A_2, and $6\lambda - 2 = 2\deg D_B$, $\deg D_B \leq 2\lambda - 1$ for B_2, both impossible. For A_1^2, with loci D_A, D_B, D_P of $A_3, B_3, A_1 \times B_2$, they give

$$4\lambda - 2 = \tfrac{3}{2}\deg D_A + 3\deg D_B + \deg D_P. \tag{5.1}$$

For a B_3 curve, count $B_4, F_4, B_3 \times A_1$ by R_B, R_F, p, and cusps $1111, 1112, 1122, 1222$ by s_i. Put $x = g-1$, $s = \sum s_i$, $(\alpha, \beta, \gamma) = (s_1+s_2+s_3+2s_4, s_1+s_2+2s_3+2s_4, s_1+2s_2+2s_3+2s_4)$. Let K_{24} be the kernel of the quotient associated with the invariant of degree 6, and let $K_2 \subset K_{24}$ be the kernel of its successive quotient associated with degree 4. The bundles K_2, K_{24} have ranks $1, 2$, respectively. We write $h_2 = \deg K_2$, $h_4 = \deg(K_{24}/K_2)$, and $h_6 = \deg(\mathcal{N}/K_{24})$. The inverse differential gives everywhere surjective maps onto these successive line quotients, with

$$(h_2, h_4, h_6) = (-2t + R_B + 2R_F + \alpha, -4t + R_B + 2R_F + \beta, -6t + R_B + 3R_F + \gamma). \tag{5.2}$$

The full characters and widths give

$$2t = 2x + s + (R_B + R_F + p)/2, \quad a = -2x - 12t + 3R_B + 7R_F + \alpha + \beta + \gamma. \tag{5.3}$$

Section 5.1.3 leaves, with cusps, only

$$(a, t, p, R_B, R_F; s_1, s_2, s_3, s_4; h_4, h_6) = (1, 3/2, 2, 0, 0; 0, 3, 1, 0; -1, -1). \tag{5.4}$$

Its plane has full generic stabilizer A_1^2, $\lambda = 3/2$, $(\deg D_A, \deg D_B, \deg D_P) = (0, 1, 1)$. Without cusps $R_B = p = 0$, also excluding B_4 through its B_3 parabolics.

At a B_3 cusp use $x_1^2 = U$, $x_2^2 = T$, $x_3^2 = T - U$, $x_4^2 = T - c_0 U$, $c_0 \neq 0, 1$. Let S be the set of squared labels. A short reflection is actual exactly at a label in S; the other lift has nontrivial fiber character. All A_1^2 fixed components lie in the same plane [15, Lemma 2.17]. Two long reflections give an excluded A_3 curve, and two short reflections give two P branches; hence their lengths differ. The invariant rings on the plane for $1112, 1122, 1222, 1111$ are $\mathbb{C}[u, v]$, $\mathbb{C}[u, v]$, $\mathbb{C}[u, uv, v^2]$, $\mathbb{C}[u^2, uv, v^2]$. Embedding dimension leaves the three 1112 and one 1122 ends in (5.4), with horizontal group of order 2, boundary degree 2, width 1. Let D_Q be the A_1^3 locus. Section 5.1.1 excludes nontransverse interior D_P, D_Q intersections: their only possible ambient group $B_3 \times A_1$ gives another coincidence plane intersecting Z in a curve. The numbers of branches of D_B, D_P, D_Q at the cusp are $(1, 0, 3)$ at 1112 and $(1, 1, 2)$ at 1122, with distinct directions $U, T, T-U, T-c_0U$. Thus $K_Z + \frac{1}{2}(D_A + D_B + D_P + D_Q) = 3L|_Z$ gives $\deg D_Q = 13$. The cusp of D_P contributes multiplicity 2, leaving 11 transverse elliptic points of order 2, contrary to $3 = 2L \cdot D_P \geq -2 + 1 + 11/2 = 9/2$. The actual $\mathcal{L}^{\otimes 2}$ has trivial cusp character. The curvature current of its invariant metric has zero Lelong number by the logarithmic cusp potential in [14, Section 2.2, equations (1)–(8)].

For A_3, count $A_4, D_4, A_3 \times A_1$ by R_A, R_D, p; all cusps are 1111. Saturate the tangent bundle to $T_C(D_A)$ at A_4. The normal quotients have degrees $(h_4, h_3, h_2) = (-4t + R_A + R_D + s, -3t + R_A + R_D/2 + s, -2t + R_A + R_D + s)$. The comparison gives $2t = 2x + s + (R_D + p)/2$, $a = -2x - 9t + 4R_A + 5R_D/2 + 3s$. The last two directions give a surjective quotient Q, $\deg\det Q = a/2 - 3t/2 - p/4 < 0$. By [15, Proposition 2.4], $Q = M^{\oplus 2}$ on a three point coincidence plane, $\deg M = -a$. Hence $10a = 6t + p$, $a + 6R_A = 18x + 4p + 3s$, $1 \leq \lambda \leq 5/3$. Section 5.1.2 excludes rank 0 by $5a/2 \leq 2\lambda - 2$; rank 1 forces $a = 1$, $x = -1$, $\lambda = 3/2$,

then $R_D = -4R_A - 1$. The plane therefore has full generic stabilizer A_1^2. At an A_3 cusp, let E be the elliptic curve in the description of its horizontal torus, with origin 0, and put $A = \ker((z_i) \mapsto \sum_i z_i : E^4 \to E)$. Coordinate permutations give $\pi : A \to A/S_4 \simeq |4[0]| \simeq \mathbb{P}^3$, and the polarization determined by the least positive central period is $\pi^*\mathcal{O}(1)$ with its induced linearization [22, Section 5.1]. The fixed components for $\langle (12), (34) \rangle$ are $(z, z, u - z, u - z)$, $u \in E[2]$; they map through $|[0] + [u]|$ to noncollinear Veronese conics, incompatible with the boundary line $\mathbb{P}(T_pZ)$ and exceptional bundle $\mathcal{O}(-1)$ of a smooth cusp.

5.1.1 Coxeter groups and the local normal directions

For $A_2 \subset B_3$, expand the power sums at $x_i = u + y_i$, $\sum y_i = 0$, with $q_k = \sum y_i^k$, $v = u^2$ and coarse frame ue_3. Removing the tangent direction gives $\left(\begin{smallmatrix} 4v & 4v \\ 12v^2 & 20v^2 \end{smallmatrix}\right)$, of determinant $32v^3$. The A_3 axis $(12u^2, -24u^3, 84u^4)$ is not immersed. For B_2 the matrix is $\left(\begin{smallmatrix} v & 1 \\ 0 & v \end{smallmatrix}\right)$. For A_1^2 the determinant character contributes $-\deg D_A/2$, and local orders at $A_3, B_3, A_1 \times B_2, A_1^3$ are $2, 3, 1, 0$, proving (5.1) without splitting.

The matrix at B_4 and power sums over the short roots at both F_4 axes give corrections $(1, 1, 1)$ and $(2, 2, 3)$ with nonzero minors (Section 4.2). The first three even Taylor coefficients of the products of sections along the branches at a cusp have a Vandermonde determinant. Thus the corrections in directions $(2, 4, 6)$ at $1111, 1112, 1122, 1222$ are $(1, 1, 1), (1, 1, 2), (1, 2, 2), (2, 2, 2)$, with width 1 at every common fixed point of trivial fiber character. Removing the kernel of the fiber character from 2222 gives 1111, replaces q by q^2, and halves the period. For $A_3 \subset A_4, D_4$, Section 4.2 gives $\det M_A = 625u^3/2$, $\det M_D = 64v^3$. The last two rows of M_D^{-1} in coarse frame ue_3 are $(0, 3/(4v), 0)$, $(0, -1/2, 1/(12v))$. These give independent coefficients with simple poles for the inverse row in degree four and the row in degree three restricted to the kernel of the row in degree four; triality, tangent saturation at A_4, and the 1111 theta coefficients complete the surjectivity and character calculation.

For the full normalizers of fixed planes, enumerate orthogonal root pairs of pointwise stabilizer A_1^2 in A_4, B_4, D_4, F_4 and all direct products of rank 4 of A_1, A_2, B_2, A_3, B_3. Restrict the other roots to the fixed plane and take invariants of $N_F(A_1^2)/A_1^2$ [10, Theorems 2.1, 3.9, 3.12, Table 5]. Roots orthogonal to the fixed space generate every stabilizer, proving exhaustiveness. Without A_3 branches or B_4 points, only $B_3 \times A_1$ gives a nontransverse P, Q pair; in F_4, equal root lengths give B_2, so A_1^2 uses both lengths.

5.1.2 Weighted vanishing on a surface

Let $\Sigma \subset X$ be a smooth coincidence plane or the strict transform in $\mathrm{Bl}_5 \mathbb{P}^3$ of a plane in $\mathbb{P}^3$, containing C, such that the projections of the inverse differential to the two coordinates of greatest invariant degree vanish along Σ. Let ℓ be a general line in the plane, or its strict transform, avoiding the cusp and blow-up centers, and put $\lambda = L \cdot \ell$, $c = \det N_{\Sigma/X} \cdot \ell$, $d = C \cdot \ell$. Thus $K_\Sigma \cdot \ell = -3$ and $\ell^2 = 1$. Retain the critical divisor R in the inequality $K_\Sigma - R \geq 3L|_\Sigma$, as in (3.1), and in the classes $5L - K_\Sigma + R$, $4L - K_\Sigma + R$ of the conormal determinants for generic ranks 0,1. In rank 1 the invariant normal line has class $2L$. For transverse order k, the vanishing orders of the curve equation and of the conormal determinant, a section of the class above for the generic rank, are at most/at least $(3k, 6k), (2k, 7k)$ for B_3 and $(2k, 5k), (2k, 4k)$ for A_3. For B_3 the f_2^3 coefficient vanishes on the mirror component of R. Thus

	rank = 0	rank = 1
B_3	$2d \leq 2\lambda + c$	$7d/2 \leq 3\lambda + c$
A_3	$5d/2 \leq 2\lambda + c$	$2d \leq 3\lambda + c$.

Retaining R prevents saturation from removing the zeros.

5.1.3 The equality cases for a B_3 curve

Put $q_0 = R_F - p$, $D_0 = 3s_1 + 2s_2 + s_3$, $U = s_1 + s_2$, $V = 2s_1 + s_2 + s_3 + s_4$. The comparison gives $a = -14x + 4q_0 + p - D_0$, $t = x + (R_B + q_0 + 2p + 2s)/4$, $h_4 = q_0 - 4x - U$, $h_6 = (3q_0 - R_B - 12x)/2 - V$. For $h_6 \geq 0$, $a \leq t$ forces $R_B = p = s_1 = s_3 = s_4 = 0$. Writing $b = s_2/3$ leaves $a = t = 2x + 2b$, $R_F = 4x + 2b$, $(h_2, h_4, h_6) = (4x + 3b, -b, 0)$. For $b > 0$, [15, Proposition 2.4] gives $b = 2a$, $x = -3a/2 < -1$; without cusps it gives $R_B = p = 0$.

For $h_6 < 0$ and two coincident labels of relative tangent degree $-e$, [15, Proposition 2.2] gives $h_4 \geq e$. The equality $R_B = -3(h_4 - e) - 2(t - a) - 2s_1 - s_2 - p - 2s_4$ forces $t = a$, $h_4 = e$, $s_3 = 5e/2$, $4 \mid e$, $a = 3e + 2m$, $g = 1 + 3e/4 + m$, where $m = R_F/4 - e$. Nonnegative bracket multidegrees force $L \equiv H$; the inverse row of the next degree on $\mathrm{Bl}_5 \mathbb{P}^3$ gives a constant row $(M^{-1})^{\oplus 4} \to M^{-1}$, $M = \mathcal{O}(-h + E)|_C$, $H = \mathcal{O}(3h - E)$. Its Euler relation shows that C lies in a plane transform with $c = 0$, $\lambda = 3$, $d = e + m \geq 4$. Section 5.1.2 excludes ranks 0,1; at F_4, degrees $2, 6, 8, 12$ exclude immersed A_1^2, B_2 planes. The remaining A_2 plane at 1122 has coordinates $qa^3, qb^3, q^2(b^2 - a^2)^3, q^2(b^2 - c_0a^2)^3$; determinant $9c_0(c_0 - 1) \neq 0$ modulo the first three products gives tangent dimension 4.

For three labels, $\mathcal{N} = J \oplus M^{\oplus 2}$, $\deg J = 3a + 2x > 0$, $\deg M = -a$, $-a \leq h_6 < 0$. The projections to the two coordinates of greatest invariant degree kill J. If $\deg J \leq h_4$, their coefficient inequalities force $a = 1$, $h_6 = -1$, $h_2 = -h_4$, then $h_4 = 1$, $s_3 = 4$, $t = 1$, contrary to $h_6 = 2$ in (5.2). Thus their quotient of rank 2 is $M^{\oplus 2}$, and

$$h_2 = 3a + 2x, \quad h_4 + h_6 = -2a, \quad h_4 \leq -a,$$
$$8t = 11a + 10x + p + s_1 + 2s_2 + 3s_3, \quad 4R_F = a + 14x + 3p + 3s_1 + 2s_2 + s_3,$$
$$4R_B = 21a - 10x - 5p - 9s_1 - 6s_2 - 3s_3 - 8s_4, \quad 4h_4 = a - 2x - p - s_1 - 2s_2 + s_3.$$

Without cusps these give $5a - 2x \leq p \leq 21a/5 - 2x$, impossible. For $R_F > 0$, the lowest invariant degrees give generic rank at most 1; Section 5.1.2, with $c = -2$, $\lambda \leq 7/2$, forces $a \leq 2$, $x = -1$. Putting $h_4 = -a - d$, $h_6 = -a + d$, $0 \leq d < a$, gives $t = 2a - 1 + d/2 + s_3/2$, $R_B = 3a/2 + 1 - 3d - p/2 - 3s_1/2 - 3s_3/2 - 2s_4$. For $(a, d) = (1, 0), (2, 0), (2, 1)$ the bounds $t \leq 3/2, 4, 7/2$ contradict $t \geq 11/6, 6, 6$. With $R_F = 0$, $x = -1$, $a \leq 2$, the plane has full generic stabilizer A_1^2. For $a = 2$, $d = 1$ is inconsistent and $d = 0$ gives $s_2 = 6$, $t = 3$, violating (5.1); for $a = 1$ that equality leaves only (5.4). □

5.2 Fields with two units and the group G_{24}

For $K \neq \mathbb{Q}(i), \mathbb{Q}(\sqrt{-3})$, the full horizontal groups are A_3, B_3, with G_{24} possible only over $\mathbb{Q}(\sqrt{-7})$ [7, 10]. The character fields of $G_{23}, \ldots, G_{27}$ are $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{-7})$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-3})$, and $\mathbb{Q}(\sqrt{-3}, \sqrt{5})$. After G_{24} is excluded, the interior groups are A_4, B_4, D_4, F_4 and [15, Lemma 2.15] restricts the groups of rank 2 to A_1^2, A_2, B_2. Their least positive invariant degree 2, descent of $\mathcal{L}^{\otimes 2}$, Lemma 3.2, and [15, Proposition 2.6] give every hypothesis of Lemma 5.2.

For G_{24}, the horizontal torus is the Jacobian of the Klein quartic with its principal polarization, and its ring of invariant sections has generators of degrees $1, 2, 4, 7$ [22]. Its common fixed point is unique by $\det(-g - I) = -1$ for an element of order 7 [12, Proposition 2.1(v) in arXiv:2107.03745v2]. The section of degree 1 is nonzero there because the quartic is non-hyperelliptic and the tangent representation has no invariant linear form. Let $\Gamma_{00} \supset \Gamma_{000}$ be the sections in $H^0(A, \mathcal{O}_A(2\Theta_0))$ vanishing at the origin of this Jacobian A to order at least 4 and 6 [18, Section 1, (1.2)–(1.5)]. For this polarization, $\dim \Gamma_{00} = 1$ and $\Gamma_{000} = 0$ [18, Section 6.2.1]; this gives a nonzero projection from theta sections of degree 2 in the polynomial invariant of degree 4 [18, equations (1.3) and (1.5), Theorems 1.1(2) and 1.3]. A determinant character needs all 21 mirrors, so it cannot occur in order 4 or 5; subtract the square of the section of degree 1. For a G_{24} curve, let R_2 count points of effective inertia 2 and $s = s_4 + s_7$ count cusps according as the direction of degree 14 appears in the derivatives of invariant sections of degree 4 or 7. Put $x = g - 1$, $j = 2t - s$. The comparison is $2t = 2x + s + R_2/2$,

$a = -24t + 13s - 2x$, $j = 2x + R_2/2$. Hence $a \le t$ gives $j > 0$, $s \le 25j + 4x \le 27j$, $a \le 14j$. The target $Q = E_{14}(4S_4 + 7S_7)$ associated with the invariant of greatest degree has degree $-7j - 3s_4 < 0$. Its lower coefficient degrees are $-5j - 3s_4$ and $-s - 4j - 3s_4$, so [15, Lemma 3.5] gives $\dim \mathrm{Hom}(f^*T_X, Q) = 1$. Then [15, Proposition 2.2] forces two coincident labels and $a \ge 21j$, a contradiction. Without cusps, $a + 2g - 2 = -24t$ gives $25a \le 2 - 2g$, also impossible.

5.3 The primitive Eisenstein groups

Put $K = \mathbb{Q}(\omega)$, $\omega^2 + \omega + 1 = 0$, $\mathcal{O} = \mathbb{Z}[\omega]$, $\delta = 1 + 2\omega$. Initially $\mathcal{L}^{\otimes 6}$ descends and $a \le 3t$ [15, Proposition 2.11], by Lemma 3.2. The possible full horizontal groups are $A_3, B_3, G(3,1,3), G(3,3,3), G(6,p,3)$ for $p = 1, 2, 3, 6$, and G_{25}, G_{26}.

For G_{25}, use the E_6 deformation $x^3 + y^3 + z^3 + a_1yz + a_2y + b_2z + a_3$ at $(a_1, a_2, b_2, a_3) = (1, -1/4, -1/4, 5/108)$. With $u = (y - z)/2$, the Kodaira–Spencer map on the slice $a_1 = 1$ is an isomorphism onto $\langle 1, u, u^2\rangle = (\mathbb{C}[x,u]/(x^2, u^3))^{\mu_3}$, where μ_3 acts by $x \mapsto \omega x$, $u \mapsto u$ [25]. Put $s = (a_2 + b_2)/2 + 1/4$, $w = (a_2 - b_2)/2$, $c = a_3 - 5/108$. The regular element of order 3 has fixed weights $6, 9, 12$, centralizer G_{25}, and linear terms $(P_6, P_9, P_{12}) = (-3s, 2w, c + s/3)$ [26]. Reversing weight 9 gives G_{26}, of degrees $6, 12, 18$. The two models of G_{26} have first theta minors $\left(\begin{smallmatrix} -2/3 & 0 \\ 1/9 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1/9 & 1 \\ 1/6 & 0 \end{smallmatrix}\right)$. For G_{25} the normal differential is $\mathrm{diag}(q^2, q^2, q^3)\left(\begin{smallmatrix} b_6 & 0 & * \\ 0 & b_9 & 0 \\ 0 & 0 & b_{12} \end{smallmatrix}\right)$, $b_6 b_9 b_{12} \ne 0$. Weight preserving equivariant changes preserve these minors. The section of degree 1 is nonzero at the three G_{25} points $(-\zeta^{-1}/4, -\zeta/4, 5/108)$, $\zeta^3 = 1$, and at the unique G_{26} origin obtained by the involution and isogeny of degree 3. Thus the theta projections have the stated orders, trivial fiber characters, and width 1.

In the following calculations, u, v, w count the points with a stabilizer that is a direct product whose effective inertia on the curve has order $2, 3, 6$, respectively, so $\sigma = u/2 + 2v/3 + 5w/6$. For G_{26}, s_1, s_2 count $1234, 1346$ cusps and $j = 6t$. There is no irreducible larger group, and

$$j/3 = 2g - 2 + s_1 + s_2 + \sigma, \quad a = 10s_1 + 14s_2 + \sigma - 19j/3, \quad h = -3j + 4s_1 + 6s_2.$$

Since $j \in \mathbb{Z}$, the first equation gives $3\sigma \in \mathbb{Z}$. These equations force $g = 0$, $h < 0$, $a = s_1 + s_2 + 2h + 2$. The lower residual degrees are at most $(h - j)/2, h/3$; for the target associated with the invariant of greatest degree, the Hom space therefore has dimension 1 and $a \ge -3h$ [15, Lemma 3.5]. Equivalently, $21s_1 + 13s_2 + 39\sigma \ge 82$ and $24s_1 + 14s_2 + 45\sigma \le 92$. The exhaustive bounds are

(s_1, σ)	$(0,0)$	$(1,0)$	$(2,0)$	$(0,2/3)$	$(1,2/3)$	$(0,1)$
lower bound on s_2	7	5	4	5	3	4
upper bound on s_2	6	4	3	4	2	3.

For G_{25}, let R count its only irreducible larger group G_{32}. The comparison, retaining the actual finite character, is

$$2t = 2g - 2 + s + 5R/6 + \sigma, \quad a = 8s - 29t + 28R/3 + \sigma,$$
$$\deg(E_6, E_9, E_{12}) = (-6t, -9t - R/2, -12t),$$
$$h = \deg E_{12}(3S + 3D_R) = 6 - 6g - 6t + R/2 - 3\sigma.$$

It gives $g = 0$, $h \le 0$. For the matrix at G_{32}, extract B as in Section 4.2, divide each column by its nonzero first entry, and set $A_{ij} = B_{i+1,j+1} - B_{i+1,1}B_{1,j+1}$. The matrix $M = \mathrm{diag}(v^2, v^3, v^4) A \,\mathrm{diag}(1, 1, v^{-1})$ and its Schur complement D have entries $A_{11} = -650/297$ and $D_{11} = (-19661200/649539)/A_{11}$. Their determinants are $\det A = -19489164500000/1162261467$ and $\det D = \det A/A_{11}$. The coefficient associated with the invariant of degree 6 has residual degree $-4t + 2 + R/6 - \sigma < 0$. The other coefficients have zeros $(2,3)$ at cusps and $(3,2)$ at G_{32}, with $D_{11}(b_{12}/v^2)(0) = D_{12}(b_9/v^3)(0)$. Surjective

evaluation at the R distinct points gives $\dim \mathrm{Hom}(\mathcal{N}, E_{12}(3S+3D_R)) = \max(\eta+1,0)+1$, $\eta = -t+2-R/3-\sigma$. Exhaustive substitution in the preceding equations, under the conditions in Table 6, gives the outputs listed there. The outputs $(4,-1,2),(2,-1,2)$ contradict [15, Proposition 2.2]. For $(3,0,3)$, the identities $M_i = L_i^2 \otimes \det(V)^{-1}$ and $a \equiv \deg V \pmod 2$ make the contribution $(m-1)(e+1)$ of each negative coincident component even when a is odd. This excludes the remaining Hom dimension, every G_{25} cusp, and the G_{32} points through their actual parabolics.

	integer inputs and bounds	$(a, h, \dim \mathrm{Hom})$
$\eta \geq 0$	$0 \leq R \leq 10$, $0 \leq s \leq 4$, $0 < 6t < 20$, $6t+R$ even, $1 \leq a \leq 3t$	$(4,-1,2)$, $(2,-1,2)$, $(3,0,3)$
$\eta < 0$	The same bounds, $a \geq -3h$, $R+39\sigma/16 \leq 3$	No solution.

Table 6: Finite conditions and outputs for G_{25} curves.

5.4 The sixth roots of unity

Rains's 16 monomial actions come from $E : z^2 = y^3 + x^6$, with $d_2 \mid 2$, $d_3 \mid 3$, $d_6 \mid 6$; no translation isogeny occurs because $1-\zeta_6$ is a unit [22, Theorem 3.6, Example 3.11]. Put $d_0 = \gcd(2d_3, 3d_2)$, $g_0 = \gcd(d_6, d_0)$, $q = 6/\operatorname{lcm}(d_2, d_3, d_6)$, $p = 6/d_6$. The full group is $G(6,q,3)$, its subgroup with trivial fiber character is $T = G(6,p,3)$, the generator degrees are $(d_6, 2d_3, 3d_2, 6)/g_0$, and the width is $\kappa = d_0/g_0$. For $p > 1$ put $B = (d_6 + 2d_3 + 3d_2 + 6)/d_0$, $\nu = 6/d_0$, $\eta = d_6/d_0$, and $\ell = 3$ for $d_0 = 1$, $\ell = \nu$ otherwise. Let R count $G(6,p,4)$, let σ be the contribution of product groups, and let s_0 count $d_0 = 1$ ends, where the curve has a semicubical cusp and saturated tangent bundle $T_C(S_0)$. We project the inverse differential to the product invariant and to the elementary symmetric invariant of greatest degree, obtaining simultaneous everywhere surjective quotients with $h_P = -18t/p + R/p + \sum \eta_c$, $h = -12t + R + \sum \nu_c$, and

$$\begin{aligned} 2t = 2g-2+s+5R/6+\sigma, \quad a = \sum B_c + (17/6+1/p)R + \sigma - (20+18/p)t, \\ ph_P = h - 6t, \qquad a = (2p+1)h_P + 18t + 2 - 2g - 2s_0. \end{aligned} \tag{5.5}$$

The other coefficient degrees for the first of these two targets are $h_P + 6t - R - \sum \ell_c$ and $h_P - h$.

To compute B_c, ν_c, η_c, use $S_1 = \sum w_i^6$, $S_2 = \sum_{i<j} w_i^6 w_j^6$, $P = (w_1 w_2 w_3)^{6/p}$. The full normalizer preserves their lines because coordinate axes contain more mirrors than other intersection lines. Primitive alternatives are excluded by the 18 reflections of order 2 that exchange coordinates, compared with zero and nine in G_{25}, G_{26}. The two theta projections onto (dS_1, dS_2) have matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$; an equivariant change $\xi_i = bw_i + O(w_i^7)$ preserves the required zeros. At upper points, $v = u^6$, $dU_2 = v\,dS_1 + dS_2$, $dU_3 = v\,dS_2$, $dU_0 = u^{6/p} dP$. A character ζ^{-r} on a coarse frame contributes $r/6$ after the allowed pole, giving the fractional η in (5.5).

The cases $p = 1, 2$ reduce to the following inequalities. For $p = 1$, $(A,b) = (4,1),(7,2),(10,3),(17,6)$, all widths are 1, and at 2366 the degree correction of the tangent cancels the removed determinant order. Put $j = 6t$, $V = \sum(A_c - 1 - 2b_c) \geq 2\sum b_c/3$. Then $g = 0$, $h = -3j + \sum b_c + R < 0$, $a = V + 2h + R + 2$, $8h \leq 3a - 6j - R - 6$, hence $11a \geq 6j + R + 6$, contradicting $a \leq j/2$. For $p = 2$, $(B, \nu, \ell) = (14,6,3),(17/2,3,3),(6,2,2),(7/2,1,1)$. Here $g = 0$, $a = 4h_P + 9t + R/2 + V_2 + 2$, $V_2 = \sum(B - 1 - 2\nu) > 0$, so $h_P < -3t/2$, $h = 2h_P + 6t \geq 0$. Both residual degrees are negative; the condition that the Hom space has dimension one gives $a \geq -3h_P > 9t/2$.

For $p = 6$, $13h_P + 11t + 2g + 5R/3 + 2\sigma \leq 2$ gives $h_P < 0$. If $g \geq 1$, then $\eta_c \leq 1$ and the logarithmic identity give $h_P \leq -t + 2 - 2g - 2R/3 - \sigma < 0$ and $h = 6h_P + 6t \leq 0$. The quotient onto both targets has negative determinant, so [15, Proposition 2.4] gives $h_P = h = -a$,

$t = 5a/6$, and $g + s_0 = 1 + a/2$. Now $\sum_c \nu_c \le 3s + 3s_0$ and (5.5) give $5a/2 + 3R/2 + 3\sigma + 9g \le 9$, a contradiction. Thus $g = 0$. If $h < 0$, the quotient of rank 2 onto both targets gives $h_P = h = -a$, $t = 5a/6$, $s_0 = 1 + a/2$. The bound $\sum \nu \le 3s + 3s_0$ gives $5a/2 + 3R/2 + 3\sigma \le 9$, hence $a = 2$, $s_0 = 2$, impossible for a line or immersed conic. If $h \ge 0$, put $h_P = -b$. The GIT kernel and negative residual degrees give $h \ge b$, $a = 3b + 2v$, $0 \le v \le h/4$, $s_0 = b + 1 + 3h/2 - v$. The five remaining sections give a base point free map $\gamma : \mathbb{P}^1 \to \mathbb{P}^3$ of degree $b + v$ whose differential vanishes at least $s_0 - v$ times. If $\gamma(\mathbb{P}^1)$ spans a linear space of dimension ρ, the Wronskian gives $\rho(s_0 - v) \le (\rho + 1)(b + v - \rho)$, $(3\rho - 1)h/4 \le b - \rho(\rho + 2)$. This contradicts $h \ge b$ for $\rho = 2, 3$; a line germ cannot be semicubical.

For $p = 3$, $a \le 26(1 - g) - 23R/3 - 12\sigma$ gives $g = 0$. The identity $a = 6h_P + 12t + R/3 + \sum(\nu - 21_{d_0=1}) + 2$ gives $h_P < -3t/2$. The residual degrees are $-2R/3 + \sum(\nu/3 - \ell) < 0$ and $-2h_P - 6t \le 0$. Equality in the second gives $h_P = h = -a$, $a = 3t$, three coincident labels, and a kernel line of degree $3a - 2 - s_0$ mapping to degree $a - 3s_0$, hence $a + s_0 \le 1$. The remaining triples $(d_2, d_3, d_6) = (2, 1, 2), (1, 3, 2), (2, 3, 2)$ give the following pairs of full groups and cusp widths: $(G(6, 3, 3), 1), (G(6, 1, 3), 3), (G(6, 1, 3), 3)$. The first has no reflection of order 6; in the others the fiber character of such a reflection is nontrivial on the connected fixed torus. [15, Lemma 2.15] excludes every actual interior reflection of order 6. Thus $a = 1$, $t = 1/3$, $s_0 = 0$, and the plane has full generic stabilizer $G(6, 3, 2)$: Lemma 3.2 excludes ranks 0,1 and the other groups of rank 2 by their least invariant degree 2.

5.5 The fixed surface and its full lattice

Let $Z \simeq \mathbb{P}^2$ have full generic stabilizer $T = G(6, 3, 2)$. The normal representation invariants have degrees 4,6; along the $G(6, 3, 3)$ curve D_6, their inverse coarse frames are $u^{-2}e_4, v^{-1}e_6$, $v = u^6$. The actual line bundles satisfy $Q_4^{\otimes 3} = \mathcal{L}^{-12}|_Z \otimes \mathcal{O}_Z(D_6)$ and $Q_6 = \mathcal{L}^{-6}|_Z \otimes \mathcal{O}_Z(D_6)$, including finite characters because the punctured plane has unchanged Picard group and trivial fundamental group. Hartogs extension of the nonzero inverse determinant gives $Q_4 = Q_6 = \mathcal{O}_Z(-1)$, $\lambda = 1/3$, $\deg D_6 = 1$, and orbifold Euler number $1/3$. Every cusp lies on D_6: the connected T fixed torus contains the origin and has constant fiber character, so lifting the entire group gives $G(6, 3, 3)$ with inertia of order 6 along the axis and actual width 1 or 3.

Lemma 5.3. The effective image of the full normalizer of Z is conjugate to the subgroup H_2 defined in Subsubsection 5.5.1. Its mirror of order 6 has signature $(0; 6, 6; 1 \text{ cusp})$ and coarse degree 1.

Proof. Put $\pi = 1 - \omega$, $h_0 = -\left(\begin{smallmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right)$. The four possibilities are the projective full unitary groups $M_{\epsilon\eta}$ of $L_{\epsilon\eta} = \mathcal{O}(2^\epsilon \pi^\eta)^{-1} e_1 \oplus \mathcal{O}e_2 \oplus \mathcal{O}e_3$, $\epsilon, \eta \in \{0, 1\}$ [27, 20]. For $M = M_{\epsilon\eta}$, let Λ_{pr} be the principal arithmetic subgroup of $\mathrm{SU}(h_0)$ normalized by the full inverse image of M, and let $\overline{\Lambda}_{\mathrm{pr}}$ be its projective image. Put $n = [M : \overline{\Lambda}_{\mathrm{pr}}]$, $j = [M : \Lambda]$ for this effective image $\Lambda \subset M$, $I = \{p : K_p \text{ is split Iwahori}\}$, and $A = \prod_p \lambda_p$, where K_p are the defining parahoric subgroups and λ_p are the factors in [27, Theorem 4.1]. Since $\mu_3 \subset \Lambda_{\mathrm{pr}}$, the group $\overline{\Lambda}_{\mathrm{pr}}$ has Euler number $A/24$, giving $jA = 8n$. By the bound $n \le 3^{1+|I|}$, the split Iwahori factors ≥ 456, and the inert factors $p^2 - p + 1 = 3, 21, \ge 111$ at $2, 5, \ge 11$, we have $I = \varnothing$, $n \le 3$, and $A \le 24$, and only the vertices at 3 and the choices at 2 remain. The determinant of $\mathrm{diag}(1, \omega, 1)$ is not a cube, giving $n = 3$. The Euler numbers are $1/72, 1/72, 1/24, 1/24$, hence $j = 24, 24, 8, 8$. Each group has one cusp since the class number and its 3 torsion are 1 and $I = \varnothing$ [28, Theorem 4.8].

The subgroup calculation uses the presentations and cusp generators in Subsection A.2. Its inputs are the four indices $24, 24, 8, 8$, the prescribed cusp reflection, and the exclusion of isolated stabilizers not generated by reflections. Table 7 gives every case passing these tests and its final obstruction; paired orders refer to a point stabilizer and its reflection subgroup. The search consists of transitive permutation actions of the prescribed degrees, so every generator has finitely many possible images. The complete relators, all extensions through the

group	cases	exclusion or surviving case
M_{00}	4	Point order 36; orders $(12,4),(6,1)$; H_2.
M_{01}	4	Mirror degrees $1/2, 3/2, 3/2, 3/2$.
M_{10}	2	Mirror degree 2; point order 36 on $\mathrm{diag}(1,2)$.
M_{11}	2	Mirror degrees $2, 1/2$.

Table 7: Outcomes for the four Eisenstein lattices.

common subgroup, and all cusp orbits are tested; the construction in Subsection A.2 proves exhaustiveness. The remaining tests concern actual point stabilizers and mirror degrees, as follows.

Here J, R, σ are the matrices of the presentation of M_{00} in Subsection A.2, where $J = RP$ and $\sigma = QP^{-1}$. To compute full stabilizers, put $D = \mathrm{diag}(1,1,-1)$, $A = D^{-1}\sigma D$, $J_+ = D^{-1}JD$, $R_+ = D^{-1}RD$, $r_2 = J_+AJ_+^{-1}$, $r_3 = J_+^{-1}AJ_+$, $b = r_2r_3r_2^{-1}$, $d = r_2J_+^2$. The full stabilizers of $[-1,0,1]$, $[\omega^2,0,1]$ are $\langle A, R_+\rangle$, $\langle A, b, d\rangle$, of orders 12,72. Enumerate their projective matrices fixing each coset and test local reflections by $\mathrm{rank}(M-\alpha I) = 1$, $Mz = \alpha z$. For mirror orbit length d and pointwise kernel of order $k \mid 6$, the effective image of the stabilizer of a mirror has index $dk/6$ in that of its full normalizer. The full normalizer $\langle A, R_+, b\rangle$ of a mirror with root of norm 1 has pointwise kernel μ_6 and effective image $\Delta(2,6,\infty)$ [19, Section 3]. The full mirror stabilizers in M_{01}, M_{11} have pointwise kernel μ_6, and their effective images contain $\mathrm{PSL}(2,\mathbb{Z})$. The trace condition permits only effective elliptic orders $2,3,6$ before the group is determined, since order 4 fails the valuation at 2 of $(\mathrm{tr}\, M)^2 = 2\det M$; hence $-\chi \geq 1/6$. Equality follows from the contained modular group, so the effective images are $\mathrm{PSL}(2,\mathbb{Z})$, of signature $(0;2,3;1\text{ cusp})$. Thus $-\chi(C) = 2L\cdot C = 2\deg C/3$ gives degrees $dk/12, dk/24$ respectively.

The remaining point stabilizers in M_{00} violate the required order 72 or the Chevalley–Shephard–Todd theorem. For the binary lattice, $|a|^2 + 2|b|^2 = 1$ gives full group μ_6^2, excluding the required effective $G(6,1,2)$ of order 72. All fractional or larger degrees contradict $\deg D_6 = 1$. For H_2, the effective image of the mirror stabilizer has index 2 in $\Delta(2,6,\infty)$, proving its signature and degree. □

5.5.1 The linear group and its preserved lattice

Use negative lines, $h_3 = \left(\begin{smallmatrix} 0&0&1\\0&1&0\\1&0&0\end{smallmatrix}\right)$, and

$$J = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -\omega & \omega \\ 1 & 1 & \omega \end{pmatrix}, \quad A = \mathrm{diag}(1, -\omega^2, 1), \quad C = \omega I, \quad S = JA^{-1}J.$$

The group $\langle J, A, C\rangle$ has the presentation $J^3 = C$, $A^6 = 1$, $S^4 = C^2$, $[A, S^2] = 1$, $C^3 = [J, C] = [A, C] = 1$. The subgroup H_2 is the inverse image of the stabilizer of 1 in the right action defined by the following disjoint cycles, with C acting trivially:

$$\begin{aligned} J &= (1,2,3)(4,6,7)(5,9,10)(8,16,12) \\ &\qquad (11,20,21)(13,22,14)(15,23,17)(18,24,19), \\ A &= (2,3,5,11,8,4)(6,12,13)(7,14,18,9,17,15) \\ &\qquad (10,19,20)(16,22)(21,24). \end{aligned} \tag{5.6}$$

Choose the first breadth first word t_i from 1 to i, in alphabet order J, A, J^{-1}, A^{-1}. The 72 edges are $e_{i,g} = t_i g t_{ig}^{-1}$ for $g = J, A, C$; inverse edges are $e_{i,g^{-1}} = e_{ig^{-1},g}^{-1}$. The 23 tree edges and the seven relations above, rewritten at all 24 points, give 191 relations, of full rank with nonunit Smith entries $6, 6$, hence abelianization $(\mathbb{Z}/6\mathbb{Z})^{\oplus 2}$. Write $\chi(e_{i,g}) = \omega^{x_{i,g}}$ and

impose $x_{i,C} = 1$. We require the character to take value 1 on reflections of orders 6 and 2 with roots of norm 1, value ω^2 on reflections of order 3 with roots of norm 1, and value 1 on reflections of order 2 with roots of norm 2. At the two points with full stabilizers of order 72, we require agreement with the normal character in degree 4. To reproduce these stabilizers, put $r = C^{-1}S^2$, $b = (JAJ^{-1})(J^{-1}AJ)(JA^{-1}J^{-1})$. Take all orbits of $\langle A, r, b\rangle$ and $\langle A, S\rangle$ in the action defined by (5.6) and rewrite the conjugates of the appropriate powers of reflections from an orbit representative. At base points 1 and 23 impose exponents $0, 0, 1$ on A, $JAJ^{-2}AJ^2A^{-1}J^{-1}$, and JAJ, conjugated by t_i. The 191 relations have rank 70 over $\mathbb{F}_3$; imposing $\chi(C) = \omega$ gives rank 71, and $\chi(A) = 1$ determines a unique character. Its values $\chi(e_{i,g}) = \omega^{x_{i,g}}$ are $x_{i,C} = 1$ for every i, and

$$\begin{aligned} x_{i,J} &= 1 \quad \text{for } i \in \{2, 6, 9, 12, 17, 20, 22, 24\}, \\ x_{i,A} &= 1 \quad \text{for } i \in \{2, 16, 17, 20, 22\}, \\ x_{i,A} &= 2 \quad \text{for } i \in \{11, 14, 19, 24\}, \end{aligned}$$

with all remaining exponents zero. Every stated local value is verified by substitution in the rewritten relators and the preceding words. Evaluate any word by summing its Schreier exponents.

On the fixed space W and its positive normal space, the actual full normalizer is

$$\begin{aligned} \rho(M) &= \operatorname{diag}(\chi(M)^{-1}M, (\det M)^{-1}, 1), \\ \iota(t) &= (\det t)\operatorname{diag}(I_3, t), \quad t \in G(6, 3, 2). \end{aligned}$$

Every block differs from $\rho(M)$ by $\iota(T)$, since the determinant condition first gives a scalar in μ_6 and the character conditions reduce it to $\chi(M)^{-1}$. Both elliptic points of order 6 of the mirror therefore have actual ambient group $G(6, 3, 4)$. [10, Theorem 3.6] excludes primitive larger groups. A product would require a reflection of order 6, already excluded. We choose coordinates in which the full stabilizer of one such point is monomial, with no condition at the second point. Put $M_1 = A$, $M_2 = JAJ$, $M_3 = JA^{-1}JAJA^2J^{-1}$. Their traces are $\omega + 3, -\omega, -\omega$. The commutant of M_1, M_2, M_3 is KI, and the corresponding intertwiner equations have dimension 1 over K. The basis $(\omega^2, 0, 1), (0, 1, 0), (-\omega, 0, 1)$ has Gram matrix $\operatorname{diag}(-1, 1, 1)$. For the first full $G(6, 3, 4)$ group, let P_i project to a positive axis and put $F_i = -I + 2P_i$. With $\zeta = 1 + \omega$, let D_{ij} act by ζ, ζ^{-1} on the i, j axes and by 1 on the other axes. The identity

$$P_i = \omega^2(D_{ij} - I)(D_{ij} - \zeta^{-1}I) - \omega(F_i + I)$$

shows that P_i lies in the integral group algebra. For $\Omega = \left(\begin{smallmatrix} \omega^2 & 0 & -\omega \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{smallmatrix}\right)$, the ideals generated by entries $12, 13$ and by entries $21, 31$ of the 72 matrices $\Omega^{-1}e_{i,g}\Omega$ are both $\delta\mathcal{O}$. For the original group, both ideals equal $\delta\mathcal{O}$ by the divisibility of all generators and the entries $\omega - 1, -2(\omega - 1), -\delta, 2\delta$ of $\Omega^{-1}M_3\Omega$ in positions $12, 13, 21, 31$. The trace and commutator calculation above gives a conjugacy over K, and actual monomial elements exchanging the normal axes give the form $h_5 = \operatorname{diag}(-1, 1, 1, 1, 1)$. The same full Γ preserves an $\mathcal{O}$ lattice by [15, Proposition 2.11]. Integral projectors from the first $G(6, 3, 4)$ group split this lattice as $d\mathcal{O} \oplus \mathcal{O}^4$. The ideals generated by the entries in its two off diagonal blocks are $\delta\mathcal{O}$, so $\delta\mathcal{O} \subset d\mathcal{O} \subset \delta^{-1}\mathcal{O}$. The three possible lattices are exchanged by duality; the unique intermediate module in $\mathcal{O}/\delta^2\mathcal{O}$ is the middle lattice. Thus every isometry of any of them preserves the middle lattice, and the original group satisfies

$$\Gamma \subset \mathrm{SU}(h_5, \mathcal{O}).$$

5.5.2 An actual reflection of order 6

For a vector p positive with respect to the form h_5 fixed above, put

$$P_p = \frac{pp^*h_5}{p^*h_5p}, \qquad B_p = -I + 2P_p.$$

The following basis is integral and has determinant 1:

$$v = (\delta, 1, 1, 1, 0), \quad q_1 = (1, 0, \omega^2, -\omega, 0),$$
$$q_2 = (1, 0, -\omega, \omega^2, 0), \quad q_3 = e_4, \quad t_0 = e_1 + \omega v.$$

Write $Q_v = (v, q_1, q_2, q_3, t_0)$, $B = \operatorname{diag}(\Omega, I_2)$, $s = (1-\omega, 2\omega^2, -1-\omega, 0, 0)$, and

$$G = B^{-1} \operatorname{diag}(\omega^{-1} M_3, (\det M_3)^{-1}, 1) B,$$
$$T_n = \operatorname{diag}(I_3, -I_2), \quad F_0 = \operatorname{diag}(1, V_0),$$
$$V_0 = \begin{pmatrix} 0 & \omega & 0 & 0 \\ -\omega & 0 & 0 & 0 \\ 0 & 0 & \omega & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then $B_s = F_0 T_n G F_0^{-1}$ belongs to the same Γ. Let $F = \operatorname{diag}(1, \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right), 1)$, and let Z be minus the interchange of the first two positive axes. For $U_1 = F B_s Z^{-1}$ and $U_2 = F U_1 F^{-1}$, the matrices in the Q_v basis are $N((-1, 1, 0), -1)$ and $N((-\omega, -1-\omega, 0), -1)$, where

$$N(b, u) = \begin{pmatrix} 1 & -\bar{b}^t & u \\ 0 & I_3 & b \\ 0 & 0 & 1 \end{pmatrix}.$$

They fix q_3 and change $q_1 + av$ to $q_1 + (a+1)v$ and $q_1 + (a+\omega^2)v$, respectively. Thus they remove any $a \in \mathcal{O}$. Every full cusp element acts monomially on the positive definite unimodular horizontal lattice; its scalar is in μ_6 and a trivial horizontal action would give $\lambda^5 = 1$, hence $\lambda = 1$. If that full action preserved q_3, it would split the horizontal representation as $2 + 1$, giving positive second Betti number by the circle bundle Gysin sequence. Thus some actual cusp element moves q_3 to $u(q_i + av)$, $i = 1, 2$. We obtain $B_{q_1} \in \Gamma$ from conjugation of B_{q_3}, the equality $Z q_2 = (1+\omega) q_1 + \omega v$, and U_1, U_2. For $r_0 = (1, 1, 1, 0, 0)$, the actual element

$$F_1 = -\operatorname{diag}\left(1, \begin{pmatrix} 0 & \omega & 0 & 0 \\ 0 & 0 & -\omega^2 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\right)$$

satisfies $F_1 q_1 = -r_0$. With $D = \operatorname{diag}(1, 1, \omega, \omega^2, 1)$ and $u_0 = (2\omega, 2\omega, \omega^2, 0, 0)$, the identity $P_{D B_{r_0} e_2} = P_{u_0}$ gives $B_{u_0} \in \Gamma$. Finally put $w_3 = J A^{-2} J$, $M_3' = w_3 A^2 w_3^{-1}$; the character above gives $\chi(M_3') = \omega^2$. The actual element

$$T_3 = B^{-1} \operatorname{diag}(I_3, \omega^2, \omega) \operatorname{diag}(\omega M_3', (\det M_3')^{-1}, 1) B = \omega I + (\omega^2 - \omega) P_{u_0} \tag{5.7}$$

has product with B_{u_0} whose eigenvalue on u_0 divided by its eigenvalue on $u_0^\perp$ is $-\omega$. All matrix identities in Subsubsection 5.5.2, including (5.7), are exact over K.

This actual reflection contradicts its prior exclusion and eliminates the three remaining triples (d_2, d_3, d_6).

5.6 The groups $G(3,1,3)$, $G(3,3,3)$, A_3, B_3 and the last cusps

Only $G(3,1,3), G(3,3,3), A_3, B_3$ remain. Their normal representation invariants give $\lambda \geq 1/2$ by Lemma 3.2, hence $a \leq 2t$ by [15, Proposition 2.6]. For $G(3,1,3)$, the generator degrees at cusps are $1111, 1133, 1333$, all widths are 1, their sums are $4, 8, 10$, and the inverse row associated with the invariant of greatest degree has pole orders $1, 3, 3$. For a cusp c, let A_c be the sum

of generator degrees and b_c the pole order of the inverse row associated with the invariant of greatest degree. Put $j = 3t$, let R count $G(3,1,4)$, $B = \sum b_c$, and $V = \sum(A_c - 1 - 2b_c)$. The comparison gives $g = 0$, $h = -3j + R + B$, $a = V + 2h + R + 2$, $V \geq B/3$, $(7/3)B \leq 7j - 3R - 2$. The lower residual degrees are negative, the first being $-2j + B - V$; thus the target associated with the invariant of greatest degree has $\dim \mathrm{Hom}(\mathcal{N}, Q) = 1$ and $a \geq -3h$. The finite conditions and their complete outputs are in Table 8; all counts and j are integers, and outputs are ordered as $(j, R, s_1, s_2, s_3, \sigma; a, h)$. Both have $a = 3t$, contradicting $a \leq 2t$. [15,

inputs and equations	outputs
$1 \leq j \leq 28$, $0 \leq R \leq 6$, $0 \leq s \leq 20$, $\sigma = u/2 + 2v/3 + 5w/6$; $2j/3 = -2 + s + 2R/3 + \sigma$, $a = \sum A_c + 11R/3 + \sigma - 20j/3$, $h = -3j + R + \sum b_c$; $1 \leq a \leq j$, $-a \leq h < 0$, $a \geq -3h$	$(6,0,1,4,1,0;6,-2)$, $(7,1,0,5,1,0;7,-2)$.

Table 8: Finite conditions and outputs for $G(3,1,3)$ curves.

Lemma 2.15] now excludes actual reflections of order 3; $\mathcal{L}^{\otimes 2}$ descends, and the least invariant degree 2 of the remaining groups gives $\lambda \geq 1$, $a \leq t$.

For $T = G(3,3,3)$, let R count $G(3,3,4)$. Its degrees are $3, 3, 6$; the full normalizer may mix the first two directions, and every cusp is 1113 of width 1. Write $P_3 = w_1 w_2 w_3$, $S_1 = \sum w_i^3$, $S_2 = \sum_{i<j} w_i^3 w_j^3$. At upper points $v = u^3$, $dU_0 = u\, dP_3$, $dU_2 = v\, dS_1 + dS_2$, $dU_3 = v\, dS_2$. At cusps the theta projections are $q\, dP_3$, $-q\, dS_1/3$, $q^3(dS_1 + dS_2)$. The projection of the inverse differential to the coordinate of greatest invariant degree is therefore $3q^{-1}dU_2 + q^{-3}dU_3$, including its lower term, and gives an everywhere surjective quotient $\mathcal{N} \to Q_6$ after pole 3. For $K_3 = \ker(\mathcal{N} \to Q_6)$, the directions of degree 3 have pole at most 1 on a finite cover trivializing their representation. Consequently every line subbundle $J \subset K_3$ satisfies $\deg J \leq b = -3t + R + s$. The quotient degrees are $\deg K_3 = 2b - 2R/3$ and $h = -6t + R + 3s$. [15, Proposition 3.4] gives $2t = 2g - 2 + s + 2R/3 + \sigma$ and $a = 6s + 3R + \sigma - 14t$. These force $g = 0$. With $j = 6t$, the exhaustive conditions and remaining cases are in Table 9. For $R = 0$,

inputs and equations	nine outputs
$1 \leq j \leq 33$, $0 \leq R \leq 7$, $0 \leq s \leq 13$, $j/3 \in \mathbb{Z}$; $\sigma = j/3 + 2 - s - 2R/3$, $a = 6s + 3R + \sigma - 7j/3$, $h = -j + R + 3s$, $b = -j/2 + R + s$; $a \in \mathbb{Z}$, $\sigma \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, $1 \leq a \leq j/6$; for $R = 0$ also $b \in \mathbb{Z}$	Three with $R = 0$; two with $R = 3$, integral t; $(a,t,b) = (1, 3/2, 1/2)$; three with $(a,t) = (2,2), (1,5/2), (2,7/2)$.

Table 9: Finite conditions and outputs for $G(3,3,3)$ curves.

the unramified representation is projectively trivial on $\mathbb{P}^1$, so $K_3 = \mathcal{O}(b)^{\oplus 2}$; the three negative rows contradict [15, Proposition 2.4]. The row $(1, 3/2, 1/2)$ has a positive normal line of degree 1 in K_3. There is no row with $s = 0$: for $R = 0, 3, 6$, $[(7R/3 + 2)/13, (7R/3 + 2)/12)$ contains no half integer.

The full projective normalizer has order 216, generated by T, $\mathrm{diag}(\omega, 1, 1)$, and $\delta^{-1} \left(\begin{smallmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{smallmatrix}\right)$. Images of a four point frame in the nine mirror lines prove completeness. Its representation on the directions of degree 3, generated by $D_2 = \mathrm{diag}(\omega, 1)$, $F_2 = \delta^{-3} \left(\begin{smallmatrix} -3 & 1 \\ 18 & 3 \end{smallmatrix}\right)$, has order 24 and $F_2^2 = -I$. Direct multiplication shows that a triple of the four conjugates of D_2^2 has product I only when its factors agree. For integral t, $\mathcal{O}(t)$ cancels the actual central μ_2 character of $\mathcal{L}^{\otimes 2}$; no such cancellation is asserted for half integral t. Thus for the $R = 3$ integral rows, K_3 has a line subbundle of degree $-3t + R/3 + s$. One is less than $-a$; the other places the positive line of degree $3a - 2 = 7$ in K_3 with $b = 0$.

For the last three rows, every immersion of degree 1 or 2 with respect to H is a line or conic in a coincidence plane, or moves one label with six fixed. The splitting $V = \mathcal{O} \oplus \mathcal{O}(k)$ gives

$a = (7-2m)k+2\sum b_i$, $k, b_i \geq 0$. A conic's positive normal line of degree 4 maps trivially to Q_6 and exceeds the bound $\deg J \leq b$. At a 1113 cusp the horizontal torus and its nine directions are

$$A = E^3/K_0, \quad E:\ x_1^3 - x_2^3 - x_3^3 = 0, \quad K_0 = \{(k_i) \in E[1-\omega]^3 : \sum k_i = 0\},$$

$$(0, \omega^j, -\omega^{-j}), \quad (1, 0, \omega^j), \quad (1, \omega^j, 0), \qquad j = 0, 1, 2.$$

They are the values of the three theta sections of degree 1 at the fixed points represented by $(u + k_i)$, $u \in E[3]/E[1-\omega]$, $\sum k_i \in E[1-\omega]$. At least two values are nonzero, proving trivial fiber character and width 1. The directions lie on a smooth cubic, so at most three lie on a line. With zero, one, or two triple points, there are at most $7, 3, 0$ complete curves moving one label and $0, 1, 2$ coincidence planes. These account for at most $7, 6, 6$ directions, a contradiction. Thus neither $G(3,3,3)$ cusps nor $G(3,3,4)$ points occur.

Only Coxeter horizontal groups remain. Their normal characters may have an unramified μ_3 factor of degree 0, but at a cusp $1 = b^5 \det F$, $b \in \mu_6$, $\det F = \pm 1$ gives $b = \pm 1$. There is no extra cusp inertia, so Lemma 5.2 applies and proves Proposition 5.1.

6 Seven points over the Gaussian field

In this section, $X = X_7$, $H = -K_X$, and $K = \mathbb{Q}(i)$. We assume that $X = (\mathbb{B}^4/\Gamma)^{\mathrm{BB}}$ for a nonuniform arithmetic lattice. All fixed quotients are formed with the full normalizer in this same group Γ.

6.1 Reflections of order four

Lemma 6.1. No smooth cusp has full horizontal group $G(4,1,3)$.

Proof. For the full fixed curve [15, Lemma 2.17], put $a = \deg f^*H, t = \deg f^*\mathcal{L}, j = 4t \in \mathbb{Z}_{>0}$; Lemma 3.2 gives $a \leq j/2$. For $T = G(4,1,3)$, the upper groups $G(4,1,4), T \times \mu_2, T \times \mu_4$ have counts R, u, v, $\sigma = u/2 + 3v/4$ [10, Theorems 2.1, 3.6]; T has normalizer $\mathbb{C}^*T$, preserving its three invariant directions of degrees $4, 8, 12$ and their characters. The generator degrees and the pole orders ν_i of the inverse rows are [22, Theorem 3.6 and Example 3.10]

degrees	1111	1122	1222	1244	1444
(ν_1, ν_2, ν_3)	$(1,1,1)$	$(1,2,2)$	$(2,2,2)$	$(2,4,4)$	$(4,4,4)$.

The cusp has width 1, measured using the least positive central period of the full cusp group, because the section of degree 1 on the axis has trivial fiber character. For the raw tuple 2244, we remove exactly the scalars μ_2 in the kernel of the fiber character. For $X_0 = \prod x_{1j}, Y = \prod y_j$ on $E: x_1^4 - x_2^4 = y^2$, $d(Y/X_0^2) = \eta(-df_1 + df_2 - df_3)/2$, $\eta \neq 0$, on both full orbits; translation $(ix_1, -ix_2, y)$ exchanges $[1:0:\pm 1]$, and the unique lift fixing the point retains full stabilizers. Put $V = \sum \nu_{1,c}, B = \sum \nu_{3,c}, h = -3j + B + R$; $V \geq B/2$ and $(b_1, b_2, b_3) = (vc_1, c_1 + vc_2, c_2 + vc_3)$. With $A_c \leq 13$, [15, Proposition 3.4] gives $2t = 2g - 2 + s + 3R/4 + \sigma$ and

$$a = \sum A_c + 15R/4 + \sigma - 26t \leq 26(1-g) - 6R - 12\sigma.$$

Consequently $C = \mathbb{P}^1$, $a = V + 2h + R + 2$, and $5h \leq 2a - 3j - R - 4 < 0$. The first residual degree satisfies $-2j + B - V \leq (a - 4j - 3R - 2)/5 < 0$. The row associated with the invariant of greatest degree and [15, Lemma 3.5] give $\dim \mathrm{Hom}(f^*T_X, \mathcal{O}_C(h)) = 1$; [15, Proposition 2.2] gives two coincident labels and $a \geq -3h$, hence $11a \geq 9j + 3R + 12$, a contradiction. □

Proposition 6.2. Every actual reflection in Γ has order 2. The square of $\mathcal{L}$ descends to an ample line bundle D on X. For every coincidence plane Z_I with $|I| = 3$, write $L|_{Z_I} = \lambda_I c_1(\mathcal{O}_{Z_I}(1))$. Then $2\lambda_I$ is a positive integer, $\lambda_I \geq 1$, and $L - H$ is nef.

Proof. Meyer's theorem in rational dimension 8 gives a cusp of any actual order 4 mirror in the original lattice, with a faithful horizontal reflection of order 4. The character fields [7, Table 1] and the trace $\operatorname{tr}(\operatorname{diag}(\zeta_m, \zeta_m^{-1}, 1)(23)) = \zeta_m$ leave $m = 1, 2, 4$ and the full horizontal groups

$$A_3, \quad B_3, \quad G(4,4,3), \quad G(4,2,3), \quad G(4,1,3).$$

The last group alone has an order 4 reflection and is excluded by Lemma 6.1. The normalized reflections generate the lift $G \subset \mathrm{U}(h, K)$ of the same Γ; its determinant ± 1 forces every scalar ζI_5 to have $\zeta = \pm 1$, so $\mathcal{L}^{\otimes 2}$ descends to D by [15, Proposition 2.11], with the full canonical frame at the cusp. For planes of ranks $0, 1$, Lemma 3.2 gives $\lambda_I \geq 1, 2/3$; the least invariant degree in rank 2 is two, giving $2\lambda_I > 1$. For $G(4,2,2)$, the invariants $(v + S, vS + P^2, u^2 P)$, $v = u^4$, give frame $(e_S, u^2 e_P)$, correction $-1/2$, and determinant order 2, hence $8\lambda_I = 2 + 3d_C/2$. The value $\lambda_I = 1/2$ would give $d_C = 4/3$, so half integrality gives $\lambda_I \geq 1$ for every plane; [15, Proposition 2.6] applied to all 35 planes gives $D - 2H$ nef. □

6.2 A lattice preserved by the original group

Proposition 6.3. After a Hermitian similarity over K, the original projective lattice Γ is the projective image of a subgroup of finite index of $\mathrm{SU}(I_{1,4}, \mathbb{Z}[i])$, where $I_{1,4} = \operatorname{diag}(-1, 1, 1, 1, 1)$. The similarity preserves the automorphic bundle, the full finite stabilizers, the full normalizers, and the indices of all central cusp subgroups.

Proof. Let A be the Gram matrix, conjugate linear in the first variable, and G the original lift generated by the normalized reflections, preserving a Gaussian lattice. Over K, irreducibility and Schur identify each full horizontal form with aI_3, $a > 0$ rational, with root norms $1, 2$; its determinant times a root norm is a norm because $a^4 = \mathrm{N}(a^2)$. Meyer's theorem in dimension 8 gives a cusp of each actual mirror, and the Witt decomposition gives

$$\det h = -\mathrm{N}(\alpha) \det h_{\mathrm{hor}}, \qquad h(r, r) = h_{\mathrm{hor}}(\bar{r}, \bar{r}).$$

Thus $(-\det A)h(r,r) \in \mathrm{N}(K^*)$; set $c = -\det A, h' = ch$, with $\det h' = -c^6$ and even valuations of root norms at odd inert primes. At odd inert p, maximize G-stable integral M; enlargement by $p^{-1}(M \cap p^2 M^{\#})$ forces valuations $0, 1$, $pM^{\#} \subset M$, and even dimensional discriminant space. For a primitive root of norm n, integrality gives $h'(r, M) \subset n\mathbb{Z}_p[i]$, $r/n \in M^{\#}$; primitivity forces $v_p(n) \leq 1$, so its even parity gives $v_p(n) = 0$. Thus $(R_r - I)M^{\#} \subset M$, and the same G acts trivially on the discriminant space. A nonzero discriminant space has an isotropic line enlarging M, so $M = M^{\#}$. At 2, put $\pi = 1 + i$ and maximize a G-stable integral lattice M. For a primitive reflection root $r \in M$ of norm n, the equality $R_r(M) = M$ gives $h'(r, M) \subset (n/2)\mathbb{Z}_2[i]$. If $v_2(n) \geq 2$, put $\Delta = \sum_{g \in G} \mathbb{Z}_2[i] gr$ and $\widetilde{M} = M + \pi^{-1}\Delta$. The inclusions

$$h'(M, \pi^{-1}\Delta) \subset \frac{n}{2\pi}\mathbb{Z}_2[i], \qquad h'(\pi^{-1}\Delta, \pi^{-1}\Delta) \subset \frac{n}{4}\mathbb{Z}_2[i]$$

show that $\widetilde{M}$ is integral and G-stable; primitivity of r makes it strictly larger than M, a contradiction. Hence $v_2(n) \leq 1$, so $(R_r - I)M^{\#} \subset M$. A maximal integral lattice $M' \supset M$ is self-dual: a vector v lifting a nonzero Gram radical has pairing in $\pi\mathbb{Z}_2[i]$ and norm in $2\mathbb{Z}_2$, so adjoining v/π would otherwise enlarge M'. Since $M \subset M' \subset M^{\#}$, every generating reflection satisfies $(R_r - I)M' \subset M \subset M'$; thus M' is stable under the original G. At split primes pair a stable module with its perfect dual; intersection patches these local self-dual modules globally, since only finitely many differ from the original lattice. A primitive isotropic vector splits an integral plane $\left(\begin{smallmatrix} 0 & 1 \\ 1 & k \end{smallmatrix}\right)$; its positive self-dual complement is I_3, by successive unit vectors. If a positive self-dual Gaussian lattice of rank $r \leq 3$ has no unit vector, its distinct points have squared distance at least 2. Any m points x_j of a translate in the unit ball then satisfy

$m\sum|x_j|^2-|\sum x_j|^2\ge m(m-1)$, hence $\sum\max(1-|x_j|^2,0)\le 1$. Integration over a fundamental domain of covolume one would give $\pi^r/((r+1)r!)\le 1$, which is false for $r=1,2,3$. A unit vector splits off an orthogonal copy of $\mathbb{Z}[i]$, and the complement is again self-dual. Make k even with a unit vector, split a hyperbolic basis v,w, and use $(v-w+e,v+e,w-e)$ to obtain $I_{1,4}$. The basis $B^*(cA)B=I_{1,4}$ and lift $g\mapsto(\det g)^{-1}g$ give a subgroup of finite index of the special unitary group whose projective image is the original Γ. Here the normalized reflections have determinant -1, so every g in their generated group has $\det g=\pm1$, and the determinant of its new lift is $(\det g)^{-4}=1$. Uniqueness uses the absence of nontrivial fifth roots of unity in K. The equivariant tautological map and rescaling by c preserve every stated stabilizer, normalizer, and bundle, and the indices of central cusp subgroups. □

6.3 Degrees on the coincidence planes

Lemma 6.4. The full generic stabilizer of a coincidence plane of rank 2 is A_1^2, A_2, B_2, or $G(4,2,2)$. An A_2 plane never occurs. A B_2 plane has $\lambda>1$.

Proof. Meyer's theorem in rational dimension 6 embeds the full generic stabilizer faithfully into a horizontal parabolic group; its orders $4,6,8,16$ exclude $I_2(6)$, and the character field and [10, Theorem 3.6] give the list. For A_2, the axes in $A_3, G(4,4,3), G(4,2,3)$ are $(u^2,u^3,u^4),(u^3,3u^4,3u^8),(3v^2,v^3,3v^4)$, $v=u^2$; their zero differentials contradict smooth embedding. The remaining B_3 has matrix $\left(\begin{smallmatrix}4u^2 & 4u^2\\ 12u^4 & 20u^4\end{smallmatrix}\right)$, giving $d_B=2\lambda-4/5$, contrary to half integrality. For B_2, put $q_2=xy, q_4=x^4+y^4$; the invariants $(uq_2, u^4+q_4, u^4q_4+q_2^4)$ give matrix $\operatorname{diag}(v,v)$, $v=u^4$, and correction $-3/4$; B_3 gives $\left(\begin{smallmatrix}v & 1\\ 0 & v\end{smallmatrix}\right)$, $v=u^2$. Hence $6\lambda-2=5d_C/4+2d_B$, whose sole integral solution at $\lambda=1$ is $(0,2)$. The direction of degree 4 is preserved by the full normalizer, has trivial divisorial character, and its residual finite character is trivial on finitely punctured $\mathbb{P}^2$; the row of the inverse differential associated with the invariant of greatest degree gives $\mathcal{O}_Z(-1)^{\oplus2}\to\mathcal{O}_Z(-4\lambda+d_C+d_B)=\mathcal{O}_Z(-2)$, a contradiction. □

Lemma 6.5. A coincidence plane with full generic stabilizer A_1^2 has $\lambda>1$.

Proof. Let d_A,d_B,d_C,d_P,d_Q,d_R be the degrees of the reduced curves with full generic stabilizers

$$A_3,\ B_3,\ G(4,2,3),\ B_2\times A_1,\ G(4,2,2)\times A_1,\ A_1^3.$$

The upper group $A_2\times A_1$ has singular axis image and is excluded. The normal determinant and the canonical class with effective branch orders give

$$4\lambda-2=\frac{3}{2}d_A+3d_B+\frac{5}{2}d_C+d_P+\frac{1}{2}d_Q,$$
$$3\lambda+3=\frac{1}{2}(d_A+d_B+d_P+d_R)+\frac{3}{4}(d_C+d_Q).$$

Lemma A.1 proves these formulas and the cusp data used below. At $\lambda=1$, the integral possibilities are

$$(d_P,d_Q,d_R)=(0,4,6),(1,2,8),(2,0,10),\qquad d_A=d_B=d_C=0.$$

The formal solution $d_A=d_Q=1$ gives $d_R=19/2$. For the first triple, count the three finite germs of Lemma A.1 by n_1,n_2,n_3, nodes and tacnodes at cusps by N_0,T_0; for $x=\sum(g_j-1)$, the full normalizers give

$$2x+2(N_0+T_0)+n_1/2+3n_2/4+3n_3/2=8,$$
$$x=2-n_3-N_0-2T_0,$$
$$n_1+2n_2+2N_0=24.$$

Thus $20 \le 3\delta + 2T_0$, $\delta = n_3 + N_0 + 2T_0$; a reduced quartic with r_D irreducible components has $\delta \le r_D + 2$ by $3 = \sum_j g_j + \delta - r_D + 1$, and degree partitions bound tacnodes by $1, 1, 2, 2, 0$, making the right side at most 19. Let $\Pi \subset \mathrm{PU}(2,1)$ be the effective image of the full normalizer of the plane, and Δ a maximal lattice containing it. For the second and third triples, the full cusp data are $(|F_\Pi|, \deg M_\Pi) = (4, 2 \text{ or } 4), (2, 2)$, each with one cusp. Euler characteristic 3 gives $[\Delta : \Pi] = 96$ and $(|F_\Delta|, \deg M_\Delta) = (4, 2 \text{ or } 4)$ by Lemma A.2. Let h be the index of the central translation subgroup of Π in that of Δ. Curvature and the unique cusp double coset give

$$[\Delta : \Pi] = \frac{|F_\Delta|}{|F_\Pi|} \frac{\deg M_\Pi}{\deg M_\Delta} h^2.$$

The factors other than h^2 are powers of 2, so the index has even 3-adic valuation, contradicting 96. □

6.4 The fixed surface with group $G(4,2,2)$

The remaining equality $\lambda = 1$ is tested through the full normalizer action on three pairs of fixed points. The associated cover has degree three and trace kernel $N_{Z/X}(-D_C) \otimes \mathcal{L}^4|_Z$; the minors of its cubic coefficients cut out the G_{31} points. Lemmas 6.6 and 6.7 make this construction precise and supply the contradiction in Proposition 6.10.

Put $T_2 = G(4,2,2)$ and $T_3 = G(4,2,3)$. Suppose that a coincidence plane $Z = Z_I$ has full generic stabilizer T_2. [15, Lemma 2.12] identifies the compactification of its fixed ball quotient with Z. Generic identifications normalize T_2, the map is finite, and Z is normal. Write D_C, D_P for the reduced curves with full generic stabilizers $T_3, T_2 \times A_1$, of effective branch orders $4, 2$.

Lemma 6.6. The interior upper groups of D_C and the germs of D_C, D_P at their points are

	D_C	D_P
$G(4,2,4)$	$y = 0$	$x^2 - 4y = 0$
G_{31}	$y^2 - x^3 = 0$	$\varnothing$
$T_3 \times A_1$	$x = 0$	$y = 0$.

At G_{31}, the image on the two indecomposable invariants of degree 4 is the full S_3. The full cusp germs and the degrees of their two surface coordinates are

(d_1, d_2, d_3)	actual degrees	$\deg(B, C)$	D_C	D_P
211	1224	$(1, 2)$	$C^2 + B^4$	$\varnothing$
221	1112	$(1, 1)$	$C^2 + B^2$	B
212	1244	$(1, 4)$	$C + B^4$	C
222	1122	$(1, 2)$	$C + B^2$	BC.

Proof. The full parabolic normalizers [10, Theorems 2.1, 3.6, 5.14 and Tables 5, 6, 8] give the first table: use $(z_1^4 + z_2^4, z_1^4 z_2^4)$ for $G(4,2,4)$, independent axes for products, and for G_{31}

$$f_8 = a^8 + 14a^4b^4 + b^8, \qquad f_{12} = a^{12} - 33a^8b^4 - 33a^4b^8 + b^{12},$$
$$J = -1728a^3b^3(a^4 - b^4)^3, \qquad f_8^3 - f_{12}^2 = 108a^4b^4(a^4 - b^4)^4.$$

The six mirrors $0, \infty, \pm 1, \pm i$ give the full G_8 of order 96; on $(S, P) = (a^4 + b^4, (ab)^2)$, generators $A = (1 - i)^{-1} \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right), B = \mathrm{diag}(i, 1)$ act by $(-S/2 - 3P, -S/4 + P/2), (S, -P)$, giving S_3. For the full torsor [15, Lemma 2.17], take all permutations and phases in $\mu_{d_1} \times \mu_{d_2} \times \mu_{d_3} \subset \mu_4(x_1) \times \mu_4(x_2) \times \mu_2(y)$, $d_j \in \{1, 2\}$. Every component and affine translate is (a_0, a_0, t), $a_0 = (0, 1, i), d_1 = 2$; the other branch supports give A_1^2, B_2 or independent squares, and 111 gives none. The sections restrict to $(0, q^{d_2} x_2^{d_2}, (-1)^{d_3} q^{2d_3} y^{d_3}, q^4 x_1^4)$, yielding the four polynomial rings and divisors after removing exactly the scalars of order $g = d_2$ in the kernel of the fiber

character, $(q, c) \mapsto (q^g, c/g)$. The unique lift fixing the point preserves full stabilizers, and every pointwise scalar is trivial on q^g; only further quotients $1224, 1244 \mapsto 1112, 1122$ are smooth, so no component or smaller period is omitted. □

Lemma 6.7. There is a finite flat morphism $p : Y \to Z$ of degree 3, with Y normal. The trace kernel of p is

$$\mathcal{B} = N_{Z/X}(-D_C) \otimes \mathcal{L}^4|_Z = \mathcal{O}_Z(-b)^{\oplus 2}.$$

Its parameters satisfy

$$b = (4\lambda - 1)/3, \qquad \deg D_C = 4b, \qquad \deg D_P = 3(5-b)/2, \qquad b \in \{1, 3, 5\}.$$

There are exactly $3b^2$ distinct G_{31} points.

Proof. Let Π_Z be the effective full normalizer of the fixed ball, and let $\Sigma \subset Z$ contain the cusps, interior upper points of D_C, and singular points of $D_C \cup D_P$. The image on the positive definite normal lattice is finite. The projective image of T_2 is the Klein four group, whose three nonidentity elements have fixed point pairs $\mathcal{P} = \{\{0, \infty\}, \{1, -1\}, \{i, -i\}\}$. Its normalizer in $\mathrm{PGL}(2, \mathbb{C})$ permutes these three pairs; the kernel is the projective image of T_2, and the preceding matrices A, B give two transpositions. Thus every normal block has the form $Q = cQ_0$, with $Q_0 \in \langle T_2, A, B\rangle$ and $c \in K^*$. Both matrices have finite order, so $c \in \mu_4$. The scalar c acts trivially on $(S, P) = (y_1^4 + y_2^4, (y_1 y_2)^2)$. The pointwise group T_2 and the central lifts also act trivially. The resulting homomorphism $\rho : \Pi_Z \to S_3$ acts by the standard representation, which is self-dual through the invariant form $S^2 + 12P^2$.

Put $U = Z \setminus (D_C \cup \Sigma)$. The inertia along D_P is trivial under ρ, so the permutation action on the three pairs defines a finite étale cover of degree 3 over U. Explicitly, if $\mathbb{B}_2^\circ$ is the inverse image of U, this cover is $(\mathbb{B}_2^\circ \times \mathcal{P})/\Pi_Z \to U$, with $\gamma(z, p) = (\gamma z, \rho(\gamma)p)$ and $[(z, p)] \mapsto [z]$. Normalize Z in the function fields of its connected components to obtain $p : Y \to Z$; this is the associated cover of degree 3, rather than the principal cover defined by $\ker \rho$. We obtain flatness from normality in dimension 2 and the Auslander–Buchsbaum formula, and trace splits $p_*\mathcal{O}_Y = \mathcal{O}_Z \oplus \mathcal{B}$.

If $j(\gamma, z)$ is the automorphy factor on the fixed ball, the normal coordinates transform by $y \mapsto Qy/j(\gamma, z)$. Hence the homogeneous normal directions transform by $\rho(\gamma)j(\gamma, z)^{-4}$, and $\mathcal{L}^4$ removes precisely this automorphy factor. The bundle $\mathcal{L}^4$ descends on all of Z because $\mathcal{L}^{\otimes 2}$ already descends on X. At a general point of D_C, put $v = u^4$. The local cover is the disjoint union of one unramified sheet and $s^2 = v$, where $s = u^2$, so the trace kernel has frames $(2, -1, -1), (0, u^2, -u^2)$. These frames identify with the invariant normal vectors $(e_S, u^2 e_P)$. After the tangent direction is removed from the invariants $(v + S, vS + P^2, u^2 P)$, their normal differential is vI_2, with image $N_{Z/X}(-D_C)$. At D_P, the cover is unramified and the normal differential is invertible. We obtain $\mathcal{B} \otimes \mathcal{L}^{-4} \simeq N_{Z/X}(-D_C)$ on $Z \setminus \Sigma$, including the regular lattices along every divisor. Normalization uniquely determines the cover at the upper points of D_C and cusps; local freeness on both sides then extends the bundle isomorphism across Σ by Hartogs. The trace discriminant and the fixed ball canonical formula give $\deg D_C = 4(\deg D_C + 1 - 4\lambda)$ and $-3 + 3\deg D_C/4 + \deg D_P/2 = 3\lambda$. Miranda's cubic $F = a_0 s^3 + 3a_1 s^2 t + 3a_2 s t^2 + a_3 t^3$, $a_i \in H^0(\mathcal{O}_Z(b))$, defines a finite map $\phi : Z \to \mathbb{P}^3$, since a common zero would give multiplicity 4 [13, Theorem 3.6]. The discriminant germ $v'^2 - u'^3$, in new local coordinates u', v', along the twisted cubic W makes the points of $\phi^{-1}(W)$ simple by the permitted local types; conversely, the minors of $\left(\begin{smallmatrix} a_0 & a_1 & a_2 \\ a_1 & a_2 & a_3 \end{smallmatrix}\right)$ characterize every G_{31} point [13, Corollary 4.6]. The Hilbert–Burch sequence

$$0 \to \mathcal{O}_Z(-3b)^{\oplus 2} \to \mathcal{O}_Z(-2b)^{\oplus 3} \to \mathcal{O}_Z \to \mathcal{O}_{\phi^{-1}(W)} \to 0$$

gives total length $3b^2$. □

Lemma 6.8. For a smooth curve C in a fixed ball quotient of dimension 2 with $2L$ descended,

$$2L \cdot C \le 2g(C) - 2 + |S| + \sum_{z \notin S} (1 - 1/e_z).$$

Here S is its cusp set, and e_z is the effective inertia on the normalization of a lifted branch. Equality makes every lifted branch a complex geodesic.

Proof. On a normal neat cover of the same linear group, kill finite characters, normalize every component of the preimage of C, and divide Riemann–Hurwitz by the degree after removing the pointwise kernel. The restricted ambient canonical frame at the cusp introduces no parameter power; for curvature -1, $c_1(2L) = \omega/(2\pi)$, and the potential $-2\log(-c\log|t| + A(t))$ for $q = t^m a(t)$ has zero Lelong number and Stokes limit. Schwarz and Gauss–Bonnet give the formula, and metric equality gives a complex geodesic. □

Proposition 6.9. A coincidence plane with full generic stabilizer $G(4,2,2)$ has $\lambda > 1$.

Proof. Suppose $\lambda = 1$; then $b = (4\lambda - 1)/3 = 1$ by Lemma 6.7. For $b = 1$, the plane $\phi(Z)$ meets the twisted cubic W in three simple points; after $\mathrm{PGL}(2,\mathbb{C})$, its cubic, discriminant normalization, and tangent map are

$$F = x(s^3 - t^3) + 3ys^2t + 3zst^2,$$
$$[3s^2t^2 : -t(2s^3 + t^3) : s(s^3 + 2t^3)], \quad [s^3 - t^3 : 3s^2t : 3st^2].$$

The syzygies $(0, t, -s), (3st, -s^2, t^2)$ identify the incidence surface $\{F = 0\} \subset \mathbb{P}^2 \times \mathbb{P}^1$ with the Hirzebruch surface $\mathbb{F}_1 = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(1))$. The tangent map identifies only $[1:0], [0:1]$, so D_C is tricuspidal with a unique bitangent $\ell = (x = 0)$. For intersection k with $D_C + D_P$, omitting a component equal to the curve, effective inertia satisfies $1 - 1/e \le k/2$: use $2\operatorname{ord} J = ek$, $\operatorname{ord} J \ge e - 1 + (\nu - 1)^2$; for a component of $D_C + D_P$ use $2(e - 1 + \operatorname{ord} g) = \operatorname{ord} g + ek$. Equality requires one transverse reflection, or two orthogonal reflections for a component. If $\ell \not\subset D_P$, the tangencies of ℓ contribute $1/2$ each, strictly less than $1/2 + (\ell \cdot D_P)_z/2$ at points with a group in the first table of Lemma 6.6 or at cusps; its second table gives intersection at least 2, and all cusps lie on D_C. Since $\deg D_P = 6$, Schwarz gives equality in $2 = 2L \cdot \ell \le -2 + |S| + \sum(1 - 1/e_z) \le -2 + 1 + 3 = 2$. Both general interior branches $v = \pm u^2$ would be geodesic with the same point and tangent, contradicting uniqueness. If $\ell \subset D_P$, the bound is again $-2 + 2(3/4) + 5/2 = 2$, with equality forcing interior $G(4,2,4)$ tangencies and no cusp. Every D_P component nevertheless has a cusp for the same Γ. On the T_2 root space, the Hermitian form is aI_2, whose determinant is a norm from the Gaussian field. The primitive root of the mirror has norm 1 or 2, by self-duality; we obtain $\mathrm{N}(\delta)$ as the negative of the determinant of the Hermitian form on the mirror. For orthogonal norms $\beta > 0, \gamma < 0$, $-\beta\gamma = \mathrm{N}(\delta)$ gives isotropic $w = (\delta/\beta, 1)$; $Q(x) = h(w,x)w$ satisfies $Q^2 = 0$, and a positive power of the integral $I + miQ$ belongs to the original Γ and centralizes T_2. □

Proposition 6.10. For every coincidence plane, $\lambda_I \ge 3/2$. In particular, $2L - 3H$ is nef.

Proof. Lemma 3.3, Lemmas 6.4, 6.5, and Proposition 6.9 exclude $\lambda_I = 1$ in every rank. Half integrality from Proposition 6.2 gives $\lambda_I \ge 3/2$; apply [15, Proposition 2.6] to $D - 3H$ on each of the 35 planes. □

7 Chern classes and the seven point quotient

We now use $N = L - 3H/2$ nef and a lattice preserved by the original Gaussian group. The proof combines three kinds of data: absolute Chern classes from all fixed components, the determinant identity along each curve, and the relations (7.14) at interior points and cusps. Only the resulting nonnegative intersection numbers are compared; no degree bound for one component is required.

7.1 Chern classes of all fixed strata

Let X be a smooth projective variety of dimension d isomorphic to the Baily–Borel compactification of an arithmetic ball quotient. We keep its lattice Γ and automorphic class L fixed. For each full stabilizer P, let V_P be its nontrivial reflection representation and let Z_P be the sum of the reduced closures of the strata with generic stabilizer P. Count each component once on its full normalizer quotient. Put $\lambda_P = \#\{g \in P : V_P^g = 0\}/|P|$, $\mathcal{C}_k = \sum_{\operatorname{rank} P = k} \lambda_P [Z_P]$, and $\mathcal{C}_0 = [X]$. If $e_{P,1}, \dots, e_{P,k}$ are the degrees of the basic invariants, then

$$\lambda_P = \frac{\prod_i (e_{P,i} - 1)}{\prod_i e_{P,i}}. \tag{7.1}$$

One has $\sum_{g \in P} t^{\dim V_P^g} = \prod_i (t + e_{P,i} - 1)$; see [24, 7].

We use [16, Section 3.5], which averages over single group elements, together with the normalization of Chern–Schwartz–MacPherson classes in [16, Theorem 1.1]. The averaging and binomial inversion are formal. To apply this averaging, we use the full normalizer of each fixed ball, all its inertia groups, and the restriction of the original automorphic class L. The coefficient calculation retains every local branch of a self-intersection. The identities for $k < d$ concern pairings with divisor classes; the cusps contribute separately in degree d.

Proposition 7.1. Assume that the quotient of every positive dimensional fixed ball by its full normalizer has finite volume and has automorphic class equal to the restriction of L. For $0 \le k < d$, the following equality holds after pairing with products of divisor classes:

$$\mathcal{C}_k = \sum_{j=0}^{k} \binom{d-k+j+1}{j} L^j c_{k-j}(T_X). \tag{7.2}$$

It is an equality of numerical cycle classes if these pairings detect numerical equivalence. If c is the number of cusps, then

$$c + \deg \mathcal{C}_d = \int_X \sum_{j=0}^{d} (j+1) L^j c_{d-j}(T_X). \tag{7.3}$$

Proof. On a normal neat cover of the same Γ, apply Burnside's formula to fixed balls. For a full generic stabilizer P, and the normalizer $\mathcal{N}$ and centralizer $\mathcal{Z}$ in the deck group of an element fixing precisely this ball, so that $\mathcal{Z}$ normalizes P and $\mathcal{Z}P$ is their product subgroup, the coefficient on the quotient of the ball by the effective image of its full normalizer is $[\mathcal{N} : \mathcal{Z}P]/|\mathcal{Z} \cap P| = [\mathcal{N} : \mathcal{Z}]/|P|$. Summing its conjugacy classes gives λ_P, with every local branch retained. For $k < d$, a general complete intersection of $d-k$ sufficiently ample divisors avoids the cusps and higher rank strata. The fixed ball of dimension $d - s$ has orbifold Chern polynomial $(1-Lt)^{d-s+1}$ by [15, Equation 3.13]. Write $A_1, \dots, A_{d-k}$ for the divisor classes of this complete intersection Y, and put $A = A_1 \cdots A_{d-k}$. Burnside's formula and Gauss–Bonnet on each normalized fixed component give

$$e(Y) = \sum_{s=0}^{k} \int_X A \mathcal{C}_s \left[\frac{(1-L)^{d-s+1}}{\prod_{i=1}^{d-k} (1 + A_i)} \right]_{k-s},$$

where brackets denote the component of the indicated codimension. On the other hand, adjunction gives $e(Y) = \int_X A[c(T_X)/\prod_i(1+A_i)]_k$. Canceling the common denominator inductively in k, and polarizing the ample divisor classes, gives

$$c_k(T_X) = \sum_{s=0}^{k} (-1)^{k-s} \binom{d-s+1}{k-s} L^{k-s} \mathcal{C}_s.$$

Binomial inversion proves (7.2). For $k = d$, use compactly supported Euler characteristics on the open quotient and finite volume Gauss–Bonnet; the left side of the displayed formula for $c_k(T_X)$ with $k = d$ becomes $\int_X c_d(T_X) - c$. Inverting gives (7.3); ends of fixed components are already counted by these Euler characteristics. □

For the applications below, [15, Lemma 2.12] gives the finite maps from the quotients of the fixed balls by their full normalizers and proves finite volume. The automorphic class on each fixed ball is the restriction of the original class L, as in [15, Equation 3.13]. We use these geometric identifications and the coefficient calculation in the proof of (7.2). For (7.3), we use compactly supported Euler characteristics on the open quotient and retain the contribution of each cusp of X.

7.2 All Gaussian fixed components

Retain the group and integral lattice of Proposition 6.3, and put $H = -K_X$, $c_j = c_j(T_X)$. Every reflection has order 2, and its primitive root has norm 1 or 2. Proposition 6.10 gives

$$N = L - \frac{3}{2}H \text{ nef.} \tag{7.4}$$

We do not exclude either root norm or the reflection subgroups $G(4,2,2)$ and $G(4,2,3)$.

For a reflection group G, put $d_G = \sum_i e_{G,i}$, where the $e_{G,i}$ are its basic invariant degrees. The full generic stabilizers of fixed surfaces are $P_D = A_1^2$, $P_A = A_2$, $P_B = B_2$, and $P_T = G(4,2,2)$. Their degrees, weights from (7.1), and the coefficients c_P are

P	$(e_{P,i})$	λ_P	c_P
P_D	$(2,2)$	$1/4$	2
P_A	$(2,3)$	$1/3$	4
P_B	$(2,4)$	$3/8$	6
P_T	$(4,4)$	$9/16$	$15/2$

,

$$c_P = \frac{4\nu_P(\nu_P-1)}{|P|}, \quad \nu_P = d_P - 2.$$

The rank 2 classification over $\mathbb{Q}(i)$ leaves these groups and $I_2(6)$ among groups generated by reflections of order 2 [7]. For $I_2(6)$, primitive root norms $a, b \in \{1,2\}$ would satisfy $4|h(r,s)|^2 = 3ab$. The only integral possibility requires a Gaussian integer of norm 3, which does not exist. The groups on fixed curves are listed in Table 10. This list follows directly from the rank 3 classification over $\mathbb{Q}(i)$; no cusp is assumed on an individual curve.

Write Z_P for the sum of the reduced fixed surfaces of generic type P, and W_k for the corresponding sum of curves of type K_k in the table. Each component is counted once on its full normalizer quotient; different local branches of a self-intersection remain distinct. The number $n_{P,K}$ counts subspaces with full pointwise stabilizer P in K. Products give the reducible rows in the table. For $G(4,2,3)$, coordinate axes and axes with two coordinates equal up to a phase give the counts $12, 16, 0, 3$; for $G(4,4,3)$, they give $0, 16, 3, 0$. Define

$$v_{P,K} = (d_K - d_P - 1)\frac{|P| n_{P,K}}{|K|}.$$

No rank 4 parabolic counts will be needed.

K_k	$(e_{K_k,i})$	$\lvert K_k\rvert$	$(n_{D,K_k}, n_{A,K_k}, n_{B,K_k}, n_{T,K_k})$	λ_{K_k}
$K_0 = A_1^3$	$(2,2,2)$	8	$(3,0,0,0)$	1/8
$K_1 = A_2 \times A_1$	$(2,2,3)$	12	$(3,1,0,0)$	1/6
$K_2 = B_2 \times A_1$	$(2,2,4)$	16	$(4,0,1,0)$	3/16
$K_3 = A_3$	$(2,3,4)$	24	$(3,4,0,0)$	1/4
$K_4 = B_3$	$(2,4,6)$	48	$(6,4,3,0)$	5/16
$K_5 = G(4,4,3)$	$(3,4,8)$	96	$(0,16,3,0)$	7/16
$K_6 = G(4,2,2) \times A_1$	$(2,4,4)$	32	$(6,0,0,1)$	9/32
$K_7 = G(4,2,3)$	$(4,6,8)$	192	$(12,16,0,3)$	35/64

Table 10: Groups on fixed curves and counts of their parabolic subgroups of rank 2.

Lemma 7.2. With the classes ψ_i, p of Lemma 2.1, one has

$$\begin{gathered}\int_X H^4 = 154, \qquad \int_X H^2 p = -14, \qquad \int_X p^2 = 10, \qquad \int_X c_4 = 38, \\ Hp \equiv -\frac{1}{11}H^3, \qquad c_3 \equiv \frac{4}{11}H^3, \qquad p^2 \equiv \frac{5}{77}H^4.\end{gathered} \tag{7.5}$$

The nef cone is generated by $U_i = (H - \psi_i)/2$ and $V_i = (H + \psi_i)/2$, $1 \le i \le 7$. For $E = \frac{1}{2}\sum_i a_i\psi_i$, $E' = \frac{1}{2}\sum_i a_i'\psi_i$, $s = \sum_i a_i$, and $s' = \sum_i a_i'$, the intersection form is

$$\int_X H^2 EE' = \frac{3}{2}ss' - 5\sum_i a_i a_i', \qquad -\int_X pEE' = \frac{1}{2}ss' - 3\sum_i a_i a_i'. \tag{7.6}$$

Proof. For a chamber divisor D of multidegree (b_i), put $q = \frac{1}{2}\sum b_i$ and $d_I = q - \sum_{i\in I} b_i$. Subtracting torus weight multiplicities 2 from 0 in the product of seven characters gives

$$h^0(X, kD) = \sum_{|I|\le 3} (-1)^{|I|} \binom{kd_I + 5 - |I|}{5}.$$

Its leading coefficient yields $\int_X D^4 = 3q^4 - 2\sum_i (q - b_i)^4 + \sum_{i<j}(q - b_i - b_j)^4$. Polarize and use $\psi_i^2 = p$: the intersections of four distinct ψ_i are 2, $p\psi_i\psi_j = -2$ for $i \ne j$, and $p^2 = 10$. Expanding $H = \sum_i \psi_i$ and the tangent sequence gives the stated numbers, (7.6), and $c_4 = (H^4 - 22H^2p + 45p^2)/24$. Divisor classes generate the rational Chow ring [5, Lemma 8, Theorem 10, Remark 3], so the symmetric pairings determine the numerical relations. The criterion of [15, Proposition 2.6], for triples of labels, gives the 14 nef generators. □

Let $\mathscr{B}$ be the reduced reflection divisor, as in Lemma 4.10. Proposition 7.1 gives

$$\begin{gathered}\mathscr{B} = 10L + 2H, \qquad \sum_P \lambda_P Z_P = c_2 + 4LH + 10L^2, \\ \sum_{k=0}^{7} \lambda_{K_k} W_k = c_3 + 3Lc_2 + 6HL^2 + 10L^3.\end{gathered} \tag{7.7}$$

The normal bundle of the fixed ball of a reflection has class $-L$. Counting ordered pairs of distinct local mirrors, including branches of one global divisor, gives $\sum_P c_P Z_P = \mathscr{B}^2 + 2L\mathscr{B}$. Solving this and the second equation in (7.7) yields

$$\begin{aligned} Z_D + \frac{2}{3}Z_A + \frac{3}{4}Z_T &= 20L^2 + 10LH + 2H^2 - 20p, \\ \frac{2}{3}Z_A + \frac{3}{2}Z_B + \frac{3}{2}Z_T &= 20L^2 + 6LH + 10p. \end{aligned} \tag{7.8}$$

Proposition 3.1 of [15], with $d = 4, k = m = 2$, gives

$$(H + (d_P + 3)L)Z_P = \sum_k v_{P,K_k} W_k. \tag{7.9}$$

Finally, if c is the number of cusps and n_J the number of points of full rank 4 type J, put

$$F(L) = \int_X (c_4 + 2Lc_3 + 3L^2c_2 + 4HL^3 + 5L^4) = c + \sum_J \lambda_J n_J. \tag{7.10}$$

7.3 The same determinant on every curve

For J as in (7.10), let $\gamma_k(J) = |K_k| n_{K_k,J}/|J|$, where $n_{K,J}$ counts its axes with full pointwise stabilizer K. At a cusp c', let $\gamma_k(c')$ be the degree of the stack points with nonzero fiber stabilizer K_k in the weighted projective quotient of the horizontal torus. Write $b(c')$ for the generator degrees measured by the least positive central period of the full cusp group, and put $s_{c'} = \sum_j b_j(c')$. With $A_k = \int_X HW_k$, $T_k = \int_X LW_k$, Proposition 3.1 of [15] gives

$$A_k + (d_{K_k} + 2)T_k = \sum_J (d_J - d_{K_k} - 1)\gamma_k(J) n_J + \sum_{c' \text{ cusp}} s_{c'} \gamma_k(c'). \tag{7.11}$$

Every local axis extends to a fixed curve for the full normalizer by [15, Lemma 2.12]. The sum over its local axis orbits is $\sum 1/e = |K| n_{K,J}/|J|$. At a cusp, [15, Lemma 2.17] gives the unique lift fixing the point in the same Γ; its disk gives the corresponding cusp of the fixed curve, with contribution $1/\kappa$ for the full fiber character of order κ. Thus every stack point counted by $\gamma_k(c')$ occurs, with its actual width. The identity includes compact fixed curves, for which the cusp sum is empty. Nefness in (7.4) gives $T_k \geq 3A_k/2$, independently of the degree and genus of each component.

The possible degree tuples of the groups J are

$$\begin{array}{cccc} (2,2,2,2), & (2,2,2,3), & (2,2,2,4), & (2,2,3,3), \\ (2,2,3,4), & (2,2,4,4), & (2,2,4,6), & (2,3,4,4), \\ (2,3,4,5), & (2,3,4,8), & (2,4,4,4), & (2,4,4,6), \\ (2,4,6,8), & (2,6,8,12), & (4,4,4,4), & (4,4,8,12), \\ (4,8,8,12), & (4,8,12,20), & (8,12,20,24). & \end{array} \tag{7.12}$$

The irreducible groups are $A_4, D_4, B_4, F_4, G(4,4,4), G(4,2,4), G_{29}, G_{31}$; the remaining tuples come from products [24, 7]. At F_4, the only parabolic types of rank 3 are $A_2 \times A_1$ and B_3, by deleting one vertex of its Coxeter diagram. Accordingly $\gamma_k(F_4) = 0$ for $k \notin \{1, 4\}$. No other vanishing of $\gamma_k(J)$ is used.

The generator degrees $b(c')$ lie in

$$1111, \quad 1112, \quad 1122, \quad 1222, \quad 1124, \quad 1144, \quad 1224, \quad 1244. \tag{7.13}$$

The full cusp group is generated by actual reflections, so its horizontal image is generated by reflections of order 2. The horizontal groups are $A_3, B_3, G(4,4,3), G(4,2,3)$; the product of the nonzero entries of a monomial matrix is multiplicative and equals ± 1 on reflections of order 2, so these reflections cannot generate $G(4,1,3)$, which has an element with value i. For $G(4,2,3)$, the five rows of [22, Section 4.4], including its torsor, have parameters

$$(d_1, d_2, d_3) = (1,1,2), (1,2,1), (1,2,2), (2,2,1), (2,2,2).$$

The model in [22, Example 3.10] has degrees $(d_1, d_2, 2d_3, 4)$. Removing exactly the scalars μ_g in the kernel of the fiber character gives $1144, 1224, 1244, 1112, 1122$, respectively; $G(4,4,3)$

gives 1124, and A_3, B_3 give the first four tuples. This operation is $q_{\mathrm{new}} = q_{\mathrm{old}}^{g}$, $c_{\mathrm{new}} = c_{\mathrm{old}}/g$, and returns the grading to the least positive central period of the full cusp group. Descent along a free translation cover retains the linearization of the full group and the unique lifts fixing the points. Any further scalar group with smooth punctured quotient has order dividing $\operatorname{lcm}_j \gcd_{i\neq j} b_i$, since the finite quotient group is generated by reflections. The only nontrivial cases are $1222 \to 1111$, $1224 \to 1112$, and $1244 \to 1122$, so no additional tuple occurs.

7.4 Quotients of projective space and abelian threefolds

Proposition 7.3. Let $M = \mathbb{P}^3$, $\Theta = \mathcal{O}(1)$, $\varepsilon = 1$, or let M be an abelian threefold with an ample line bundle Θ, $\varepsilon = 0$. Let a finite group act on (M, Θ), faithfully on the punctured total space of Θ^{-1}, with a polynomial ring of invariant sections whose generators have degrees $b = (b_1, \dots, b_4)$. Assume that the punctured quotient is smooth, that every actual reflection has order 2, and that the stabilizers of ranks 2 and 3 have the types above. On the weighted projective stack of weights b, let β_P be the degree of the reduced curve images of generic type P, measured by $\eta = c_1(\mathcal{O}(1))$, and let γ_k be the degree of its stack points of type K_k. The group at a point is the kernel of the full character on the nonzero fiber, and all residual stabilizer orders are retained in these degrees. Put $s = \sum b_i$, $v = \prod b_i$, and let e_j denote the elementary symmetric polynomial of degree j. Then $\beta_P, \gamma_k \geq 0$ and

$$\begin{aligned} \sum_P \lambda_P \beta_P &= \frac{e_2(b-\varepsilon)}{v}, \qquad \sum_P c_P \beta_P = \frac{4(s-4\varepsilon)(s-5\varepsilon)}{v}, \\ \sum_k \lambda_{K_k} \gamma_k &= \frac{e_3(b-\varepsilon)}{v}, \qquad (s-\varepsilon(d_P+2))\beta_P = \sum_k v_{P,K_k}\gamma_k. \end{aligned} \tag{7.14}$$

Here $b - \varepsilon = (b_i - \varepsilon)_i$; any common divisor of the b_i is retained.

Proof. The sections give a finite map from M to the weighted projective stack, pulling $\mathcal{O}(1)$ back to Θ. The sections have no common zero on M; after base change by the atlas $\mathbb{A}^4\backslash\{0\}$, this map is the finite quotient of the punctured Θ^{-1} torsor by the stated group. For a horizontal point with stabilizer F_x and fiber character χ_x, its residual order is $r = |\operatorname{im}\chi_x|$, and orbit counting assigns weight $|\ker\chi_x|/|F_x| = 1/r$. Thus a point contributes $1/r$ to γ_k; a curve contributes its degree against η, with the same weight at its generic point, to β_P. For example, a weighted projective coordinate point of residual order r has stack degree $1/r$, rather than coarse degree one. Apply Proposition 7.1 to general complete intersections. The fixed components are projective linear spaces for $\varepsilon = 1$ and translates of subtori for $\varepsilon = 0$, with Chern polynomials $(1+\varepsilon\eta)^{4-a}$ in codimension a. The weighted Euler sequence gives $c_j = e_j(b)\eta^j$ and $\int \eta^3 = 1/v$. Binomial inversion yields the first and third equations. For the reduced branch divisor, ramification gives $\mathscr{D} = 2(s-4\varepsilon)\eta$. Counting distinct ordered mirrors, whose normal class is $\varepsilon\eta$, gives $\sum_P c_P \mathcal{Z}_P = \mathscr{D}^2 - 2\varepsilon\eta\mathscr{D}$ for the cycle classes $\mathcal{Z}_P$ on the stack of the curve images of type P; since $\eta\cdot\mathcal{Z}_P = \beta_P$, this proves the second equation. On a normalized curve of type P, the tangent and the two invariant normal directions have total first Chern class $\varepsilon(d_P+2)\eta$. The same determinant calculation as in (7.9), with its full characters, gives the last four equations. □

7.5 A comparison of all components

We first explain the geometric inputs to the final inequality. The classes H, p come from the GIT quotient X and L from the automorphic bundle, while Z_P and W_k record every fixed surface and curve with its full normalizer. Equations (7.8), (7.9), and (7.11) give linear equalities among their intersection numbers. Nefness of $L - 3H/2$ and the generators of the GIT nef cone give inequalities among the same numbers. The quotients of $\mathbb{P}^3$ at interior points and of horizontal tori at cusps satisfy (7.14); their degrees β_P, γ_k include the finite fiber characters and all local

branches. Thus the global equalities, nef inequalities, and (7.14) constrain one collection of intersection numbers. The rational linear combination below separates these constraints: it is nonnegative by nefness and strictly negative after the equalities and local estimates are substituted. Its coefficients are a certificate of this incompatibility, rather than an additional geometric assumption.

The same method on another smooth projective variety would require its intersection relations and nef inequalities, finite maps from all relevant fixed balls by their full normalizers with the same restricted automorphic class, and the local degree formulas at every interior point and cusp. Proposition 7.1 alone does not supply these inputs or a nonexistence theorem for another variety.

For any two nef classes E, E', the fourteen generators in Lemma 7.2 give

$$0 \le -\int_X pEE' \le \frac{1}{7}\int_X H^2EE'. \tag{7.15}$$

Put $l = \int_X L^4 > 0$ and normalize all intersection numbers by l:

$$\begin{aligned}(z_P, h_P, j_P) &= l^{-1}\int_X (L^2Z_P, HLZ_P, H^2Z_P),\\ (a_k, w_k) &= l^{-1}(A_k, T_k),\\ (\rho, x, u, y, z_4) &= l^{-1}\int_X (HL^3, H^2L^2, -pL^2, H^3L, H^4).\end{aligned}$$

All these numbers are nonnegative. Nefness of N, L, H and (7.15) imply

$$\rho, x, y, z_4 \le 1, \qquad u \le 1/7, \qquad z_P, h_P, j_P \le 64, \qquad a_k, w_k \le 160, \qquad F(L)/l < 13. \tag{7.16}$$

One has $\rho \le 2/3$, $x \le 2\rho/3$, $y \le 2x/3$, and $z_4 \le 2y/3$. The two equations in (7.8), paired with L^2, bound their positive left sides by $35, 26$, respectively, and give $z_P < 64$. The inequalities $z_P \ge 3h_P/2 \ge 9j_P/4$ give the bounds on h_P and j_P. The last equation in (7.7), paired with L, gives

$$\sum_k \lambda_{K_k} w_k = 10 + 6\rho + \frac{3}{2}x + \frac{15}{2}u + \frac{4}{11}y < 19.$$

Since $\lambda_{K_k} \ge 1/8$ and $w_k \ge 3a_k/2$, the bounds on a_k, w_k follow. Finally,

$$F_0 = F(L)/l = 5 + 4\rho + \frac{3}{2}x + \frac{15}{2}u + \frac{8}{11}y + \frac{19}{77}z_4 < 13.$$

Lemma 7.4. Equations (7.7)–(7.11) and (7.14) contradict (7.4).

Proof. Let $n_\nu, c_\nu \ge 0$ be the numbers of interior points and cusps, respectively, with the degree tuple b and the vector (β_P, γ_k) of (7.14) indexed by ν, divided by l. For interior points put $\Lambda_\nu = \prod_i (b_i - 1)/\prod_i b_i$. Then $F_0 = \sum_\nu \Lambda_\nu n_\nu + \sum_\nu c_\nu$.

Let R_1, R_2 be the left sides minus the right sides of (7.8). For $(D_1, D_2, D_3) = (L^2, HL, H^2)$, put $E_{2j-1} = l^{-1}\int_X R_1 D_j$ and $E_{2j} = l^{-1}\int_X R_2 D_j$. The next four relations, $E_7, \dots, E_{10}$, are (7.9) paired with L, in the order D, A, B, T. Let E_{11} be the last equation in (7.7) paired with L, and let $E_{12} = F_0 - \sum \Lambda_\nu n_\nu - \sum c_\nu$. Let $E_{13}, \dots, E_{20}$ be (7.11), for $k = 0, \dots, 7$, with the actual γ_k and the counts n_ν, c_ν. Finally, $E_{21}, \dots, E_{24}$ are (7.9) paired with H, and E_{25} is the last equation in (7.7) paired with H. All are left sides minus right sides, divided by l, and all vanish.

The following nine quantities are nonnegative:

$$\begin{gathered}\mathcal{G}_P = h_P - \tfrac{3}{2}j_P \quad (P = D, A, B, T),\\ \mathcal{G}_5 = \tfrac{1}{7}x - u - \tfrac{6}{77}y, \qquad \mathcal{G}_6 = \tfrac{1}{7}x - u - \tfrac{12}{77}y + \tfrac{9}{77}z_4,\\ \mathcal{G}_7 = w_0 - \tfrac{3}{2}a_0, \quad \mathcal{G}_8 = w_5 - \tfrac{3}{2}a_5, \quad \mathcal{G}_9 = w_7 - \tfrac{3}{2}a_7.\end{gathered}$$

The first four follow from $NHZ_P \geq 0$; the next two follow from (7.15) applied to (N, L) and (N, N), respectively. Use the order $\mathcal{G}_D, \mathcal{G}_A, \mathcal{G}_B, \mathcal{G}_T, \mathcal{G}_5, \ldots, \mathcal{G}_9$. Define α_j, μ_i by the following integer arrays, divided by 10^4:

$$10^4(\alpha_1, \ldots, \alpha_{25}) = \begin{matrix} 125599 & -26317 & 838943 & -156924 & -2996 \\ -26435 & -17943 & -8274 & 4386 & 0 \\ -1024187 & -1649243 & 13946 & 14520 & 14040 \\ 17758 & 17386 & 31752 & 18397 & 26604 \\ -106826 & -49999 & 26631 & -16462 & -1519032 \end{matrix},$$

where the entries are read successively along the rows, and

$$10^4(\mu_1, \ldots, \mu_9) = \begin{matrix} 73215 & 46413 & 8681 \\ 38908 & 6326668 & 1136405 \\ 10462 & 157397 & 79931 \end{matrix},$$

with the same row order. Put $Q = \sum_i \mu_i \mathcal{G}_i - \sum_j \alpha_j E_j \geq 0$. Expanding Q gives the constant term $-2003/2000$. Among its global variables, the positive coefficients of z_P, h_P, j_P have sum at most $29/30000$, and those of a_k, w_k sum to $437/320000$. The only other positive terms are $13x/14000 + 2y/9625$. Thus (7.16) bounds the total positive global contribution by

$$64 \cdot \frac{29}{30000} + 160 \cdot \frac{437}{320000} + \frac{1}{880} = \frac{5806}{20625} < \frac{3}{10}.$$

For a degree tuple b, put $\Lambda = \prod(b_i - 1)/v$ for an interior point, or $\Lambda = 1$ for a cusp. The coefficient in Q of the normalized count of b is

$$\omega_\varepsilon(b, \gamma) = \alpha_{12}\Lambda + \sum_{k=0}^{7} \alpha_{13+k}\big(s - \varepsilon(d_{K_k} + 1)\big)\gamma_k.$$

The seven equations in (7.14) give a common bound for this expression. Set

$$(r_0, r_1, r_2) = 10^{-4}(173706, -35568, 1857),$$
$$U = r_0 s - 191\varepsilon, \qquad V = r_1 s^2 + \tfrac{213}{2}\varepsilon s - 536\varepsilon^2, \qquad W = r_2 s^2 - 7\varepsilon s + 39\varepsilon^2,$$
$$\zeta_P = (r_1 \lambda_P + r_2 c_P)s + \varepsilon\eta_P, \qquad (\eta_D, \eta_A, \eta_B, \eta_T) = (19/2, 9/2, -19/5, 7/5).$$

Direct substitution of the four groups P and the eight groups K_k gives

$$U\lambda_{K_k} + \sum_P \zeta_P v_{P,K_k} \geq \alpha_{13+k}(s - \varepsilon(d_{K_k} + 1)),$$
$$V\lambda_P + W c_P \geq \zeta_P(s - \varepsilon(d_P + 2)).$$

For $\varepsilon = 1$, both differences are affine in s; the quadratic terms in the second cancel. It suffices to substitute $s = 8, 64$, where their respective minima are $687/5000$ and $39/125$. For $\varepsilon = 0$, the second difference is zero and the first is s times a positive coefficient, whose minimum over the eight rows is $1/80000$. Multiply the first inequalities by γ_k, apply the last four equations in (7.14), multiply the second by β_P, and apply the remaining three equations. This proves

$$\omega_\varepsilon \leq \alpha_{12}\Lambda + \frac{Ue_3(b - \varepsilon) + Ve_2(b - \varepsilon) + 4W(s - 4\varepsilon)(s - 5\varepsilon)}{v}.$$

For the eighteen degree tuples of interior points other than F_4, the fraction divided by Λ is at most $2075924/13125 < -\alpha_{12}$. For the eight tuples in (7.13) its maximum is $6597/40$, giving $\omega_0 \leq 7/10000$. At F_4, the vanishing of γ_k for $k \notin \{1, 4\}$ and the seven equations give uniquely

$$(\beta_D, \beta_A, \beta_B, \beta_T) = (1/4, 1/6, 1/8, 0), \qquad \gamma_1 = \gamma_4 = 1, \quad \gamma_k = 0 \ (k \neq 1, 4).$$

Substitution gives the common bound

$$\max \frac{\omega_\varepsilon(b,\gamma)}{\Lambda} = \frac{493}{154000} < \frac{1}{250}. \tag{7.17}$$

Thus (7.17) follows from the inequalities at $s = 8, 64$ and substitution of the degree tuples at interior points and cusps, without enumerating vertices. Consequently the positive local contribution is at most $F_0/250 < 13/250$. We obtain

$$0 \le Q \le -1 + \frac{3}{10} + \frac{13}{250} = -\frac{81}{125} < 0,$$

a contradiction. □

Theorem 7.5. The hypothetical Gaussian arithmetic ball quotient with coarse compactification X_7 does not exist.

Proof. Propositions 6.2, 6.3, and 6.10 give all hypotheses of Lemma 7.4, in the same original group. The comparison uses neither integrality of L nor the exclusion of a root norm, and imposes no index or degree bound on an individual fixed component. □

Theorem 4.18, Proposition 5.1, and Theorem 7.5 prove Theorem 1.1.

A Local calculations and subgroup presentations

A.1 Gaussian cusp calculations

Lemma A.1. The two formulas and every local input in Lemma 6.5 use the full normalizers. In the second and third rows of Lemma 6.5, the effective image of the full normalizer has exactly one cusp, with the holonomy and the positive line bundle on the elliptic curve stated there.

Proof. Substitute axes $(u+x, u-x, -u+y, -u-y)$, $(u+x, u-x, y)$ into the basic invariants, with frames (e_+, ue_-), $(u^2 e_{x^2}, e_{y^2})$ for A_3, T_3. In the order of the six groups in Lemma 6.5, determinant orders $(2,3,3,1,1,0)$, corrections $(-1/2, 0, -1/2, 0, -1/2, 0)$, and inertia $(2,2,4,2,4,2)$ give both formulas. The full list of parabolic subgroups [10, Tables 5, 6, 8 and Theorem 3.6] leaves finite types $T_2 \times A_1^2, T_3 \times A_1, T_2^2$, with normalizers $\mu_4 \times \mu_2, G(4,1,2), \mu_4^2$, inertia $2, 4, 4$, and intersections $1, 2, 0$. At A_3, let E denote the elliptic curve of this Gaussian cusp, with the sum map and polarization used above. The components $(x, x, t-x, t-x)$, $t \in E[2]$, map to conics in $|4[0]|$, excluding a smooth surface germ. For exchanges at B_3, two long reflections give a curve of D_A, and a long and a short reflection give a curve of D_B, on the same fixed surface, contrary to $d_A = d_B = 0$. For different coordinate labels, $(x_j^2) = (U, T, T-U, T-c_0U)$ gives invariant rings $\mathbb{C}[u,v]$ and

ambient degrees	D_P
1122	$(u^2-v^2)(c_0u^2-v^2)$
1222	$(v-u^2)(c_0v-u^2)$
1111	$(u-v)(c_0u-v)$

.

Only $2222/\mu_2$ fits row three: its two lines meet at one cusp, with the central period of the ambient cusp, order 2, and a line bundle of degree 2; none fits row two. The phases in μ_4 likewise leave different coordinate labels; for $D_4 = q_{\mathrm{amb}}^2 x_2^4$, row one gives

parameter	C	D_Q	D_R
221	$iq_{\mathrm{amb}}y$	$D_4(D_4 - C^2)$	$\varnothing$
222	$-q_{\mathrm{amb}}^2 y^2$	$D_4(D_4 - C)$	$C.$

These are the required tacnode and node; row three excludes components with phases of order 4, and row two excludes 212 by intersection $4 > 2$ and 221 by contact 2. For 222, put $a_0 = (0, 1, i), a_1 = (1, 0, 1), r = (1, 1, 0)$; the components $(a_0, a_1, t), (a_0, r, t)$ give the pairs $(D_4(D_4 - C), C), (D_4 - B^2, D_4)$, each exhausting the intersection and hence giving exactly one cusp. Restoring the phases gives both scalars $(-1)^{b_1+a_2+u}$ for the first component and, by $a_2 = b_2, b_1 + b_2 + u = 0 \bmod 2$, the trivial scalar for the second. Thus $q_Z = q_{\mathrm{amb}}^2, q_{\mathrm{amb}}$, respectively; the full holonomy μ_4 and the actual degrees $(1, 1), (1, 2)$ give line bundles of degrees $4, 2$. We obtain the stated central periods using the full cusp groups, including the freely acting deck transformations. □

Lemma A.2. A Gaussian Picard lattice with orbifold Euler characteristic 3 lies in a maximal lattice of index 96. The maximal lattice has one cusp and horizontal group of order 4. The positive line bundle on its elliptic curve has degree 4 or 2.

Proof. Let Π be the Gaussian lattice in the statement, Δ a maximal lattice containing it, and $\widetilde{\Delta} \subset \mathrm{SU}(2, 1)$ its full inverse image. For the principal arithmetic subgroup Λ_{pr} normalized by $\widetilde{\Delta}$, put $n = [\widetilde{\Delta} : \Lambda_{\mathrm{pr}}]$, $m = [\Delta : \Pi]$. Use $A = \prod_p \lambda_p$ and I for its parahoric subgroups over $\mathbb{Q}(i)$, with the local factors and split Iwahori primes defined above. Here $\Lambda_{\mathrm{pr}} \cap \mu_3 = 1$, but $\mu_3 \subset \widetilde{\Delta}$. By the volume and normalizer formulas [27, Sections 3.3–4.1], [20, Section 2.3], with $-L_{\mathbb{Q}(i)}(-2)/48 = 1/96$, we have

$$3 \le n \le 3^{1+|I|}, \qquad \chi_{\mathrm{orb}}(\mathbb{B}^2/\Delta) = 3A/(32n), \qquad mA = 32n. \tag{A.1}$$

The nonminimal factors $(p^2 + p + 1)(p + 1)$, $p \ge 5$, $p^2 - p + 1$, $p \ge 3$, force $I = \varnothing, A = 1, n = 3$. Class number 1 [20, Proposition 5.3] leaves the two ramified vertices, with Euler number $1/32$, index 96, one cusp [28, Theorem 4.8], and models

$$J = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad M_0 = \mathbb{Z}[i]^3,$$
$$M_s = \mathbb{Z}[i]e_1 + \mathbb{Z}[i]e_2 + \mathbb{Z}[i]\rho e_3, \qquad \rho = (-1 + i)/2.$$

The vertex bases $((1 - i)^{-1}e_1, e_2, \rho e_3), (e_1, e_2, \rho e_3)$ identify the standard group and the other maximal lattice by Gram reduction modulo $1 - i$ [30, Section 3.11]. For

$$U(z, t) = \begin{pmatrix} 1 & -\bar{z} & (-|z|^2 + it)/2 \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix},$$

the full Heisenberg lattices are

$$N_0 = \{U(z, t) : z \in (1 - i)\mathbb{Z}[i],\ t \in 2\mathbb{Z}\},$$
$$N_s = \{U(z, 2a) : z \in (1 - i)\mathbb{Z}[i],\ a \equiv |z|^2/2 \bmod 2\}.$$

The full groups are $N_0 \rtimes \mu_4, N_s \rtimes \mu_4$, with least positive central periods $t_0 = 2, 4$, area 2; commutator $U(0, 4\,\mathrm{Im}(z\bar{w}))$ gives degrees $4 \cdot 2/t_0 = 4, 2$. □

A.2 Presentations and enumeration for four Eisenstein lattices

In this subsection, $K = \mathbb{Q}(\omega)$, $\omega^2 + \omega + 1 = 0$, $\mathcal{O} = \mathbb{Z}[\omega]$, and $\delta = 1 + 2\omega$. We retain the groups and indices from Lemma 5.3. Write E_{ij} for matrix units of size 3. For M_{00} use $P = I + E_{12} - \omega E_{13} + (\omega - 1)E_{22} + \omega E_{23}$, $Q = I + E_{12} - \omega E_{13} - 2E_{22} - E_{23}$, $R = -E_{13} - E_{22} - E_{31}$, $J = RP$, $\sigma = QP^{-1}$. The group M_{00} has the presentation $J^3 = \sigma^6 = (J\sigma^{-1}J)^4 = 1$,

$[\sigma,(J\sigma^{-1}J)^2]=1$; its cusp group is $\langle P,Q\rangle$ [4, Theorem 5.9, Proposition 5.11]. For M_{01} use $S=I+E_{12}+\overline{\omega}\delta E_{13}/3-2E_{22}-E_{23}$, $T_0=I-\delta E_{13}/3$, $I_s=-\delta E_{13}/3-\omega E_{22}-\delta E_{31}$. With $j=S^{-1}I_s$, $r=T_0^{-1}I_s$, $z=jr^{-1}j=I_s$, the group M_{01} has the presentation $j^3=r^3=z^{12}=1$, $[r,z^2]=1$; its cusp group is $\langle z^2,S\rangle$ [32]. These presentations come from fundamental domains.

Set $H_\eta=M_{0\eta}\cap M_{1\eta}$. Reduction modulo 2 has image of order 216, transitive on nine isotropic lines; H_η fixes the first line and has indices 9,3. Reidemeister–Schreier rewriting of this action on nine points gives ten generators and 36 relations. For $U=I+\delta E_{13}/2$, generators of M_{10} are $h_1=J^{-1}\sigma J$, $h_2=\sigma$, $h_3=(J\sigma J^{-1})\sigma^3(J\sigma^{-1}J^{-1})$, U; those of M_{11} are $k_1=S^{-1}$, $k_2=r^{-1}I_sr$, U. The right coset representatives of H_η in $M_{1\eta}$ are $1,U,UC$, with $C=h_3,k_2$ respectively. To obtain a presentation, rewrite $r_ig=w_{i,g}r_{\kappa(i,g)}$ by selecting the unique representative whose quotient fixes the first isotropic line modulo 2. Enumerate $w_{i,g}$ in h_1,h_2,h_3 or k_1,k_2 until the projective matrices agree. The presentation obtained from the action on nine points verifies that these generate H_η, guaranteeing termination; the rewritings and its 36 relations present $M_{1\eta}$.

Enumerate transitive actions by extending the first undefined generator image, propagating inverses and all relators, rejecting contradictions or excess points, and relabeling breadth first from every base point. A branch assigns at most n images for each of the finitely many generators, where $n=24$ or 8; each unassigned image has at most n choices. Propagation only fills these entries or rejects the branch, so the enumeration terminates without a bound on word length or running time. Every complete transitive action extends a branch, and relabeling from every base point identifies precisely simultaneous conjugacy, so the procedure is exhaustive. At index 24 in $M_{0\eta}$, require a conjugate of σ and exclude $J,(\sigma J)^3$ or j respectively; their isolated full stabilizers of orders 3,4 contain no reflection [3, Section 7, Tables 9–10]. This leaves eight and six actions. At index 8 in $M_{1\eta}$, $\Delta=\Lambda\cap H_\eta$ satisfies $3[H_\eta:\Delta]=8[\Lambda:\Delta]$ and $[\Lambda:\Delta]\le 3$, hence the indices are 8,3. There are 20 actions of H_0 and 12 actions of H_1 on eight points. Induce them to $(a,i)g=(aw_{i,g},\kappa(i,g))$ on 24 points; extensions are the 64 blocks of size 3 through the base point, one point per fiber. Testing the translates by every generator and its inverse for equality or disjointness and canonically relabeling leaves four actions for each η.

For $\epsilon=1$ the full cusp groups are $\langle P,Q,U\rangle$, $\langle z^2,S,U\rangle$. Their horizontal lattice is $\mathcal{O}$ and central period is $\delta/(2^\epsilon 3^\eta)$. Since $1+\overline{\omega}$ is a unit, every element of order 6 in the cusp stabilizer is conjugate to $\sigma^{\pm1}$. Requiring a σ fixed point in every cusp orbit leaves the full $4,4,2,2$ cases.

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